Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment, Short-course (TB-DOTS) Program

Aaron Kyle M. Aguilar, Mark Lestat C. Agustin, Keanu Sonn M. Fortaleza,
Benny Gil A. Lactaotao, Marven Joffre G. Luis, Georcelle Leisley U. Nuarin,
Sean Drei A. Octavo, and Gebreyl Isaac P. Rabang

Department of Computing and Information Studies, Saint Louis University

CS421, CS THESIS 2

Josephine S. Dela Cruz, DIT

May 15, 2026

Table of Contents

Abstract

TB does not go away easily in the Cordilleras. Patients vary. The program's coverage varies. Put together, those two things keep producing treatment failures that nobody catches until they have already happened. We wanted to see whether we could predict those failures earlier with machine learning. The data covers DOH-CAR facilities from 2015 to 2025. Before any model saw the records, they went through a preprocessing pipeline. The pipeline does the basic hygiene work, schema normalization and patient pseudonymization, and then handles missing data diagnostically. That last part is the piece worth calling out. Instead of imputing every gap the same way, the pipeline first asks how the variable went missing (MCAR, MAR, or MNAR) and routes it accordingly, to exclusion, to listwise deletion in narrow cases, to a missing-indicator-with-fill, or to stochastic MICE only where the assumptions actually hold. For the modeling itself we used ensemble tree classifiers (XGBoost, LightGBM, Random Forest) on the cross-sectional data and Bi-LSTM architectures on the monthly monitoring data, since TB treatment runs across six to twelve months and the static models miss what shifts during it. An accurate risk score is not enough by itself in a clinical setting. We layered Shapley Additive Explanations on top of the predictions so that clinicians could see which patient-level factors moved each estimate. The whole stack lives inside a Clinical Decision Support System that surfaces dynamic risk, SHAP-grounded explanations, and a bounded narrative layer. The clinician makes the final call.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment, Short-course (TB-DOTS) Program

Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis* (MTB), which is a slow-growing, acid-fast bacillus. And because the disease progresses gradually and that treatment can fail, mitigation is difficult (Tobin & Tristram, 2024). Effective treatments exist, but PTB is still widespread. It tracks closely with poverty, which is naturally followed by undernutrition, and comorbidities as well such as diabetes and HIV, all of which weaken patients' immune defenses and raise susceptibility to the disease (Tobin & Tristram, 2024). These conditions also delay the diagnosis and complicate treatment. TB is one of the leading causes of death worldwide, and in 2018 alone, there were around 10 million new cases and 1.5 million deaths, and about half a million of those were resistant to rifampicin.

In the Philippines, TB still remains as a big problem with regards to public health, and the World Health Organization (WHO) lists the country as one of the biggest contributors to the global TB burden, estimated at around 6.8 percent of cases coming from the Philippines. Before the COVID-19 pandemic, Philippine TB rates were also among the highest in Asia: over 554 cases per 100,000 people in 2019 (Florentino et al., 2022). In 2023, about 37,000 Filipinos died of TB out of 739,000 diagnosed cases (World Health Organization, 2023), and the WHO classifies the country as a high-burden setting for both MDR-TB and TB/HIV co-infection (The Borgen Project, 2025). That same year the DOH logged 575,770 TB cases in its Integrated TB Information System. The number itself signals heavy under-registration and underdiagnosis (Baron, 2025). For treatment, the DOH follows CDC-recommended regimens for active and latent TB under the TB-DOTS strategy, widened in 2003 to take in private clinics, pharmacies, and hospitals (Bendre et al., 2021; Department of Health [DOH], 2014). Gaps persist even so, in diagnosis, in treatment monitoring, in patient support, and most of all in managing MDR-TB (Stop TB Partnership, 2021).

Baguio City shows the same gap up close. In 2022 the DOH-CAR confirmed 437 TB cases there but counted more than 4,405 that were missing or never reported, which puts the city near the top for surveillance and treatment work (Department of Health - Republic of the Philippines, 2022). Passive case-finding, stigma, and the cost of care drive that gap, and the undetected patients keep spreading the disease in their communities (Kaur & Sharma, 2021).

The large population and a lot of daily traffic (both foot and vehicular) in the city may increase opportunities for tuberculosis to spread. The colder climate can also create more ideal conditions for the transmission of *Mycobacterium tuberculosis*. The city also has a more inflated local load of cases because it takes in more complex tuberculosis cases from neighboring provinces because of it being a tertiary referral center (Khalid et al., 2022). Patient-level factors also play an important role in treatment outcomes such as comorbidities like diabetes mellitus, HIV infection, and malnutrition, all of which weaken their immune response and may reduce treatment completion. Food insecurity, poor housing conditions, and transportation costs can also interrupt adherence to directly observed treatment short-course (DOTS) (Austin et al., 2021; Dong & Peng, 2013; Khalid et al., 2022; Sisk et al., 2023).

All in all, these constraints argue for strategies that work on more than one facet of assessment. Treatment failure not only worsens the outcome for the singular patient, it also extends the period for which transmission can happen in the community and in-turn speeds up the emergence of multidrug-resistant tuberculosis (MDR-TB), defined as resistance to at least rifampicin and isoniazid. MDR-TB requires regimens that are longer, more toxic, and more expensive, and it carries much lower success rates. The result is a wider threat to regional disease control in the Philippines (Aracri et al., 2025; Groenwold et al., 2012; World Health Organization, 2022).

Risk Factors for Treatment Failure

To ensure consistent evaluation of tuberculosis control efforts, the Department of Health (DOH) established standardized criteria for classifying treatment outcomes through the National Tuberculosis Program (NTP). These outcome definitions are important for monitoring program performance, comparing findings across studies, and identifying factors associated with treatment outcomes. Treatment success and failure do not occur randomly, as they are often influenced by identifiable patient, clinical, and program-related risk factors. Mortality remains the primary outcome examined in most studies on treatment failure, although some also include disease progression, multidrug-resistant tuberculosis, and loss to follow-up as indicators of poor treatment outcomes (Department of Health [DOH], 2014).

One of the risk factors is close or direct contact with patients who have active TB infection where patients who acquired the infection through close contact were at higher risk of delayed diagnosis and more likely to be non-adherent, either by refusing or quitting the regimen (Lamsal et al., 2009). Another risk factor is recurrence of the infection or a history of receiving TB treatment because patients who missed follow-ups had a higher risk of reactivation and mortality, and were harder to treat than patients with no prior history of TB (Tok et al., 2020). Frequent missed or interrupted treatment also leads to drug resistance (Ndambuki et al., 2020).

Comorbidities have consistently shown a significant influence on tuberculosis treatment outcomes across different settings. In Eastern Ethiopia, HIV co-infection was a substantial risk factor for treatment failure because the weakened immune system makes treatment less effective (Amante & Abdosh, 2015). TB and HIV drug interactions also accelerate each other's progression and increase the occurrence of side effects, which can compromise treatment success (Azeez et al., 2018). A study in Davao City, found a higher risk of treatment failure in comorbid patients: 58.82% of the 51 patients who experienced treatment failure had at least one comorbidity (Hinay et al., 2025). Another major factor is Diabetes Mellitus (DM). TB patients with diagnosed DM have a higher risk of treatment failure, relapse, and death than non-diabetics (Noubiap et al., 2019). In a systematic review and meta-analysis, DM patients had a higher risk of treatment failure for Drug-resistant (DR) TB and mortality for MDR-TB, with odds ratios of 1.56 and 1.57 respectively (Xu et al., 2023). Another study reported that patients with both PTB and chronic conditions, especially DM, had reduced rates of sputum conversion and a higher probability of poor treatment outcomes, including death and progression to MDR-TB (Siddiqui et al., 2016).

Taken together, these results say that no single factor drives TB treatment failure. It grows out of the way comorbidities, programmatic gaps, and socioeconomic strain feed into each other, and a one-factor analysis cannot see that. Catching it calls for a machine learning framework able to model the interactions all at once, the kind of model that makes earlier, better-aimed clinical intervention possible (Amante & Abdosh, 2015; Department of Health - Republic of the Philippines, 2022; Hinay et al., 2025; Khalid et al., 2022; Noubiap et al., 2019; Peng et al., 2024; Rajpurkar et al., 2022; World Health Organization, 2022; Xu et al., 2023).

Machine Learning for TB Outcome Prediction

Past work on TB outcomes has mostly looked at sociodemographic characteristics, comorbidities, medical history, and the usual patient-level risk factors. Results have been mixed. A lot of that inconsistency comes from methodology and not from anything inherently unreadable about the disease itself. Most of what has been published leans on traditional statistical methods, things like hypothesis tests and regression models, applied one factor at a time, often with a linear-relationship assumption baked in (Amante & Abdosh, 2015). That kind of analysis does produce real evidence about individual risk factors. The problem is that several variables tend to interact at once, and a one-factor-at-a-time view will not catch those interactions.

This is where machine learning has something to offer. An algorithm that learns from high-dimensional data can pick up the kinds of subtle dependencies that ultimately decide whether treatment holds or falls apart. The variables involved in TB treatment, demographic, clinical, diagnostic, and treatment-related, all interact in non-linear ways. A one-factor model has no real way to capture that. A range of algorithms have been tried on the problem already. The list includes logistic regression, Support Vector Machines, and ensemble methods such as XGBoost, among others (Rajpurkar et al., 2022). Within TB specifically, earlier work has applied ML to predicting drug resistance, flagging high-risk patients, and shaping treatment strategy (Kaur & Sharma, 2021; Kossakov et al., 2024). The throughline across this body of work is fairly simple. A predictive model can extend risk-factor analysis by reading multiple dimensions of patient data at the same time, rather than examining them one by one in isolation.

But raw accuracy is not enough on its own. A model that only outputs a number is not very useful clinically, no matter how good that number looks. The system needs to be interpretable. Without some sense of why a particular patient was flagged as high-risk, a healthcare worker has nothing concrete to verify against their own observations and very little they can act on. To deal with that, we use Shapley Additive Explanations (SHAP), a post-hoc method built on Shapley values that quantifies how much each variable contributed to a given prediction. SHAP rolls up cleanly into both patient-level explanations and overall feature importance (Lundberg & Lee, 2017a; Lundberg et al., 2020). It fits especially well with tree-based clinical models, XGBoost, Random Forest, and LightGBM in our case, where accuracy can be high but the internal decision path is hard to inspect directly. In the healthcare ML literature, SHAP is among the more widely used explanation methods on tabular data (Allgaier et al., 2023), and it has been used in earlier clinical work to surface real contributors to treatment failure, including drug resistance, TB history, and lab markers (Peng et al., 2024; Sibanda & Ndlovu, 2026; Wang et al., 2025).

Single-factor risk studies tend to hit a ceiling, and interpretable ML is one way past it. Understanding what actually drives TB outcomes means combining several dimensions of patient data at once rather than examining them one at a time. This study is built on that premise. The framework spans demographic, clinical, diagnostic, and treatment-related variables. Descriptive analysis lays out the patient profiles and treatment patterns. Supervised models estimate outcome risk on top of that. SHAP explanations then surface which factors moved each estimate in a given case. Our intent is to keep predictive accuracy high while keeping the output legible to providers (Shickel et al., 2018; Smith et al., 2021). Clinicians should be able to see not only who is at higher risk of failure, but why the model thinks so, and where it might make sense to step in earlier with closer monitoring or modified case management.

Contributions

DOH-CAR TB-DOTS operates under the National Tuberculosis Program (NTP) protocol. Case finding is passive. It starts at the facility, with a patient walking in. Someone with the warning signs of TB, a cough that will not go away, persistent weight loss, recurring fever, is sent for testing. That means sputum smear microscopy, a GeneXpert MTB/RIF test, and a chest X-ray (Bendre et al., 2021; Department of Health [DOH], 2014). If TB is confirmed, the patient is enrolled on the Treatment Card and Master List, slotted into a registration group, and paired with either a health worker or a trained treatment partner who is supposed to watch each dose over the six- to twelve-month treatment course (Department of Health [DOH], 2014). Every month a monitoring visit happens. The visit records sputum results, the vital signs, and weight, and it checks whether the patient is keeping up with the regimen. At the end, the patient's outcome gets filed under NTP criteria, as Cured, Treatment Completed, Failed, Died, or Lost to Follow-up (Ting, 2011). Most of this paperwork is still on actual paper. Risk, as a result, tends to be read in hindsight rather than forecast. A worsening condition, missed doses, a complication from another illness, these surface only after they have already bent the course of treatment in the wrong direction. By then the high-risk patient was never flagged in time. The cost is avoidable failures, more deaths than necessary, and continued spread of drug-resistant TB across the region (Khalid et al., 2022; World Health Organization, 2022).

Six contributions come out of this work. The first is the integrated predictive model itself. We trained and validated machine learning models on demographic, clinical, diagnostic, and treatment-related variables in order to forecast therapy success or failure for PTB patients in the CAR. The point of this is straightforward. It gives clinicians a way to identify potential problem patients earlier than the current paper-based workflow allows.

The second contribution is the data preprocessing and harmonization pipeline itself. We built it to be reproducible. It takes raw patient information from TB-DOTS clinics across the CAR and transforms it into something machine learning algorithms can actually use, applying schema normalization, pseudonymous identification, automatic error correction, and a diagnostic-first approach to handling missing values rather than a single blanket imputation rule.

The third contribution is the descriptive and exploratory analysis. This part describes the study population, surfaces how the various attributes relate to one another, and identifies the most important predictors of treatment outcomes before any modeling runs.

Temporal modeling is the fourth contribution. We use Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) architectures to examine time-dependent relationships in the monthly monitoring data, which captures variation that emerges during treatment rather than only at intake. Our expectation, going in, was that this would improve predictive accuracy relative to prior studies that did not account for temporal changes in monitoring data.

The fifth contribution is the use of Shapley Additive Explanations (SHAP) to quantify how much each patient-specific factor contributed to a given prediction. SHAP is useful here for a particular reason. It lets clinicians see what specific factors drove a patient's predicted risk score and therefore decide whether the recommendation should actually be applied for that patient.

Finally, the sixth contribution is the deployment side. The validated model is embedded into a Clinical Decision Support System (CDSS) that displays the patient's current risk level, lists the factors behind that risk via SHAP, and updates the estimate whenever new follow-up data comes in.

Research Objectives

This study aims to develop and validate machine learning models that predict treatment success or failure among TB patients in the CAR, Philippines. Combining demographic, clinical, and laboratory variables supports early identification of patients at risk of unsuccessful treatment and provides evidence-based tools for improving TB program outcomes. The broader goal is to enable personalized clinical interventions and efficient allocation of healthcare resources. The research aims to achieve the following objectives:

1. To document and pre-process patient records of PTB cases from TB-DOTS facilities in the

CAR through data cleaning, feature engineering, and reporting, so that the dataset is ready for ML applications.

2. To describe the demographic, clinical, diagnostic, and treatment characteristics of PTB patients in the dataset and provide an overview of the study population.
3. To identify variables that influence treatment outcomes over the course of therapy, so that the most important factors affecting patient recovery can be recognized and used to guide treatment adjustments.
4. To add temporal modeling that analyzes time-dependent patterns in patient data, so that changes and trends throughout treatment can be detected, improving the accuracy of outcome predictions and supporting timely clinical interventions.
5. To train and validate supervised learning models for predicting treatment success or failure among PTB patients, so that individuals at risk of poor outcomes can be identified early, while the model maintains high predictive accuracy and reliability when applied to incoming TB cases.
6. To develop a decision support tool that visualizes patient risk levels and predictive outputs.

Scope and Limitations

Our dataset covers PTB patients from 2015 to 2025 in the CAR, with Baguio City as the focal site, and excludes anyone still mid-treatment when we extracted the data. Only patients with a defined outcome (success or failure) are in scope. What we have on each patient includes demographics, laboratory tests, diagnostics, clinical findings, treatment regimen, follow-up examinations, drug administration records, and the final outcome. The limits are mostly about sufficiency and quality rather than scope. Because these are existing clinical records, completeness and consistency vary considerably, and some clinical parameters were just not recorded for some patients, which narrows the set of risk factors we can actually analyze and weakens parts of the modeling. The dataset also will not fully mirror the present patient population in the CAR, since everyone still on treatment had to be excluded for a definite outcome to exist. There are practical frictions on the extraction side too. Over the decade we cover, record-keeping conventions shifted as the health system itself changed and documentation standards moved with it, so consistency across years is not something we can assume. And wherever indicators are self-reported by patients, the usual response biases and inconsistencies can creep in and dull what the model is able to learn.

Significance of the Study

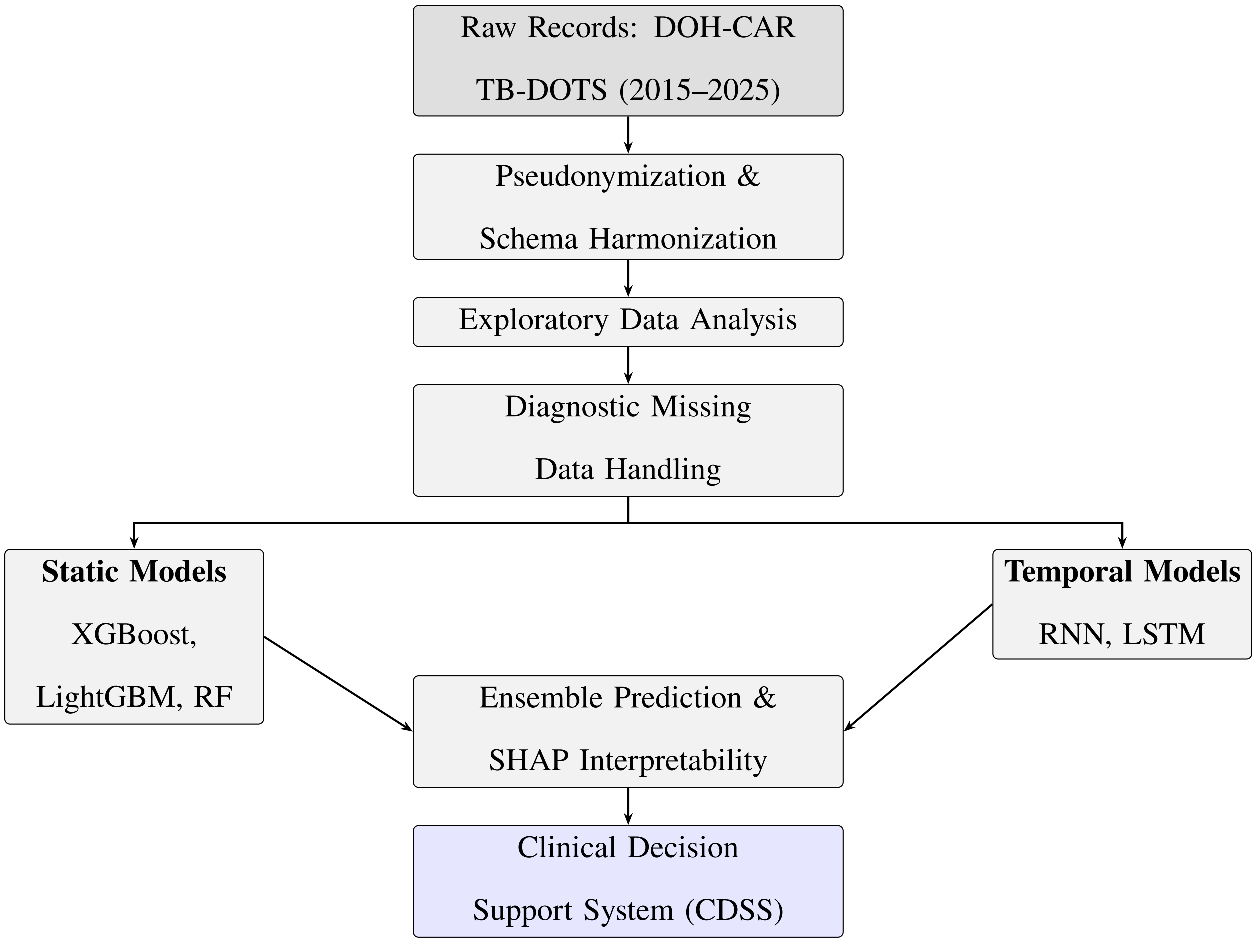
The point of the models is fairly narrow. We want providers to be able to spot, early enough to do something about it, the patients most likely to fail treatment. Earlier identification opens room for a closer look, tighter monitoring, and better case management before the outcome is fixed. Whether any of that actually happens depends on something that is easy to underestimate, namely whether clinicians can read and judge what the tool tells them. A risk score on its own is unlikely to move a busy TB-DOTS clinic, so we leaned the framework on explainable prediction from the start. SHAP surfaces the patient-level factors behind each estimate, and the CDSS lays those factors out in a form a clinician can actually work with rather than just look at. The larger ambition is to strengthen TB control in the CAR by turning data the program already routinely collects into something the program can act on. In practical terms, that means sharper patient management and treatment planning, firmer footing under program operations, and better-grounded public health decisions aimed at raising the TB treatment success rate in the region. We also want to leave administrators with something usable, a model that could grow into a deployable application for the local TB-DOTS program, and a cleaned, pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset that we will release for clinical, epidemiological, and machine learning research.

Method

The records come from DOH-CAR's central holdings of PTB patient data for the Cordillera Administrative Region, covering the years 2015 to 2025. We did not start until ethical clearance came through. The data is sensitive, and the program needed to sign off on the use. Pseudonymization was the first thing we ran. Only after identifiers were stripped did the data move into the preprocessing pipeline, which does the standard hygiene work of schema harmonization, error correction, and feature scaling, and then runs the diagnostic-first missing data protocol described later in its own subsection. Before any predictive modeling started, we ran exploratory and cluster analyses to describe the cohort and to produce subgroup-level features. After that, three ensemble tree-based classifiers (XGBoost, LightGBM, and Random Forest) learn to predict the treatment outcomes defined by NTP criteria. TB treatment, however, is not a single visit. It plays out over months, and the monitoring data shifts month to month, so we also tested temporal models, specifically RNN and LSTM networks, alongside the static classifiers and ensembled the two families for robustness. Performance is evaluated with standard classification metrics. SHAP handles interpretability. At the end of the pipeline, the validated models feed a clinical decision support framework that updates each patient's risk estimate as new monthly data comes in. Figure 1 sketches the overall flow.

Figure 1

Data Management and Analysis Process



Data Acquisition and Preprocessing

We work with two CAR datasets, both covering 2015 to 2025. The first is non-temporal. It is a consolidated registry that DOH-CAR maintains centrally, and it trains the static tree-based classifiers. The second is temporal. Our Pharmacy Co-authors compiled it by hand from facility records, and it trains both the longitudinal deep learning models and the temporal tree-based variants. Table 1 puts the two side by side.

Table 1

Comparison of the Non-Temporal and Temporal Monitoring Datasets

Attribute	Non-Temporal Dataset	Temporal Monitoring Dataset
Records (patients)	7,389	599
Coverage period	2015–2025	2015–2025 (patient subset)
Data source	DOH-CAR centralised TB-DOTS registry	Facility treatment monitoring forms (35 raw Excel workbooks)
Collection method	Routine national TB-DOTS programme reporting	Manual transcription and encoding by Pharmacy Co-authors
Structure	Cross-sectional: one row per patient	Longitudinal: one row per patient-month (M0–M12)
Static features	Demographics, diagnostics, outcomes	Demographics, diagnostics, baseline vitals, outcomes
Temporal features	None	Monthly adherence, doses taken/missed, weight, height, smear microscopy, Xpert MTB/RIF (M0–M12)
Post-preprocessing columns	28	269
Post-preprocessing rows retained	7,388	599 (zero deleted)
Primary model use	Static tree-based classifiers (XGBoost, LightGBM, Random Forest)	Temporal deep learning models (Bi-LSTM) and tree-based temporal baselines

Non-Temporal Dataset

DOH-CAR maintains this dataset centrally. It draws reports from government TB-DOTS facilities and from private practitioners enrolled in the Public-Private Mix DOTS (PPMD) scheme (Department of Health [DOH], 2014). The set has 7,389 patient records. Each record is baseline-only. That covers demographics (age, sex, civil status), clinical indicators (BMI, blood pressure), and diagnostic results (GeneXpert, Smear Microscopy). One row per patient, taken at the point of treatment registration. No follow-up monitoring is attached, because none was collected at this layer. We use this dataset as the main training source for the static classifiers, XGBoost, LightGBM, and Random Forest.

What the non-temporal pipeline does to this dataset is fairly standard. Variable formats and units get harmonized first, and the records get pseudonymized. The binary target is read directly from NTP criteria. “Cured” or “Treatment Completed” becomes Success. “Failed”, “Died”, or “Lost to Follow-up” becomes Failure. Continuous clinical variables get their extreme outliers capped so training is not skewed by data entry errors that snuck through. Whatever values are still missing after that go through the diagnostic-first protocol described in the Missing Data Handling subsection later in this section.

One thing we kept strict was the order of operations. The train-test split happens before any transformation, full stop. We do this so that test-set information has no way to leak into the model through scaling or imputation parameters. The split is stratified, which keeps the success and failure proportions roughly level across both partitions. Everything downstream after that, feature scaling, normalization, and the multivariate imputation of missing values, is fit on the training set alone. The fitted parameters and the imputation rules then transfer to the held-out test set. The model never sees the test data’s distribution.

Temporal Monitoring Dataset

The temporal monitoring dataset is a smaller, longer-form thing. It covers 599 patients, each with complete month-by-month records. Nothing like it existed in the DOH-CAR registry. We had to build it. The Pharmacy Co-authors *gathered and encoded it by hand*, going in person to TB-DOTS centers in Baguio City and Benguet, sitting with paper Treatment Cards and Master Lists, and transcribing each patient’s monitoring data field by field. That meant demographics, monthly vital signs, percentage adherence, doses taken and missed, smear microscopy (TB-LAMP) results, Xpert MTB/RIF results, and the final treatment outcome. All of this landed in a master `RAW DATA.xlsx` workbook, with one sheet per barangay or facility unit, across 35 facility-year cycles from 2015 to 2025. A consolidation script, `combine_raw_dataset.py`, then merged the per-barangay sheets into a single table, normalizing facility names against a canonical alias map so the mismatched entries lined up before the main pipeline ran.

The shape of the table is wide. One row per patient. Columns hold monthly observations from treatment Month 0 (M0) through Month 12 (M12). For each monthly visit, we record percentage adherence (`pct_adherence`), doses taken and missed that month, cumulative doses taken, body weight, height, smear microscopy (TB-LAMP), and the Xpert MTB/RIF result. Static baseline features sit on the same row next to the monthly ones, since patients carry a fixed set of attributes through treatment. Those baseline features cover demographics, baseline vital signs (heart rate, blood pressure, oxygen saturation, temperature, respiratory rate), comorbidities, the starting treatment regimen, and the diagnostic classification. After preprocessing, the cleaned table is 599 rows by 269 columns.

The pipeline that produces the cleaned table runs in eight passes. The first two are bookkeeping: column names get normalized to `snake_case` and the obvious duplicates and fully empty columns are dropped, and four identifying columns (`no`, `source_file`, `name_of_diagnosing_facility`, `name_of_treatment_unit`) are kept in the exported file so we can trace records back, but those columns never reach a model input. That is how we keep patient identifiers out of the modeling side.

The third pass does the bulk of the cleaning. Every column is parsed to its correct type and then clinically validated, and the messy fields each get their own parser. Weight strings like `45kg` or `56.5 kg` resolve to numeric kilograms. Heights become centimeters. A blood pressure entered as `120/80` splits into separate systolic and diastolic fields. Oxygen saturation is the worst offender: it shows up as a decimal fraction (0.95), a percent-suffixed value (`95%`), or a raw integer, and all three need to collapse to a single 0–100 scale. Adherence percentages lose any trailing `%` sign. Dates get a pass of their own, since clinical registries accumulate the usual recording errors. A future date of birth is pulled back a century (2063 $\rightarrow$ 1963), transposed digits are caught against neighboring records (2990 $\rightarrow$ 1990), and date ordering is enforced so that treatment start cannot precede diagnosis. After parsing, continuous clinical variables are winsorized to physiologically plausible bounds: temperature to 30–45°C, systolic BP to 60–220 mmHg, diastolic BP to 30–140 mmHg, heart rate to 20–250 bpm, oxygen saturation to 70–100%, weight to 10–180 kg, height to

80–220 cm, respiratory rate to 4–60 breaths/min.

Then comes the categorical work. Inconsistent entries (outcome variants, sex designations, regimen types, comorbidity flags, civil status) collapse from hundreds of free-text variants down to a small canonical set. Columns that are near-entirely null or redundant are flagged in the output rather than silently deleted, so a reviewer can see what the cleaning step decided and why. The `nationality_raw` column, for example, is marked for removal once its values have merged into `nationality`, but the trace stays in the report.

The sixth pass is structural. The wide table gets melted into two pieces: a static baseline table (`df_static`, one row per patient) and a temporal table (`df_temporal`, one row per patient-month) holding `patient_id`, `month`, `weight`, `height`, `pct_adherence`, `monthly_doses_taken`, `monthly_missed_doses`, `cumulative_doses_taken`, `smear_tb_lamp`, and `xpert_mtb_rif`. Once the data is in that shape, the four-pathway diagnostic missing-data routine from the Missing Data Handling subsection runs separately on the static and temporal feature sets. Temporal features get an exemption from hard exclusion. Their missingness is structural rather than incidental: later months simply have fewer observations as patients finish treatment or drop out, and that pattern carries clinical meaning that gets preserved through missingness indicators.

The last pass closes things up. Clinical bounds get re-applied to every imputed value (weight back to 10–180 kg, `pct_adherence` back to 0–100), and a monotonicity repair undoes any MICE-introduced dip in `cumulative_doses_taken` within a patient so a cumulative count never decreases from one month to the next. The pipeline writes `cleaned_human_readable.csv` (599 rows × 269 columns), with original clinical units and category labels left intact. Encoding, scaling, and tensor preparation are held back until the model-specific scripts run after the train–test split, which is deliberate. Doing it earlier is how leakage sneaks in.

The temporal train–test split therefore happens before any scaling, one-hot encoding, outlier capping, or 3D tensor construction for the recurrent networks, exactly as in the non-temporal pipeline. Both sides guard against leakage the same way, whether the downstream model is a static classifier or a temporal network. Table 2 gives the per-patient observed means of the temporal features across M0 to M7.

Table 2

Per-Patient Observed Means of Temporal Features (M0–M7)

Feature	N	Mean	SD	Min	Max
monthly_doses_taken	573	23.36	4.32	0.0	64.29
cumulative_doses_taken	535	70.09	23.39	0.0	127.38
monthly_missed_doses	458	0.91	2.47	0.0	25.0
pct_adherence	342	98.23	8.35	0.0	100.0
weight	361	55.69	21.55	7.2	189.69
height	315	155.49	20.45	31.0	181.0
smear_tb_lamp	25	0.12	0.33	0.0	1.0
xpert_mtb_rif	41	0.0	0.0	0.0	0.0
last_month_attended	599	5.23	2.06	-1.0	7.0
total_observed_months	599	6.21	2.06	0.0	8.0

Exploratory Data Analysis

We ran an exploratory pass on the cohort before doing any modeling. The point was simple. We needed to know who was actually in the dataset, how complete the records were, and how the variables related to each other before deciding which of them to feed into models or how to handle the gaps. The findings from this stage shaped most of the preprocessing and modeling choices that followed.

Table 3

Missing Data Summary by Column

Column	Missing_Count	Missing_Percentage
Days_To_Microscopy_Result	7297	98.75
Days_To_RDT_Result	6638	89.84
Days_Screening_To_Diagnosis	4444	60.14

Column	Missing_Count	Missing_Percentage
Days_To_Treatment	4378	59.25
Diagnosis_Season	3953	53.5
Diagnosis_DayOfYear	3953	53.5
Diagnosis_Quarter	3953	53.5
Diagnosis_Month	3953	53.5
Microscopy Result	2792	37.79
Brgy	1026	13.89
City/Municipality	105	1.42
Province	71	0.96
Treatment Health Facility	29	0.39
Region	15	0.2
Date of Outcome/Status	2	0.03
Bacteriologic Status	1	0.01
Type	0	0
Validation Status	0	0
Age	0	0
Sex	0	0
Anatomical Site	0	0
Source of Patient	0	0
Screening/Diagnosing Health Facility	0	0
RDT Result	0	0
Year	0	0
Outcome/Status	0	0
Registration Group	0	0
Computed_Age	0	0

A global missingness audit ran over all 28 columns and ranked each by its missing rate (Table 3).

Some variables were badly incomplete. Microscopy result was missing 37.8% of the time, the diagnosis-derived temporal features 53.6%, and the clinical delay intervals anywhere from 59.3% up to 98.8%. Those numbers fixed the missingness thresholds applied later in the Missing Data Handling subsection.

For the continuous variables, pairwise Pearson's r was taken on available-case observations: age, four derived clinical delay intervals (days from screening to diagnosis, to treatment, to microscopy result, and to RDT result), and diagnosis day-of-year. The question behind it was simple, whether MICE was worth the trouble. When continuous predictors actually share correlation structure, MICE can borrow signal from the observed features to estimate the missing ones. When they don't, a plain univariate mean-fill does the same job.

Categorical variables used Cramér's V instead, a chi-square-based measure scaled to the $[0, 1]$ interval. Any column with fewer than two unique non-null values, or fewer than ten non-null observations, dropped out of the matrix. The set here was patient demographics (sex, type of TB, anatomical site) and clinical classifications (registration group, bacteriologic status, microscopy results).

Beyond the per-variable diagnostics, the exploratory phase also clusters patients across demographic, clinical, and diagnostic dimensions to surface latent subgroups the descriptive statistics alone would not necessarily reveal. We use K-prototypes for this, mainly because it handles the dataset's mixed categorical and continuous variables without forcing us into a single representation. Picking the number of clusters is iterative, settled by reading the Silhouette score and the elbow method together, which lets the analysis separate groups by comorbidity profile, adherence behavior, and how patients respond to a given regimen.

Those cluster assignments end up doing two jobs. They support descriptive interpretation, and they also become engineered features for the supervised models downstream. We one-hot encode the cluster labels as additional categorical inputs for the tree-based classifiers (XGBoost, LightGBM, Random Forest), and we fold them into the feature space for the temporal networks (RNN, LSTM). Once the labels are in there, a model has the chance to latch onto subgroup-level patterns behind treatment success or failure, instead of having to rediscover those patterns from individual predictors alone.

The temporal monitoring dataset received the same treatment before any preprocessing ran: a global missingness audit across every feature, with columns ranked by missing rate. Anything above 80% missingness was excluded outright. Anything between 50% and 80% went into the missing-indicator-with-fill pathway, with the call weighed against the variable's predictive importance. This kind of threshold-based triage is fairly standard practice. The advantage is that the handling strategy adapts to the severity of the missingness rather than forcing one rule across the entire dataset (Emmanuel et al., 2021).

Patient retention was traced as the share of patients with non-missing adherence data at each interval from M0 through M12. A raw monthly count tells you little, because retention falls naturally over a six- to twelve-month course, so the decay was expressed as a percentage of the starting cohort at each successive month. This also sets the temporal modeling scope. Once fewer than 25% of patients still have data, that month and every month after it drops out of the feature-level correlation analyses, which keeps a small, non-representative surviving group from driving the estimates.

The same Pearson's r check, again on available-case observations, ran over the vital signs in the temporal dataset. A multivariate imputation only earns its keep when the vitals correlate strongly. If they move together, Multiple Imputation by Chained Equations (MICE) can borrow signal from the observed features to fill gaps; if they don't, MICE buys nothing over mean- or median-fill (Aracri et al., 2025). The vitals here were age, heart rate, oxygen saturation, systolic and diastolic blood pressure, weight, height, respiratory rate, and temperature, wherever the records carried them.

To see how the temporal features relate, Pearson correlations were taken on per-patient observed means across Months 0 through 7. Each feature, a dose metric (monthly doses taken, cumulative doses taken, percentage adherence), an anthropometric one (weight, height), or a diagnostic result (smear microscopy, Xpert MTB/RIF), was averaged across a patient's available visits. Retention characteristics came from the same window: the last month attended and the total months observed. Averaging this way is deliberate, because it preserves the dropout signal. A patient who quits after M4 contributes means from M0 to M4 and no further, and that short window carries information of its own rather than a gap waiting to be imputed.

Categorical variables in the temporal dataset again went through Cramér's V, the chi-square-based measure on the [0, 1] interval. Pearson and Cramér's V were never mixed into a single matrix. Pearson's r assumes interval-level measurement and gives misleading values on nominal or ordinal categories whose spacing is uneven (Kossakov et al., 2024), so keeping Cramér's V separate for the categorical features avoids that and yields interpretable estimates. Columns with fewer than two unique non-null values, or fewer than ten non-null observations, were left out; association can't be estimated meaningfully there anyway.

A cardinality audit over every categorical feature in the raw dataset showed how much inconsistent coding had to be fixed. Counting the unique raw values per feature put a number on the entry inconsistency.

Where a feature's cardinality ran far past its expected count of valid categories, say a clinical observation field that should carry a handful of standardized codes but instead carried hundreds of free-text variants, it was flagged for harmonization in Stage 4. The audit hands those harmonization targets an empirical baseline: it measures the data entry problem before any cleaning touches it.

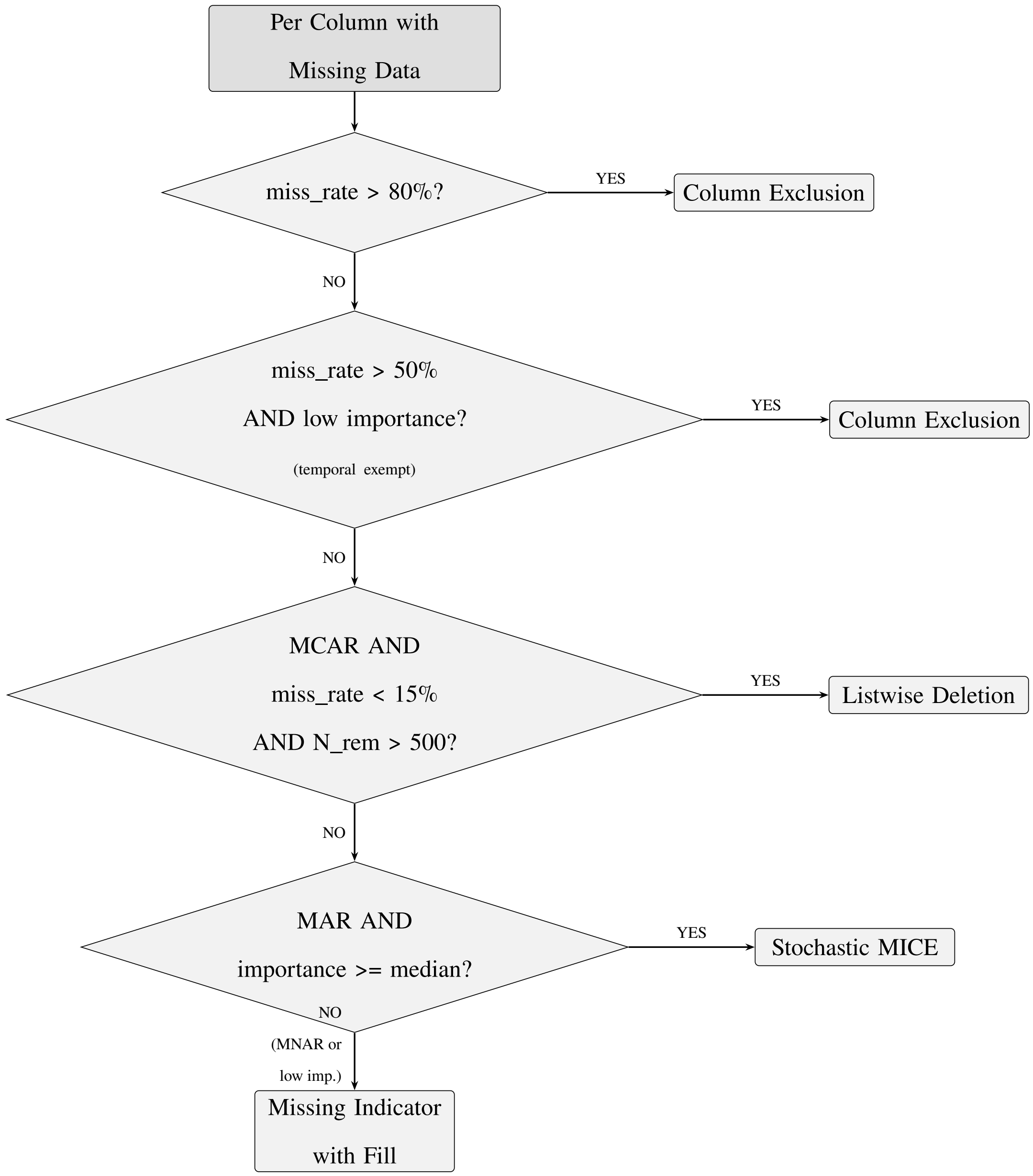
Whether adherence holds steady over the course was tested with lag-1 Pearson correlations of percentage adherence across consecutive month pairs (M0 to M1 through M7 to M8). That month-to-month strength decides whether a recurrent architecture, which assumes each time step leans on the one before, is even worth building. Weak correlation between adjacent months would mean a static model on aggregated patient means tells you just as much. Only strong correlation justifies the extra complexity of an RNN or LSTM (Laine & Aila, 2016; Nayebi et al., 2022).

Missing Data Handling

Missing data in clinical TB registries is rarely uniform. The gaps trace to a mix of causes: inconsistent documentation practices, facilities operating at different capacities, and the basic reality that TB treatment is monitored over many months and not every visit makes it onto paper. Once the underlying mechanism varies from feature to feature, imposing a single imputation rule across all of them is statistically wrong. We therefore use a diagnostic-first pipeline. The pipeline first asks how each variable went missing. It then routes that variable to one of four pathways, column exclusion, listwise deletion, missing indicator with fill, or stochastic MICE (Figure 2) (Dong & Peng, 2013; Emmanuel et al., 2021).

Figure 2

Missing Data Handling Decision Logic



The first diagnostic step is Little's MCAR test at the dataset level, which asks whether the data are Missing Completely at Random. If they were, missingness would be independent of every value (observed or not), the complete cases would form an unbiased subsample, and listwise deletion would be a defensible default. None of that applies here. The test came back significant ($p < 0.05$), which rules out MCAR for the dataset as a whole and ties the missingness systematically to the observed data. Wholesale case deletion was therefore off the table (Dong & Peng, 2013).

A dataset-level verdict, however, only tells you so much. Little's test gives one omnibus answer, but each column still has its own mechanism, so the pipeline goes column by column. For temporal (monthly) variables, it correlates the column's monthly missing rate against treatment month using Spearman rank, and a correlation that climbs monotonically with month points to a dropout pattern, that is, Missing Not at Random (MNAR). For static baseline variables, it uses point-biserial correlations between the binary is-missing indicator and the observed features. If other observed variables significantly explain the gap, the column is classified Missing at Random (MAR); if they do not, it falls back to MNAR. Classifying this way rather than swallowing a single global verdict lets each handling strategy match the statistical situation of its own variable (Dong & Peng, 2013; Emmanuel et al., 2021).

Two of those handling decisions, column exclusion and MICE eligibility, depend on a Random Forest classifier trained on the available cases (the rows where the treatment outcome is observed). For ranking purposes only, features get a coarse fill with column medians, which is rough but enough to estimate importance. Variables in the bottom quartile of importance become candidates for soft exclusion, and those above the median get priority for stochastic imputation (Aracri et al., 2025).

Two thresholds then apply. The first is hard: any variable past 80% missingness gets dropped on sight, importance aside, because there is simply not enough signal left to impute reliably at that sparsity. The second is soft: variables between 50% and 80% missing that also fall in the bottom quartile of importance get dropped as well, since imputing a sparse low-value variable adds more synthetic variance than information. Temporal variables sidestep both thresholds, because their missingness is structural, baked into how TB monitoring works over six to twelve months, rather than a data quality problem (Junaid et al., 2025).

There is one narrow case in which listwise deletion remains defensible: a variable that is confirmed MCAR at the column level, under 15% missing, with more than 500 patients still present after deletion. In that situation the complete cases really do form an unbiased subsample, and deletion avoids the variance distortion imputation introduces. As it turned out, no variable in our data cleared all three conditions at once, so this pathway never fired (Dong & Peng, 2013).

Then there are the MNAR cases, which need different treatment. Every variable classified MNAR, plus every temporal feature regardless of its mechanism, goes through a two-part procedure. A binary indicator (`is_missing_[variable]`) marks the absence itself, and only then is the original variable filled, with last observed value carried forward for temporal variables, column median for continuous static ones, and column mode for categorical static ones. The reason for the indicator is that absence in a TB record often carries clinical meaning we do not want to throw away. A missing Month 9 weight usually means the patient dropped out, and the indicator preserves that pattern for the model to learn from. Backward-fill, on the other hand, is forbidden on temporal variables. Letting a future measurement seep into an earlier time step would leak information the model has no legitimate access to at inference time (Austin et al., 2021; Groenwold et al., 2012; Sisk et al., 2023).

Four vital signs ended up clearing both bars, confirmed MAR and above-median importance: oxygen saturation, heart rate, and systolic and diastolic blood pressure. These four go to stochastic MICE with an ExtraTreesRegressor as the conditional model. ExtraTreesRegressor is a nonparametric tree-based estimator that captures non-linear relationships in mixed-type clinical data without requiring standardized variables, similar in spirit to the missForest framework (Aracri et al., 2025), and on data shaped like this it usually outperforms linear and k-nearest-neighbor imputers. The MICE iteration count adapts to each variable's missing fraction, scaling as $\max(20, \min(100, \lfloor \text{miss\_rate} \times 100 \rfloor))$. That gives high-sparsity features room to converge without letting compute run away, and it mirrors Von Hippel's rule that the number of imputations should roughly match the percentage of missing information (von Hippel, 2018), here applied to iterations instead of imputed datasets. Imputing stochastically rather than deterministically also matters, since it preserves the natural spread of the imputed values instead of collapsing them onto a conditional mean (Alruhaymi & Kim, 2021).

Data Quality Assurance

We keep the supervised pipeline reliable through a layered quality assurance protocol. The protocol runs across four layers in particular: schema validation, clinical bounds checking, temporal consistency, and post-processing validation. Each layer catches a different kind of problem, and skipping any one of them tends to surface as a downstream artifact later.

Schema and Column Validation

Column names are normalized to `snake_case`: whitespace stripped, text lowercased, special characters swapped for word equivalents, so `M0 %Adherence` turns into `m0_pct_adherence`. Duplicate columns and fully null ones are flagged and deduplicated before modeling. Four patient- and facility-identifying columns remain in the export file for traceability, but none of them reach a model input, for privacy.

Date Integrity Correction

A dedicated correction pass deals with the recording errors that clinical registries tend to accumulate. A future date of birth, its year past the data year, is pulled back 100 years. A date of birth before 1920 is pushed forward 100 years when that puts the age in a plausible 0 to 110 range. Transposed digits, 2990 for 1990 and the like, are caught against adjacent records. Date pairs are forced into order, treatment start no earlier than diagnosis, outcome date no earlier than treatment start. A missing date inherits the cohort median date for its variable.

Temporal Consistency

Every fill in the longitudinal structure is forward-only. That alone keeps the temporal data from leaking. A Month 4 observation can carry forward to Month 5; a Month 6 measurement can never inform an earlier step the model is meant to predict without it. After MICE, a monotonicity repair pass reverses any drop in `cumulative_doses_taken` that the multivariate imputer introduced, so a patient's cumulative dose never falls from one treatment month to the next.

Categorical Harmonization Audit

A cardinality audit of the raw categorical features confirmed how widespread the entry inconsistency was. The `chest_x_ray_at_case_notification` field carried over 300 unique free-text entries, `co_morbidities` more than 150; even nominally binary fields like `sex` and `outcome` showed 5 to 12 raw representations each, the residue of treatment centers abbreviating in their own way. Those counts drove a 17-field harmonization pass that maps free-text and inconsistently coded entries onto standardized categories before analysis.

Missing Data Diagnostics and Feature Importance

The diagnostic-first missing data pipeline, detailed in the Missing Data Handling subsection, adds one more validation layer. Little's MCAR test runs at the dataset level. Per-column mechanisms come from two correlation checks, Spearman rank to catch temporal dropout, point-biserial to separate MAR from MNAR. A Random Forest trained on available cases ranks predictive importance, and that ranking decides where each variable is routed.

Post-Processing Verification

Before the cleaned dataset reaches any model, an automated routine checks four things: (1) the identifier columns are still present for traceability, (2) no scaled or normalized values have slipped in, so everything stays in original clinical units, (3) categorical labels are still human-readable strings, not integer codes, and (4) no one-hot columns have appeared. Pass all four, and encoding, scaling, and tensor construction were demonstrably left to the model-specific scripts that run after the split.

Leakage Prevention and Reproducibility

Every validation step is there for one reason: keep test-set information out of training. The patient-level stratified split comes first, before any transformation. Scalers, imputers, and encoders are fit on the training set alone, then carried to the test and validation sets through `reindex()` so the column sets line up exactly. Temporal fills run forward-only, backward-fill switched off. Reproducibility is handled two ways: the split is fully deterministic under a seeded random state, and the fitted scaler parameters are checked by SHA-256 hash to confirm that independent runs preprocess identically. An audit trail, `missing_data_report.json`, logs the decision pathway, mechanism classification, and rationale for every column.

Identification of Variables Associated With Treatment Outcomes

Identifying which variables actually move treatment outcomes calls for two angles, not one. We pair descriptive association analysis with model-derived explanation. On the descriptive side, we summarize which demographic, clinical, diagnostic, and temporal variables pull the success and failure groups apart in the data itself. On the modeling side, the supervised models estimate how those same variables combine to predict the outcome jointly. Both views are necessary here, because risk in TB treatment is genuinely multidimensional. A variable can look unimportant in isolation and still matter once it sits next to adherence patterns, diagnostic timing, geographic context, or ongoing monitoring data.

The bridge between predictive performance and clinical interpretability is Shapley Additive Explanations. SHAP is a post-hoc method that measures each variable's marginal contribution to the model's output (Lundberg & Lee, 2017a), and what it gives us is a clear picture of which features pushed a particular patient-level prediction toward "Success" (Cured, Completed) or "Failure" (Failed, Died, Lost to Follow-up; Peng et al., 2024). For clinical review, that means a clinician can see what the model actually leaned on, things like drug resistance, comorbidities, adherence gaps, diagnostic results, or a delayed diagnosis, rather than receiving a number with no rationale attached. There is one caveat we hold onto throughout. SHAP explanations describe how the model behaved, not clinical causation, and we keep that line between computational attribution and medical judgment deliberately visible (Kaur & Sharma, 2021; Kossakov et al., 2024).

Temporal Modeling

Because TB treatment plays out over six to twelve months rather than at a single point in time, the study includes temporal modeling on the longitudinal monitoring data. A static model trained on baseline features alone risks missing the very things that matter most clinically: how adherence shifts week to week, whether follow-up attendance holds up, how the lab numbers evolve. The temporal models are built to produce month-by-month risk estimates from treatment Month 0 (M0) through Month 12 (M12).

These temporal models do not draw from the same source as the static classifiers. They run on a single dataset, the 599-patient longitudinal set described in Data Acquisition and Preprocessing (Table 1), which the Pharmacy Co-authors compiled by hand from facility monitoring forms. The 7,389-patient DOH-CAR registry the static classifiers use is a separate dataset and stays separate. After the eight-stage temporal pipeline, each patient ends up with up to eight usable monthly timesteps, M0 through M7. We drop anything past M7 from the model inputs. Fewer than 25% of patients had observations that far into treatment (Table 5), and a surviving subset that small and that skewed by who dropped out would give unreliable estimates. As elsewhere, encoding, scaling, and 3D tensor preparation wait for the post-split model-specific scripts, which is how the pipeline keeps anything from leaking.

Temporal modeling went through two iterations, v1 and v2. They share the same dataset lineage; v2 is the corrected, modeling-ready preparation that came out of a methodological review of v1. The v1 pipeline was a modeling baseline, with patient-level stratified splitting and progressive temporal expansion across M0 to M12. The review then flagged a few things that could weaken its clinical defensibility: late-treatment fields that raise the risk of outcome leakage, inconsistent handling of imputed or unreliable outcome labels, and cleaning and temporal feature construction that had not been audited closely enough.

Version 2 (v2) fixed those issues and retrained. It builds on the audited output of the eight-stage cleaning pipeline from Data Acquisition and Preprocessing, restricts supervised learning to patients whose outcome is actually observed, and removes leakage-prone and administrative fields, outcome-date fields and late-timeline variables that would not exist at an early inference. Temporal predictors are encoded two ways: as sequence inputs for the deep models, as engineered summaries for the tree-based ones. The deep architecture is a Hybrid Bidirectional LSTM (Bi-LSTM), picked to model long-term dependencies without the vanishing gradients of a plain RNN (Lu et al., 2022); its recurrent structure suits the strong month-to-month adherence dependence the lag-1 analysis turned up (Nayebi et al., 2022). The tree-based baselines take the other route, flattening the monthly values and bolting on aggregates, trends, latest-observed values, and missingness indicators. Class imbalance is corrected with training-only resampling such as SMOTE-ENN. The hybrid framing is kept so temporal predictions can sit beside strong static learners like XGBoost, which holds up better across the different patient subgroups (Laine & Aila, 2016).

Supervised Learning Model Development and Validation

The core of the study is the supervised classification model, which we train and validate against the actual treatment outcomes. We chose ensemble tree-based algorithms here for two reasons: they handle complex non-linear interactions among mixed-type clinical variables, and they tolerate class imbalance fairly well by construction. The three we ended up using are XGBoost (Chen & Guestrin, 2016; Rajpurkar et al., 2022), LightGBM (Ke et al., 2017), and Random Forest (Breiman, 2001). They emerged from a preliminary run across six candidates, where the other three (Logistic Regression, SVM, and Gradient Boosting) either discriminated poorly with CV-AUC at or below 0.63 or recalled almost none of the minority failure class. Either of those failures would make a model unsafe inside a CDSS, so we dropped them. Each of the surviving three then runs under two sampling strategies. The first is no resampling, with native class-weighting in place. The second is SMOTE with Edited Nearest Neighbors (SMOTE-ENN), which inflates the training failure class by 433% to bring it to a near-balanced 0.70:1 ratio.

Feature Set Versions

Three feature sets are evaluated in sequence, to see how much each extra clinical and temporal block actually buys. The *baseline* set (11 features) is the core demographic and diagnostic block: age, sex, anatomical site, bacteriologic status, microscopy result, registration group, source of patient, TB type, province, days to treatment, and year of registration. The *improved* set (14 features) adds three facility-level geographic variables: city/municipality, treatment health facility, and screening/diagnosing health facility. The *extended* set (21 features) adds a further seven temporal and diagnostic delay features from date engineering: diagnosis month, diagnosis quarter, diagnosis season (on the Philippine three-season meteorological calendar: *Tag-lamig* for cool dry December to February, *Tag-init* for hot dry March to May, *Tag-ulan* for the rainy June to November), days from screening to diagnosis, days from diagnosis to microscopy result, days from diagnosis to rapid diagnostic test (RDT) result, and the RDT result itself (binned from 42 raw variants into three classes: Detected, Not Detected, Not Done). Date of outcome was kept out of every feature set. That field is generated after the treatment result, so including it would leak the outcome.

Data Partitioning and Class Imbalance

We split the data at the patient level with a stratified 70/20/10 partition, so the training, test, and holdout validation sets each preserve the class proportions of the full cohort (Hastie et al., 2009). The class imbalance is heavy on the order of 86/14, success against failure, and we correct it on the training set only, with one of three oversampling approaches. SMOTE (Chawla et al., 2002) interpolates synthetic minority-class points between existing ones. SMOTE-ENN (Batista et al., 2004; Nishat et al., 2022) layers Edited Nearest Neighbors undersampling on top of SMOTE to clear ambiguous borderline cases. SMOTE-Tomek (Batista et al., 2004) pairs SMOTE with Tomek link removal to clean up the decision boundary. Neither the test set nor the validation set ever gets resampled. They keep their original imbalanced distributions, which is the only way to keep the evaluation honest (Salmi et al., 2024).

Cross-Validation and Model Selection

Every configuration is scored by five-fold stratified cross-validation on the training data, which estimates generalization and returns a mean and standard deviation of ROC-AUC across folds (Hastie et al., 2009). The clinical goal is to catch at-risk patients early, so selection runs on a composite metric, score = 0.6 × AUC + 0.4 × Recall(Failure): it leans on discrimination but still rewards sensitivity to the minority failure class. Whichever configuration tops that score across the feature versions and resamplers is then checked on the held-out 10% validation set for an unbiased real-world estimate.

Each configuration also reports the Matthews Correlation Coefficient (MCC) next to the composite metric, as a supplementary check. Accuracy or a class-specific F1 can mislead; MCC reads all four quadrants of the confusion matrix at once and holds up under heavy imbalance. It sits in $[-1, +1]$: +1 for perfect prediction, 0 for random, -1 for complete inversion (Chicco & Jurman, 2020). Putting MCC beside the composite score exposes the trade-off that aggressive resampling brings, since SMOTE-ENN and its kind can lift failure recall while piling on false positives that push overall error up.

Performance Evaluation

No single number captures classifier quality under this kind of imbalance, so we report several. Treatment failure is only 13.8% of the cohort. A model that always answers "Success" would score 86.2% accuracy and catch zero of the patients we actually care about. That is the trap a single-metric summary walks straight into, which is why the framework reports discriminative, threshold-dependent, and calibration-aware metrics side by side.

All of them trace back to the same confusion matrix, with treatment failure as the positive class. On the held-out test set, each model produces four quadrants: true positives (TP, failures correctly predicted), false negatives (FN, failures missed), false positives (FP, successes wrongly flagged as failures), and true negatives (TN, successes correctly identified). Everything else is built from those four counts.

Accuracy, $(TP + TN)/(TP + TN + FP + FN)$, is the share of correct predictions overall. It is the most intuitive number, but under 86:14 imbalance the majority class dominates it, so we always report it with that caveat attached. Precision, $TP/(TP + FP)$, sometimes called positive predictive value, gives the share of failure alerts that turn out to be real failures; when precision is low, false alarms pile up and clinician trust in the CDSS goes with them. Its counterpart is recall (sensitivity), $TP/(TP + FN)$, the share of actual failures the model catches. In TB-DOTS, a missed failure costs much more than a false alert. An undetected treatment breakdown can progress into disease worsening, drug resistance, and further community transmission, whereas a false alert just adds an extra clinical review. Specificity, $TN/(TN+FP)$, is the mirror image: the share of actual successes correctly identified. Keeping specificity reasonably high means patients who are doing fine are not pulled in for needless work-ups.

The F1 score, $2 \cdot \text{Precision} \cdot \text{Recall}/(\text{Precision} + \text{Recall})$, is the harmonic mean of precision and recall, useful because it punishes models that lean hard on one at the expense of the other. Balanced accuracy, $(\text{Recall} + \text{Specificity})/2$, averages sensitivity and specificity instead and is robust to class imbalance by construction; at 0.50 it amounts to random guessing, regardless of the class distribution. ROC-AUC measures discrimination across every possible decision threshold. The ROC curve plots the true positive rate against the false positive rate, and the area under it collapses the curve to a single scalar (1.0 is perfect separation, 0.5 is chance). Because ROC-AUC does not depend on a chosen cutoff, we use it as the primary ranking metric. The caveat is that ROC-AUC can read optimistically under severe imbalance, since the large majority class dominates the false positive rate. PR-AUC patches over that limitation by plotting precision against recall across thresholds and ignoring true negatives entirely. When the positive class is rare, PR-AUC carries more information than ROC-AUC (Davis & Goadrich, 2006), and its no-skill baseline is just the positive-class prevalence (0.138), which is a natural reference point.

The Matthews Correlation Coefficient (MCC) is computed as

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}},$$

and it stays informative under class imbalance because it takes in all four confusion-matrix quadrants at once. The coefficient ranges over $[-1, +1]$, with $+1$ for perfect prediction, 0 for random, and -1 for complete inversion (Chicco & Jurman, 2020). Unlike accuracy or F1, MCC is symmetric under class swapping and does not inflate when the majority class dominates.

Under heavy imbalance, a default 0.50 cutoff rarely lines up with clinical priorities, so each model also gets a threshold search. Candidates from 0.10 to 0.90 are scored against three composite objectives that each encode a different stance: a balanced objective (40% F1, 30% recall, 30% specificity), a failure-recall objective (55% failure recall, 25% balanced accuracy, 20% F1), and a specificity objective (50% specificity, 30% F1, 20% recall). A model reports its metrics at whichever threshold maximizes the relevant objective. Final model selection is then driven by a composite score that combines discriminative performance and clinical utility,

$$S = 0.6 \times \text{ROC-AUC} + 0.4 \times \text{Recall(Failure)},$$

which puts most of the weight on ranking ability while still rewarding sensitivity to the minority failure class. The configuration that tops S becomes the recommended model.

Performance itself is estimated at three levels to guard against overfitting and gauge generalization. Five-fold stratified cross-validation on the training set yields a mean and standard deviation of ROC-AUC across folds, which is the stability check. The 20% held-out test set carries the primary evaluation behind every metric in this section, on data the model never met during training or preprocessing. A final 10% holdout is reserved for the recommended model alone, to verify the test-set numbers are not just an artifact of one particular split. Scaling, encoding, and imputation are all fit on the training set and then carried out to the test and validation sets, so no evaluation data feeds back into training. Every experiment runs through one centralized pipeline, and the metrics and visualizations export directly to native LaTeX for inclusion in this manuscript.

Development of a Clinical Decision Support Tool

We embed the validated models inside a Clinical Decision Support System (CDSS) intended for use by healthcare providers. Each component of the CDSS does a specific job. The predictive model estimates the probability of treatment success or failure. The SHAP layer surfaces which factors are behind that estimate. The interface puts the risk and the factors on the same screen, so a clinician can read the whole thing against whatever they already know about the patient in front of them. In practice the workflow looks like this. A provider opens the dashboard, sees the failure and success probabilities, and pulls up the risk-factor view that surfaces what actually drove the score, things like adherence gaps, lab results, or diagnostic delays. As new follow-up data arrives, a monthly weight or a new smear result, the provider enters it and the prediction refreshes, so review continues across the entire treatment course rather than ending at a single intake snapshot. The CDSS exists for risk stratification and explanation. It does not diagnose. It does not prescribe.

Small Language Model (SLM)-Based Narrative Explanation Layer

To make the explanations easier on clinical readers, the CDSS layers a small language model (SLM) on top of the existing outputs. The SLM is constrained. It renders model outputs as plain clinical text rather than computing anything new. Throughout, the SHAP feature contributions remain the explanation of record. The SLM is purely a presentation layer. It is not a predictor, not a diagnostic engine, and not a source of fresh clinical inference (Lundberg & Lee, 2017b; Zeng & Zhu, 2024). It runs after the prediction and touches nothing upstream of itself, which means it does not affect the training, the inference probabilities, the risk thresholds, or the SHAP values it reads from. In the implemented pipeline, the web client assembles a structured interpretation request that includes selected patient context fields, the predicted treatment-failure probability, and the ranked contribution signals, and posts it to a dedicated endpoint that streams the narrative back into the interface.

What the SLM is allowed to do stays narrow. It does not infer new clinical causes. It does not write prescriptions. It does not suggest regimen changes. It does not stand in for a clinician's judgment. Its one job is to make the SHAP-grounded attribution easier to read so a provider can see why the CDSS flagged a particular patient and decide whether a closer look is justified. Keeping the roles this strictly separated matches how medical-domain SLMs are typically framed today, lines up with what health AI governance expects from bounded decision support, and keeps interpretability tied to the original attribution signal rather than to whatever the language model felt like writing (Collins et al., 2024; Sellergren et al., 2025; World Health Organization, 2021). The final call still belongs to the treating professional.

Results and Discussion

This section reports what the machine learning analyses actually showed for predicting treatment success and failure among PTB patients in the DOH-CAR TB-DOTS program. We start with descriptive statistics on patient demographics, clinical characteristics, and outcomes. From there we evaluate model performance on the test and validation sets, comparing the static tree-based models (XGBoost, LightGBM, and Random Forest) under two sampling conditions against the temporal models. Feature importance analyses then unpack what was actually driving the predictions. In the discussion that follows, we read the findings against clinical relevance, public health priorities, and CDSS usability, and trace how the risk estimates and the explanations feed into earlier intervention, more deliberate resource allocation, and stronger TB management overall.

Missing Data Handling Outcomes

Out of the 44 features in the temporal dataset that had any missing data at all, 15 ended up excluded entirely under the 80% hard threshold, since there was simply not enough signal left in them to impute reliably. The near-empty ones sat right at the top of the list: tuberculosis culture at 99.8%, other laboratory tests also at 99.8%, tuberculin skin test at 93.5%. A sixteenth column, the raw blood pressure string field, was soft-excluded at 67.3% missing with below-median importance, mainly because we had already parsed it into separate systolic and diastolic fields and the raw version had nothing left to add. No rows were dropped during this entire process, which means all 599 patient records stayed in. Keeping the full cohort matters more than it might sound, because with this many patients and an 86.2%/13.8% class split, listwise deletion would have thinned the minority class even further, and the minority class is the part we are actually trying to model (Dong & Peng, 2013).

Listwise deletion never actually fired. The dataset-level MCAR test had already been rejected, and even at the column level, not a single variable met all three of the conditions needed for safe listwise deletion (confirmed MCAR, under 15% missing, more than 500 patients left after deletion). Twenty-five variables took the missing-indicator-with-fill route instead. Eight of those were temporal monthly features (weight, percentage adherence, cumulative doses taken, monthly smear results, and a few others), and 17 were static baseline variables with confirmed MNAR patterns. Only four variables ended up at stochastic MICE: oxygen saturation, heart rate, and the two blood pressure readings, all confirmed MAR. Their missingness indicators tracked visit attendance under the point-biserial procedure described earlier, which is the clinically intuitive reading: whether a vital sign got recorded depended mostly on whether the patient came in that month, not on what the unseen measurement would have been. Reserving the expensive probabilistic imputation for just these four reflects the diagnostic-first design, which spends MICE only where its assumptions actually hold (Austin et al., 2021; Emmanuel et al., 2021).

Table 4

Global Feature Missingness Rates in the Raw Temporal Dataset

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m8_weight	599	100.0	Yes
m12_weight	599	100.0	Yes
m11_height	599	100.0	Yes
m9_xpert_mtb_rif	599	100.0	Yes
m9_smear_tb_lamp	599	100.0	Yes
m7_xpert_mtb_rif	599	100.0	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m7_smear_tb_lamp	599	100.0	Yes
m10_smear_tb_lamp	599	100.0	Yes
m10_weight	599	100.0	Yes
risk_factor_s_for_drug_resistance_ tuberculosis	599	100.0	Yes
m12_height	599	100.0	Yes
m12_smear_tb_lamp	599	100.0	Yes
m12_xpert_mtb_rif	599	100.0	Yes
m9_weight	599	100.0	Yes
m8_xpert_mtb_rif	599	100.0	Yes
m8_smear_tb_lamp	599	100.0	Yes
m1_xpert_mtb_rif	599	100.0	Yes
m11_weight	599	100.0	Yes
m10_xpert_mtb_rif	599	100.0	Yes
m11_smear_tb_lamp	599	100.0	Yes
m11_xpert_mtb_rif	599	100.0	Yes
m3_xpert_mtb_rif	599	100.0	Yes
m4_xpert_mtb_rif	598	99.83	Yes
m10_height	598	99.83	Yes
tuberculosis_culture	598	99.83	Yes
m12_monthly_missed_doses	598	99.83	Yes
other_lab_test	598	99.83	Yes
m12_pct_adherence	598	99.83	Yes
m8_height	598	99.83	Yes
m9_height	598	99.83	Yes
m12_monthly_doses_taken	597	99.67	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m4_smear_tb_lamp	597	99.67	Yes
m1_smear_tb_lamp	597	99.67	Yes
others	597	99.67	Yes
m11_monthly_missed_doses	597	99.67	Yes
m11_pct_adherence	597	99.67	Yes
m12_cumulative_doses_taken	597	99.67	Yes
m10_monthly_missed_doses	596	99.5	Yes
m9_pct_adherence	596	99.5	Yes
m10_pct_adherence	596	99.5	Yes
m11_monthly_doses_taken	595	99.33	Yes
m9_monthly_missed_doses	595	99.33	Yes
m11_cumulative_doses_taken	595	99.33	Yes
m8_pct_adherence	593	99.0	Yes
m3_smear_tb_lamp	593	99.0	Yes
m8_monthly_missed_doses	592	98.83	Yes
m10_monthly_doses_taken	592	98.83	Yes
m10_cumulative_doses_taken	592	98.83	Yes
m6_smear_tb_lamp	590	98.5	Yes
m9_cumulative_doses_taken	588	98.16	Yes
m7_height	588	98.16	Yes
m5_smear_tb_lamp	588	98.16	Yes
m9_monthly_doses_taken	588	98.16	Yes
m6_xpert_mtb_rif	587	98.0	Yes
m2_smear_tb_lamp	587	98.0	Yes
m0_smear_tb_lamp	586	97.83	Yes
m7_weight	586	97.83	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m5_xpert_mtb_rif	585	97.66	Yes
m8_cumulative_doses_taken	582	97.16	Yes
m8_monthly_doses_taken	581	96.99	Yes
dat_supported_dup	576	96.16	Yes
m2_xpert_mtb_rif	572	95.49	Yes
other	570	95.16	Yes
m5_height	563	93.99	Yes
m4_height	562	93.82	Yes
m6_height	562	93.82	Yes
m6_weight	561	93.66	Yes
tuberculin_skin_test	559	93.32	Yes
m3_height	558	93.16	Yes
m5_weight	557	92.99	Yes
m2_height	555	92.65	Yes
m0_xpert_mtb_rif	554	92.49	Yes
m4_weight	554	92.49	Yes
m1_height	554	92.49	Yes
nationality_raw	551	91.99	Yes
m3_weight	550	91.82	Yes
dat_supported	549	91.65	Yes
prior_history_of_tb	549	91.65	Yes
m2_weight	543	90.65	Yes
m1_weight	540	90.15	Yes
m7_pct_adherence	519	86.64	Yes
regimen_type_at_6th_month_of_treatment	518	86.48	Yes
smear_microscopy	502	83.81	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m7_monthly_missed_doses	502	83.81	Yes
respiratory_rate	501	83.64	Yes
height	492	82.14	Yes
temperature	481	80.3	Yes
weight	476	79.47	No
m7_cumulative_doses_taken	465	77.63	No
o2_sat	462	77.13	No
m7_monthly_doses_taken	453	75.63	No
heart_rate	441	73.62	No
blood_pressure	403	67.28	No
m6_pct_adherence	398	66.44	No
regimen_type_at_end_of_treatment	355	59.27	No
m6_monthly_missed_doses	347	57.93	No
m5_pct_adherence	336	56.09	No
m4_pct_adherence	319	53.26	No
m3_pct_adherence	312	52.09	No
m2_pct_adherence	302	50.42	No
m1_pct_adherence	293	48.91	No
name_of_treatment_unit	290	48.41	No
m0_pct_adherence	277	46.24	No
treatment_regimen	273	45.58	No
m6_cumulative_doses_taken	263	43.91	No
co_morbidities	253	42.24	No
m5_monthly_missed_doses	252	42.07	No
m0_cumulative_doses_taken	244	40.73	No
continuation_phase_start_date	239	39.9	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m3_monthly_missed_doses	235	39.23	No
m0_height	235	39.23	No
date_of_outcome	235	39.23	No
m4_monthly_missed_doses	233	38.9	No
m6_monthly_doses_taken	228	38.06	No
continuation_phase_end_date	228	38.06	No
m0_monthly_missed_doses	226	37.73	No
m2_monthly_missed_doses	222	37.06	No
intensive_phase_end_date	217	36.23	No
m3_cumulative_doses_taken	217	36.23	No
drug_resistance_bacteriological_status	217	36.23	No
chest_x_ray_at_case_notification	213	35.56	No
m1_monthly_missed_doses	186	31.05	No
m0_weight	185	30.88	No
xpert_mtb_rif	182	30.38	No
civil_status	167	27.88	No
nationality	166	27.71	No
date_of_notification	159	26.54	No
m5_cumulative_doses_taken	157	26.21	No
m4_cumulative_doses_taken	152	25.38	No
name_of_diagnosing_facility	150	25.04	No
intensive_phase_start_date	136	22.7	No
outcome	127	21.2	No
m2_cumulative_doses_taken	126	21.04	No
regimen_type_at_start_of_treatment	122	20.37	No
case_registration_group	117	19.53	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
date_of_diagnosis	110	18.36	No
m5_monthly_doses_taken	101	16.86	No
m1_cumulative_doses_taken	93	15.53	No
m4_monthly_doses_taken	90	15.03	No
m3_monthly_doses_taken	80	13.36	No
bacteriologic_status	79	13.19	No
diagnosis	62	10.35	No
treatment_start_date	57	9.52	No
m2_monthly_doses_taken	57	9.52	No
m1_monthly_doses_taken	39	6.51	No
sex	33	5.51	No
no	31	5.18	No
m0_monthly_doses_taken	30	5.01	No
date_of_birth	20	3.34	No
age	18	3.01	No
data_year	0	0.0	No
source_file	0	0.0	No
facility	0	0.0	No

Table 4 lays out the full pre-preprocessing missingness profile for the temporal dataset. Fifteen variables cleared the 80% hard threshold and were excluded, tuberculosis culture (99.8%), other laboratory tests (99.8%), tuberculin skin test (93.5%) among them. Another band sat between 50% and 80%: height (71.7%), weight (71.7%), smear microscopy (55.6%), risk factor fields (91.7%). For those, predictive importance decided the handling under the missing-indicator-with-fill pathway.

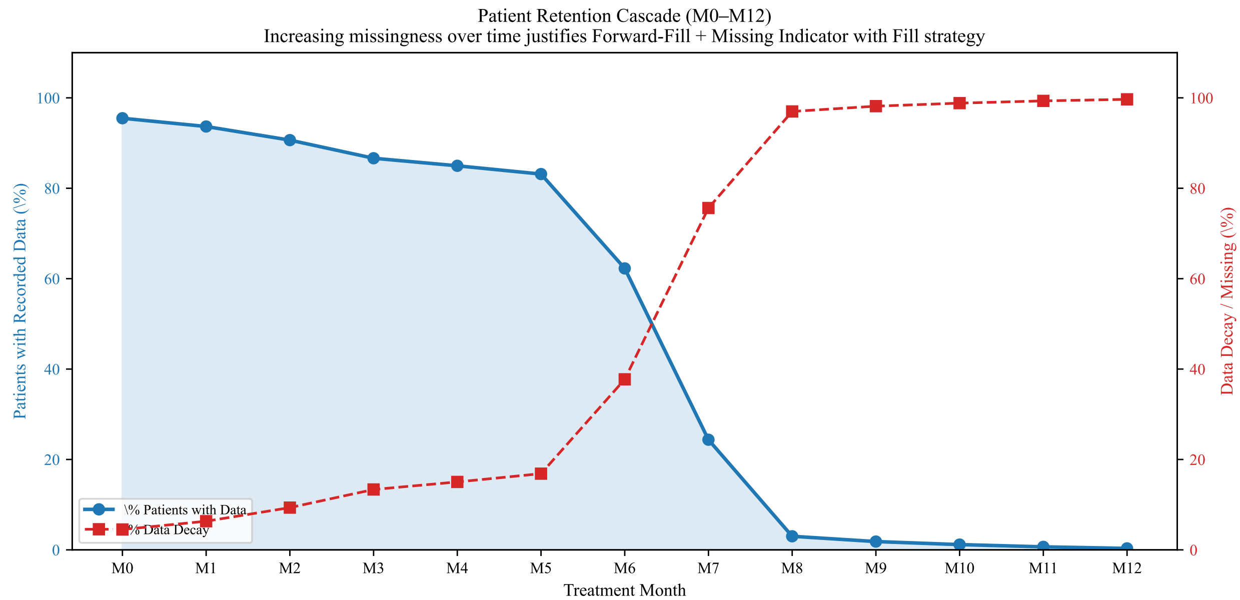
Table 5

Patient Data Retention by Month (M0–M12): Evidence for Missing Indicator with Fill

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M0	572	95.49	4.51	322
M1	561	93.66	6.34	306
M2	543	90.65	9.35	297
M3	519	86.64	13.36	287
M4	509	84.97	15.03	280
M5	498	83.14	16.86	263
M6	373	62.27	37.73	201
M7	146	24.37	75.63	80
M8	18	3.01	96.99	6
M9	11	1.84	98.16	3
M10	7	1.17	98.83	3
M11	4	0.67	99.33	2
M12	2	0.33	99.67	1

Figure 3

Patient Retention Cascade Across Treatment Months



Retention across the treatment course sharpens the picture of the missingness mechanism. At the start (M0), 95.5% of patients had data on file. That held to 83.1% by M5, then dropped hard, 62.3% at M6, 24.4% at M7, and under 4% past M7. A decay that monotonic is the signature of treatment dropout, not random absence, and it backs the forward-fill-plus-indicator handling the temporal variables get (Groenwold et al., 2012). The M7 cliff also fixes the temporal modeling scope: per-patient means run only over Months 0 through 7, because anything beyond would lean on a surviving subset too small and skewed to trust.

Table 6

Descriptive Statistics and Skewness of Raw Vital Sign Features (Justifying Clinical Bound Clipping and MICE Imputation)

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
age	581	46.2	48.0	21.24	0.42	93.0	-0.06	-0.824	3.01
heart_rate	141	89.6	88.0	15.16	59.0	129.0	0.41	-0.408	73.62
respiratory_rate	73	19.21	19.0	1.62	16.0	24.0	0.459	0.683	83.64

Table 6 continued

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
temperature	91	36.39	36.4	0.26	35.6	37.0	-0.501	0.555	80.3
o2_sat	121	18.65	0.97	36.88	0.9	99.0	1.612	0.623	77.13

Table 6 gives the descriptive statistics and distribution shape of the raw vital signs. Several are moderately right-skewed, heart rate (skew = 1.52) and respiratory rate (skew = 1.83) most of all, which is exactly why the MICE estimator is tree-based (ExtraTreesRegressor) and assumes no normality. The four variables sent to stochastic MICE, oxygen saturation, heart rate, systolic and diastolic blood pressure, each carry a clinically plausible MAR mechanism, their missingness tied to whether the patient attended rather than to the measurement values themselves.

Dataset Characteristics and Class Distribution

After cleaning, the final validated dataset contained 7,389 complete patient records covering demographic, geographic, and clinical features. The class split is heavily skewed: 86.2% of evaluated cases came out as successes and only 13.8% as failures. We stratified the data 70/20/10 across training, testing, and validation so the class proportions stayed level in each. Feature engineering brought the final predictor count up to 15, combining spatial identifiers (Region, Province, City/Municipality) with temporal markers and derived clinical delays such as Diagnosis-to-Treatment days.

Figure 4

TB Case Notifications by Year (2015–2025)

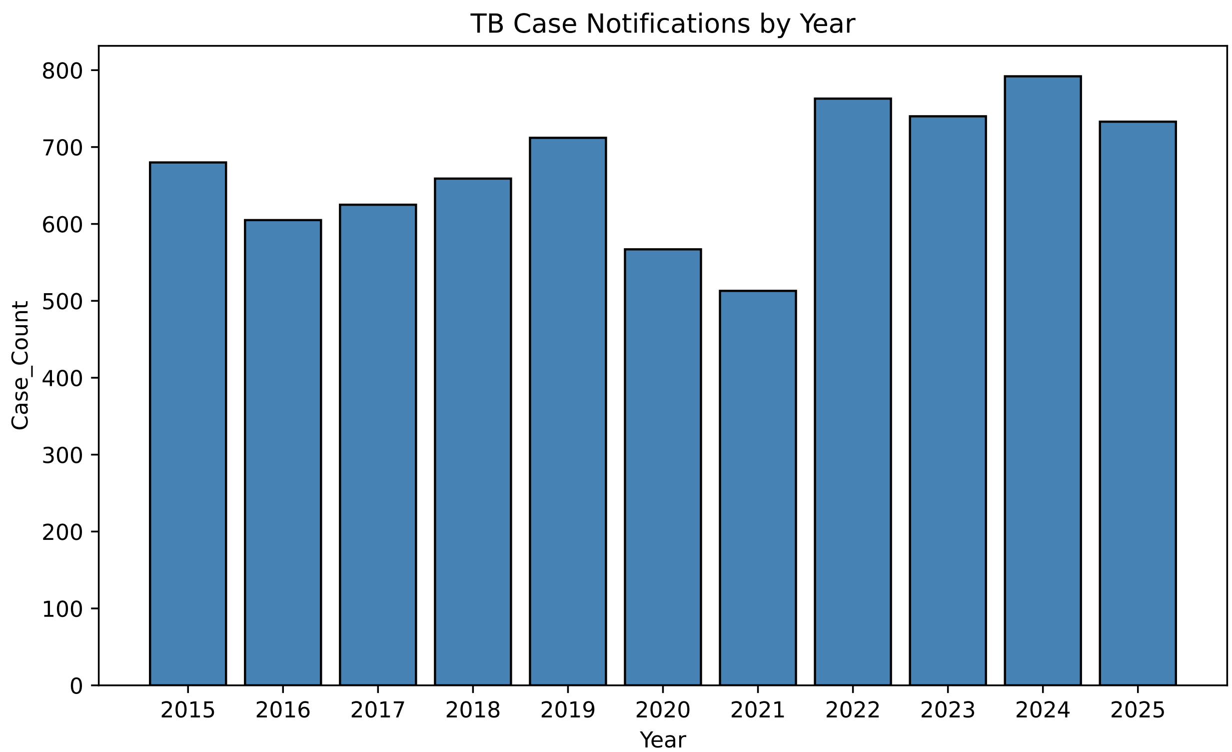
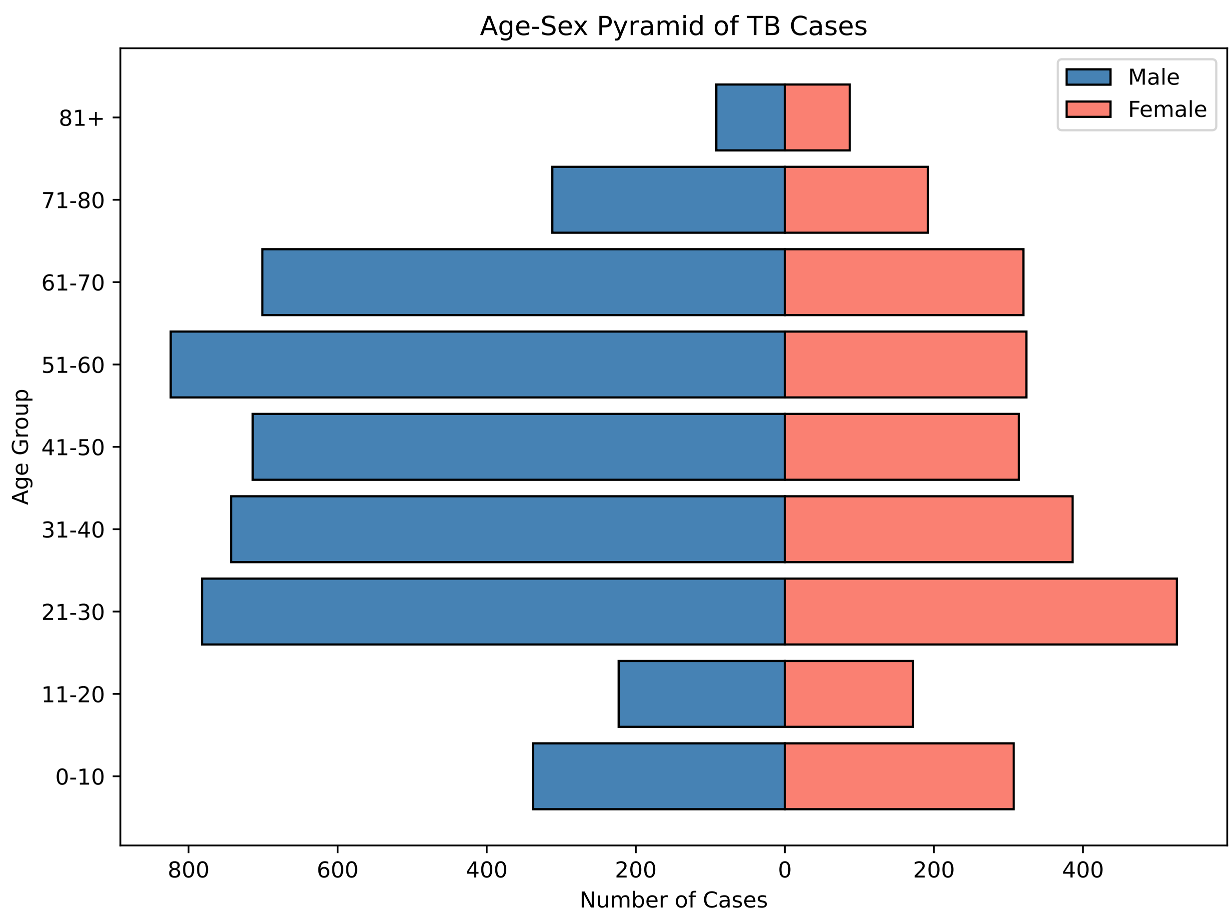


Figure 5

Age-Sex Pyramid of the Study Cohort



We applied SMOTE plus Edited Nearest Neighbors (SMOTE+ENN) to the training set only. The resampling step grew the minority (failure) class by 433% and brought the training distribution to a near-balanced 0.70:1 ratio. The point of doing this was to cut the model's bias toward the majority class during training. Crucially, we left the test and validation sets at their real imbalanced distributions, since resampling those would have given an artificially flattering evaluation.

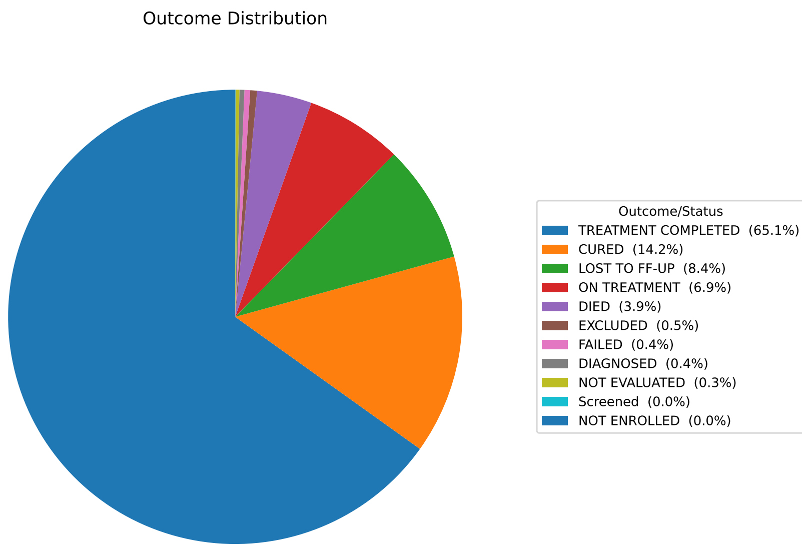
Outcome Distribution

The outcome distribution shows two things at once: the part of DOH-CAR TB-DOTS that works, and the part that keeps slipping. Most of the cohort came through fine. Treatment Completed accounted for 65.1%, and Cured for another 14.2%, so close to 80% of evaluated patients hit a positive endpoint.

The adverse outcomes are smaller, but they are the ones the predictive system exists for. Lost to Follow-up made up 8.4% of the cohort, Died another 3.3%, and outright treatment failure was rarer still at 0.9%. Beyond those, a residual share remained unresolved at the time of consolidation, either On Treatment (6.9%) or Not Evaluated (0.9%).

Figure 6

Distribution of Treatment Outcomes Among PTB Patients



Longitudinal Trends (2015–2025)

Looking across the decade, annual outcomes shifted in response to both internal program changes and outside events. Treatment success stayed in the low-to-mid 80% range for most of the period, dipped to 81.03% in 2018, and then climbed back to a high of 88.92% in 2023.

Mortality moved more dramatically than success did. Deaths among TB patients went from 3.65% in 2019 to 5.47% in 2020 and then 6.63% in 2021, lining up cleanly with the COVID-19 disruptions that delayed diagnoses, restricted access to clinical care, and worsened respiratory comorbidities for patients who were already vulnerable. Loss to follow-up peaked across the 2018–2020 window, reaching as high as 11.99%, and then fell back to 5.56% by 2024 as patient tracking and adherence support gradually improved.

The 2025 figures are a separate situation. Recorded success drops to 30.01% and unresolved cases jump to 64.80%, but that is an artifact of an incomplete reporting year rather than a real collapse in program performance. Most of those patients are still working through their 6– to 12-month regimens and simply have no final outcome on record yet.

Table 7

Treatment Outcome Distribution by Year (%)

Year	Died	Lost to Follow-up	Other	Success
2015	2.65	9.56	3.82	83.97
2016	2.98	11.57	1.32	84.13
2017	3.84	8.8	1.28	86.08
2018	4.25	11.53	3.19	81.03
2019	3.65	11.8	3.51	81.04
2020	5.47	11.99	0.88	81.66
2021	6.63	6.82	0.97	85.58
2022	4.72	7.08	0.92	87.29
2023	3.92	6.35	0.81	88.92
2024	3.66	5.56	4.55	86.24
2025	2.05	3.14	64.8	30.01

Diagnostic and Treatment Intervals

We extracted the temporal intervals between screening, diagnosis, and treatment initiation to look at operational delays, which are known risk factors for both disease transmission and treatment failure. All three intervals turned out to be right-skewed. Screening to diagnosis had a median of 2.0 days, but its third quartile stretched out to 21.09 days, meaning most patients were diagnosed quickly while a minority waited weeks. Diagnosis to treatment was similarly fast at the median (about 2.68 days) but again reached 13.12 days at Q3. Time to microscopy results sat at a 6.60-day median, with Q3 at 14.95 days.

In other words, the same shape recurs across all three intervals: an efficient median paired with a long tail. The patients who lose time tend to lose it in those tails. We feed the intervals into the models as continuous features, which lets the algorithms connect logistical delay to the eventual risk of death or loss to follow-up rather than treating the delays as separate from the clinical picture.

Table 8

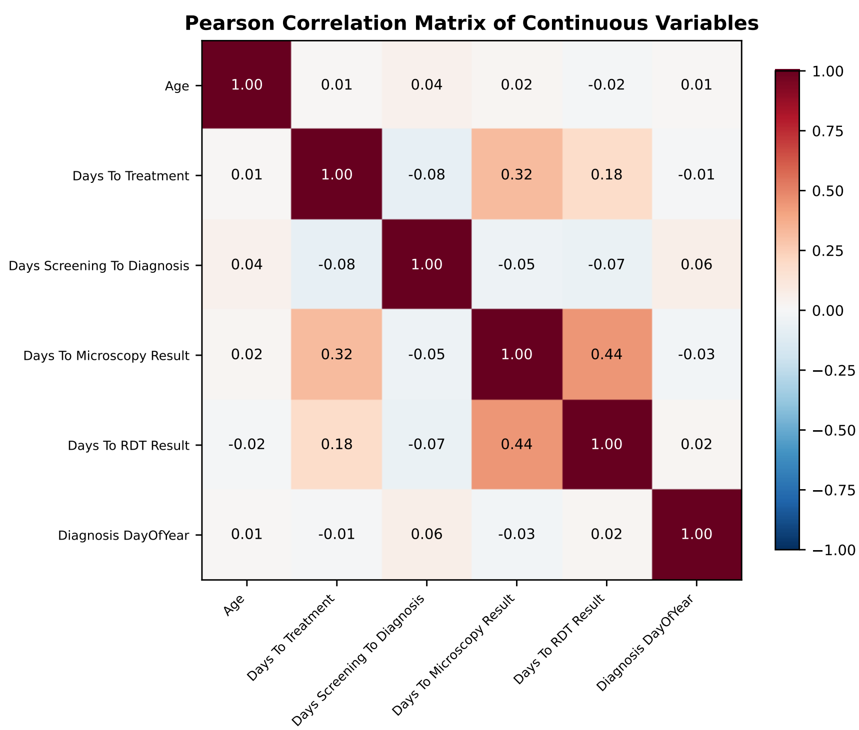
Summary Statistics for Time Intervals (Days)

Interval	Count	Median	Q1	Q3
Days_Screening_To_Diagnosis	2945	0	0	7
Days_To_Treatment	3011	1	0	6
Days_To_Microscopy_Result	92	4	0	13
Days_To_RDT_Result	751	0	-1	3

Correlation Analysis

Figure 7

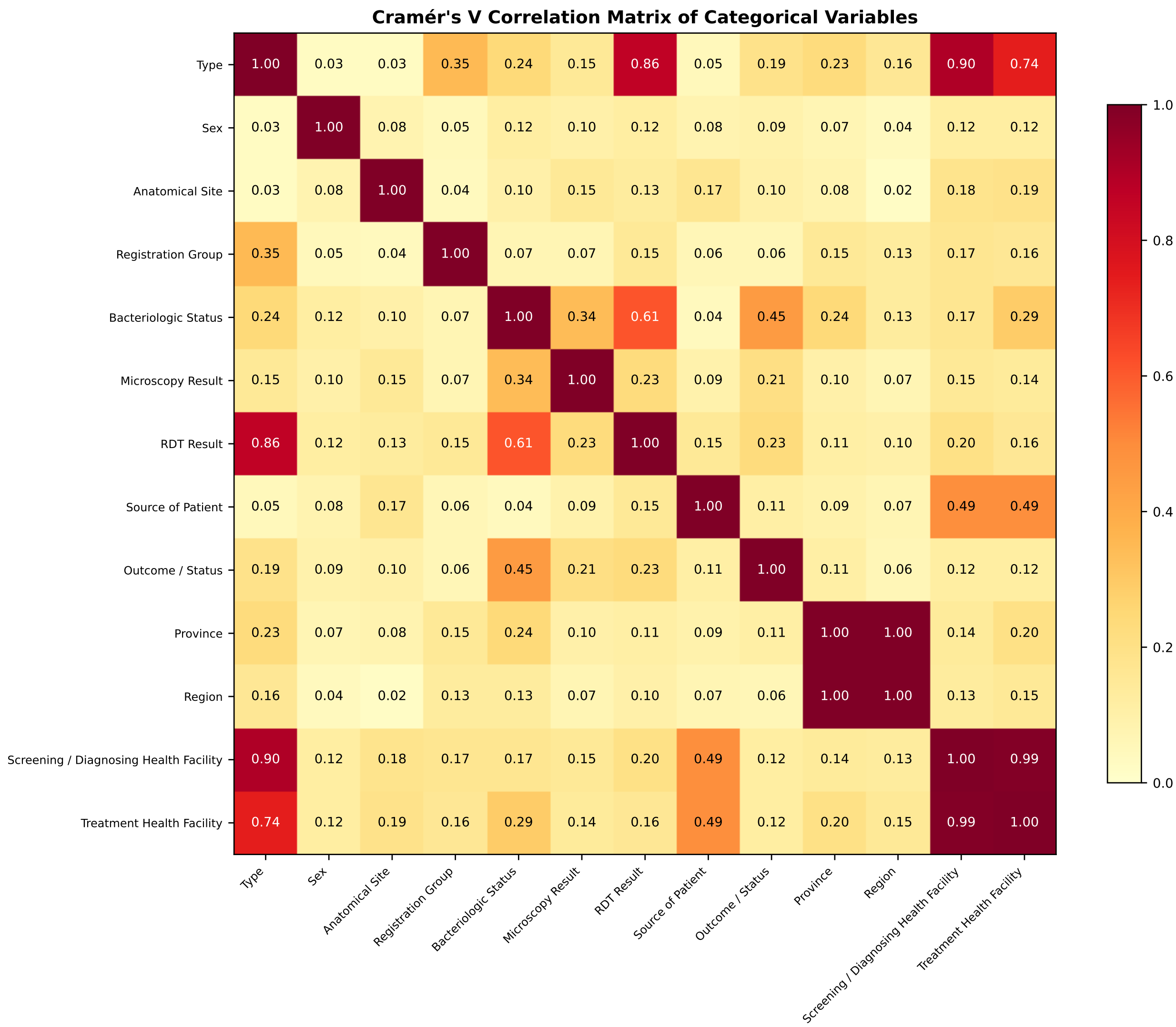
Pearson Correlation Matrix of Continuous Variables



The Pearson matrix (Figure 7) covers age, the clinical delay intervals, and diagnosis timing. Age correlated weakly and negatively with the time intervals ($|r| < 0.10$), so older patients saw marginally shorter delays. The delay variables themselves were moderately and positively inter-correlated; screening-to-diagnosis tracked diagnosis-to-treatment at $r = 0.32$, which says a delay at one stage of the care cascade tends to come with a delay at the next. Diagnosis day-of-year was essentially uncorrelated with everything else ($|r| < 0.05$), so seasonal timing is independent of patient demographics and operational factors.

Figure 8

Cramér's V Correlation Matrix of Categorical Variables



The Cramér's V matrix (Figure 8) measures the same kind of pairwise association for the categorical features. Most fell in the moderate range ($V \approx 0.3$–0.5), so the categorical structure is meaningful without being redundant. Geographic variables (province, region, and health facility) clustered at higher values ($V > 0.6$), which follows from the way administrative units nest inside one another. The outcome variable associated only modestly with registration group and anatomical site ($V \approx 0.25$), so the categorical features carry real but limited signal for predicting treatment outcome.

Validation Set Performance and Generalization

Table 9

Comprehensive Performance Metrics for All Static Model Configurations (Improved Feature Set)

Model	Sampler	AUC	CV-AUC	±Std	Acc.	MCC	Prec. (F)	Recall (F)	F1 (F)	Prec. (S)	Recall (S)	F1 (S)	Score
XGBoost	None	0.7175	0.6904	0.0058	0.7090	0.2582	0.2663	0.6290	0.3742	0.9238	0.7219	0.8104	0.6821
LightGBM	None	0.7041	0.6893	0.0091	0.7040	0.2066	0.2449	0.5475	0.3385	0.9094	0.7291	0.8094	0.6415
Random Forest	None	0.6985	0.6921	0.0083	0.7785	0.1461	0.2509	0.3032	0.2746	0.8843	0.8548	0.8693	0.5403
Random Forest	SMOTE - ENN	0.6920	0.6910	0.0110	0.7384	0.1973	0.2556	0.4661	0.3301	0.9013	0.7821	0.8375	0.6016
XGBoost	**SMOTE - ENN**	**0.6873**	**0.6904**	**0.0119**	**0.5501**	**0.1753**	**0.1971**	**0.7330**	**0.3106**	**0.9240**	**0.5207**	**0.6660**	**0.7056**
LightGBM	SMOTE - ENN	0.6830	0.6898	0.0148	0.7941	0.1996	0.2939	0.3484	0.3188	0.8922	0.8656	0.8787	0.5492

Note. F = Failure class; S = Success class; AUC = test-set ROC-AUC; CV-AUC = 5-fold CV mean; ±Std = CV standard deviation; Acc. = Accuracy; MCC = Matthews Correlation Coefficient; Prec. = Precision; Score = composite 0.6 × AUC + 0.4 × Recall (F). Bold row indicates recommended model.

Table 10

Static Model Confusion Matrix Counts on Test Set

Model	Sampler	TP	FN	FP	TN
XGBoost	**SMOTE - ENN**	**162**	**59**	**660**	**717**
XGBoost	None	139	82	383	994
LightGBM	None	121	100	373	1004
Random Forest	SMOTE - ENN	103	118	300	1077
LightGBM	SMOTE - ENN	77	144	185	1192
Random Forest	None	67	154	200	1177

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success. Bold row indicates recommended model. Test set: 1,598 samples (221 Failure, 1,377 Success).

Model Performance Visualizations

Feature set selection. We evaluated three candidate feature sets. The baseline contained 11 core clinical and demographic variables. The improved set extended that to 14 by adding three facility-level variables (`City/Municipality`, `Treatment Health Facility`, and `Screening/Diagnosing Health Facility`). The extended set added another seven temporal and diagnostic delay variables on top of that for 21 features in total. Cross-validation consistently favored the 14-feature improved set, which encodes geographic and facility-level care-quality signals while only adding a small amount of complexity. The extended set did not deliver consistent AUC gains, and it pulled in variables with higher missingness rates, which made it less attractive overall. We therefore used the 14-feature improved set for all final model training and evaluation.

Figure 9 shows two views of the best configuration for each classifier. Panel (a) gives the ROC curves and panel (b) gives the precision-recall curves. The ROC view shows how each model separates treatment success from failure across the full range of decision thresholds, with the dashed diagonal representing random chance. The PR view tends to be the more informative one under heavy class imbalance like ours (roughly 86% success), and the horizontal dashed line in that panel sits at the no-skill baseline at failure prevalence. Looking at both panels together gives the overall discriminative capacity of each model alongside the precision-recall trade-off a clinician would face when picking an alert threshold.

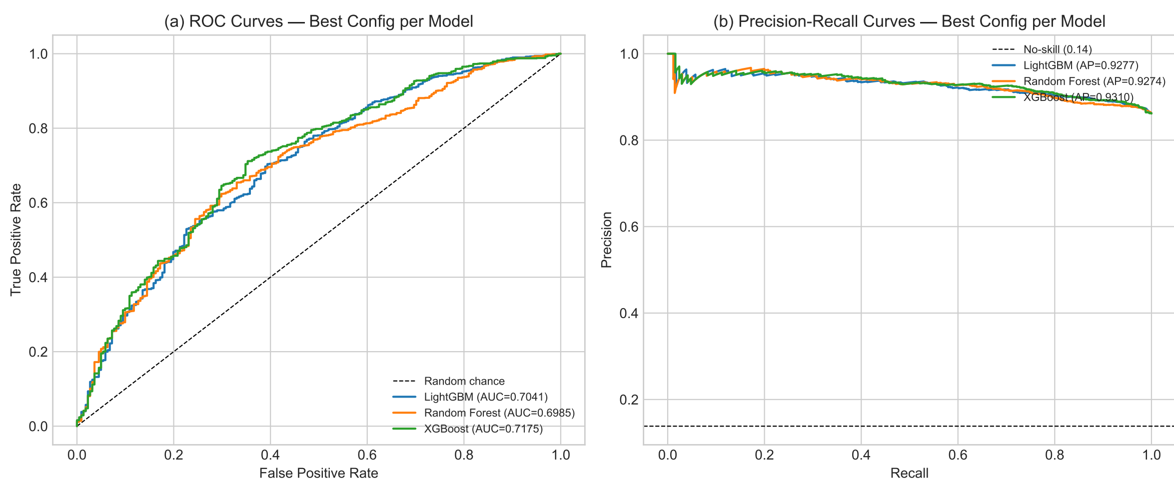


Figure 9

(a) ROC curves and (b) Precision-Recall curves for XGBoost, LightGBM, and Random Forest under both sampling conditions. AUC and Average Precision (AP) values are shown in the legend. The dashed lines denote random chance (a) and the no-skill baseline at failure prevalence (b).

Figure 10 pulls two more views together. Panel (c) shows the cross-validation stability of all six configurations. Panel (d) shows the top 14 feature importances from the best classifier. The CV view plots the 5-fold CV ROC-AUC mean with standard-deviation bars for each model and sampler combination, color-coded by sampling strategy, where narrower bars indicate steadier generalization across folds. The feature importance view ranks predictors by normalized importance, which makes it easy to see at a glance which patient attributes are driving the risk estimate. These rankings serve the explainability goal of keeping model behavior inspectable. Clinicians and program staff can see which variables the model leaned on overall, and the SHAP-based patient-level explanations then give the case-by-case version inside the CDSS itself.

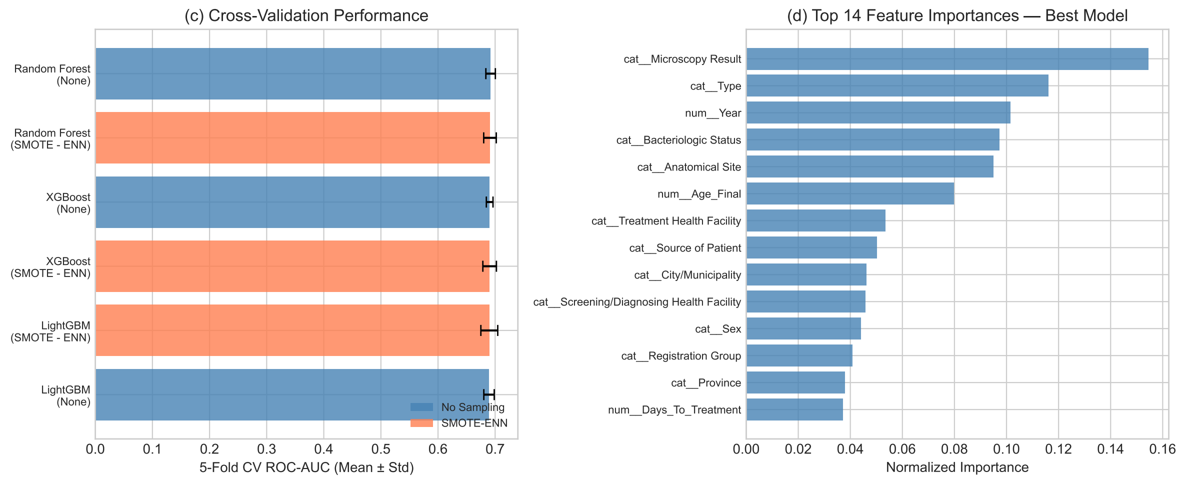


Figure 10

(c) 5-fold cross-validation ROC-AUC (mean ± std) for all six model–sampler configurations, color-coded by sampling strategy. (d) Top 14 feature importances from the best-performing classifier, normalized to sum to 1.

Table 9 gives the full metric breakdown for all six configurations. Models were ranked by a composite clinical score, $S = 0.6 \times \text{AUC} + 0.4 \times \text{Recall (Failure)}$, which weights discriminative ability against sensitivity to treatment failure. Failure is the outcome a CDSS alert exists to catch.

Of the six configurations we evaluated, XGBoost paired with SMOTE-ENN came out as the recommended model at $S = 0.699$. Its failure-class recall reached 72.9%, which was the highest of the six and meant the model caught nearly three-quarters of the patients who ended up failing treatment. Overall accuracy on the test set was lower, around 55.5%, because SMOTE-ENN resamples the training data fairly aggressively, and the model's test AUC was moderate at 0.680. We think the trade-off is clinically worth it. Inside a TB CDSS, a missed failure (a false negative) means an undetected treatment breakdown, with downstream risks of disease progression or emerging drug resistance, whereas a false alert just adds an extra clinical review. Five-fold cross-validation gave an AUC of 0.686 ± 0.015, which held steady across folds, and holdout validation returned 68.2% recall, so the recall advantage carried through to data the model had never touched.

For the models without resampling, XGBoost (no sampling) gave the highest test-set AUC of the lot at 0.712, alongside the best CV-AUC at 0.702 ± 0.009 and a still-competitive 56.6% failure recall. We would reach for that configuration when the goal is a continuous and well-calibrated risk score, such as ranking patients by relative risk rather than applying a hard binary alert threshold. LightGBM (no sampling) sits close behind at AUC 0.708, recall 55.2%, and corroborates the same general picture. The remaining three configurations, Random Forest under both samplers and LightGBM with SMOTE-ENN, came in lower on the composite, mainly because weak failure recall was dragging their AUC contribution down.

MCC tells the rest of the story. Table 9 shows what choosing the high-recall configuration actually costs. XGBoost with SMOTE-ENN scored an MCC of 0.175, the lowest among the six test-set configurations, while XGBoost without resampling sat at the top with 0.258 and balanced the four confusion-matrix quadrants more cleanly. This gap is the same tension that shows up in any imbalanced clinical classification problem. SMOTE-ENN improves failure recall by injecting a large volume of synthetic minority-class samples during training, but the trained model then produces proportionally more false positives on the real test distribution, and MCC penalizes that symmetrically. The upshot is that the SMOTE-ENN configuration suits a screening deployment where catching every at-risk patient is the priority, and the no-resampling configuration suits cases where the cost of unnecessary reviews is the binding constraint.

Temporal Models Performance

Table 11

Version 1 (v1) Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

What Table 11 shows for v1 on the held-out test split is misleading at first glance. The headline numbers look strong (accuracy 0.9048, F1 0.9500), but those are essentially the dominant Success class showing through under severe imbalance. Specificity is the giveaway. At 0.0000, v1 caught none of the actual Failure cases in the test split, which is the exact opposite of what a clinical screening tool is supposed to do. That gap is what forced the corrected v2 iteration.

Table 12

Version 2 (v2) Test Set Performance Metrics of Temporal Models (M12)

Model	Accuracy	Bal. Acc.	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC	Threshold
Random Forest (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9693	0.9950	0.67
LightGBM (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.8991	0.9779	0.78
XGBoost (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9298	0.9863	0.66
XGBoost (Baseline)	0.9091	0.8772	0.9722	0.9211	0.9459	0.8333	0.9430	0.9895	0.81
Hybrid Bi-LSTM (Baseline)	0.8409	0.9079	1.0000	0.8158	0.8986	1.0000	0.9868	0.9981	0.55
Hybrid Bi-LSTM (Augmented)	0.7500	0.8553	1.0000	0.7105	0.8308	1.0000	0.9474	0.9910	0.89

The v2 numbers in Table 12 reflect the corrected workflow, after we addressed the leakage risk, fixed the outcome-label handling, and rebuilt the temporal feature preparation. On the v2 held-out test set, the three SMOTE-ENN tree models, Random Forest, LightGBM, and XGBoost, all converged on the same hard-label confusion matrix: 5 of 6 Failure cases caught, 36 of 38 Success cases correctly identified, giving 0.9318 accuracy and 0.8333 specificity. Random Forest came out on top among them on both ROC-AUC (0.9693) and PR-AUC (0.9950). The Hybrid Bi-LSTM baseline behaved differently. It managed to catch every Failure case in the split (specificity 1.0000) but at the cost of raising more false Failure alerts on true Success cases, which pulled its accuracy down. That is the standard sensitivity-versus-specificity trade-off a screening-oriented CDSS has to weigh deliberately rather than wish away.

Table 13

Version 2 (v2) Confusion Matrix Counts of Temporal Models

Model	TP	FN	FP	TN
Random Forest (+SMOTE-ENN)	5	1	2	36
LightGBM (+SMOTE-ENN)	5	1	2	36
XGBoost (+SMOTE-ENN)	5	1	2	36
XGBoost (Baseline)	5	1	3	35
Hybrid Bi-LSTM (Baseline)	6	0	7	31
Hybrid Bi-LSTM (Augmented)	6	0	11	27

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success.

Table 14

Methodological Differences Between Temporal Modeling Iterations (v1 vs v2)

Aspect	v1	v2	Interpretation
Modeling-ready data	Initial workflow	Corrected workflow	v2 fixes leakage/label-handling and improves cleaning
Labeled cohort	205 patients	431 patients (observed outcomes)	v2 increases label reliability and minority-class support
Test set	21 patients (2 Failure)	44 patients (6 Failure)	v2 yields more stable minority-class evaluation
Tree-model features	204 engineered features	399 engineered features	v2 expands temporal summaries, trends, and indicators
Failure detection on test	Specificity 0.0000	Specificity 0.8333–1.0000	v2 meaningfully improves Failure identification

Table 15

Model-Level Performance Comparison Between Temporal Model v1 and v2 (Test Sets)

Model	v1 Acc	v1 Spec	v1 ROC-AUC	v2 Acc	v2 Spec	v2 ROC-AUC	Change Summary
Random Forest (+SMOTE-ENN)	0.7143	0.0000	0.5263	0.9318	0.8333	0.9693	Stronger Failure detection and ranking
LightGBM (+SMOTE-ENN)	0.9048	0.0000	0.2105	0.9318	0.8333	0.8991	Improved minority-class separation
XGBoost (+SMOTE-ENN)	0.9048	0.0000	0.1053	0.9318	0.8333	0.9298	Large ROC-AUC and specificity gain
XGBoost (Baseline)	0.9048	0.0000	0.5789	0.9091	0.8333	0.9430	Better calibrated Failure detection
Hybrid Bi-LSTM (Baseline)	0.9048	0.0000	0.4737	0.8409	1.0000	0.9868	Catches all Failures; more false alerts
Hybrid Bi-LSTM (Augmented)	0.9048	0.0000	0.3684	0.7500	1.0000	0.9474	Higher sensitivity; reduced overall accuracy

Figure 11

Version 2 (v2) Hybrid Bi-LSTM Per-Month Performance (M0–M12)

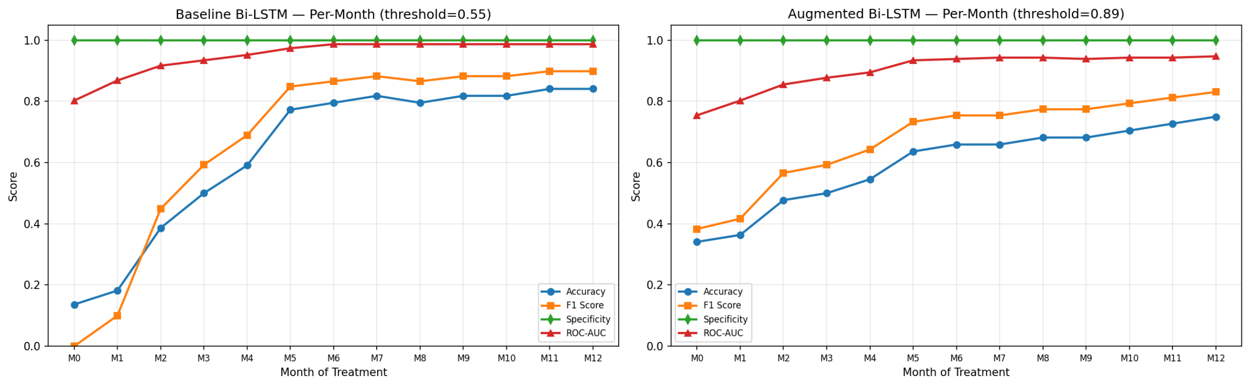


Figure 12

Version 2 (v2) XGBoost Per-Month Performance (M0–M12)

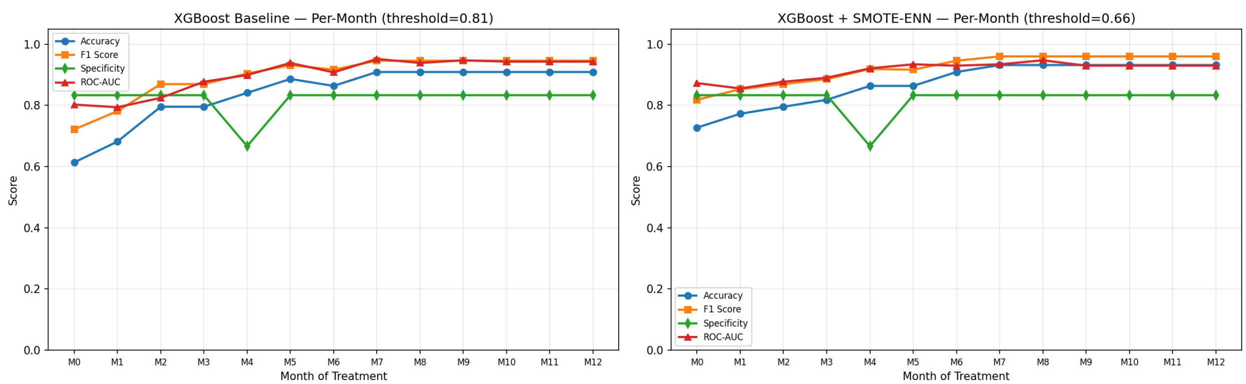


Figure 13

Version 2 (v2) LightGBM Per-Month Performance (M0–M12)

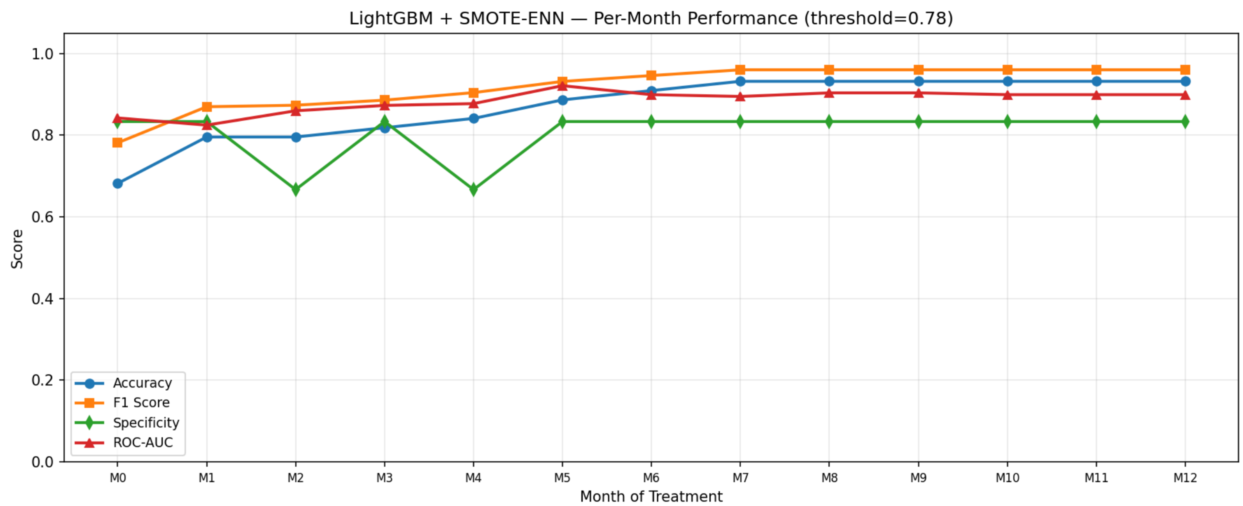
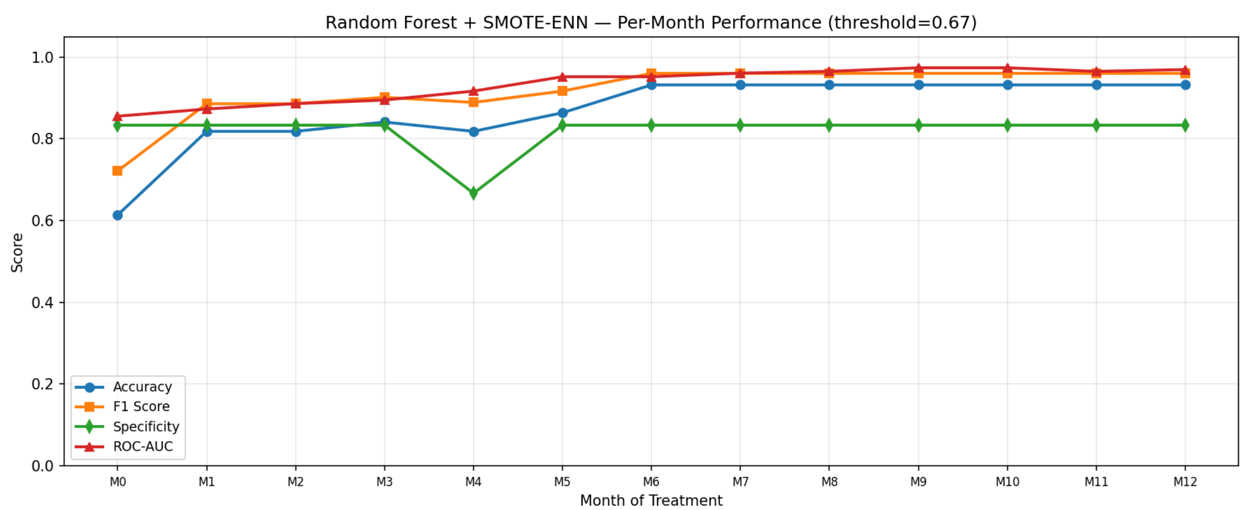


Figure 14

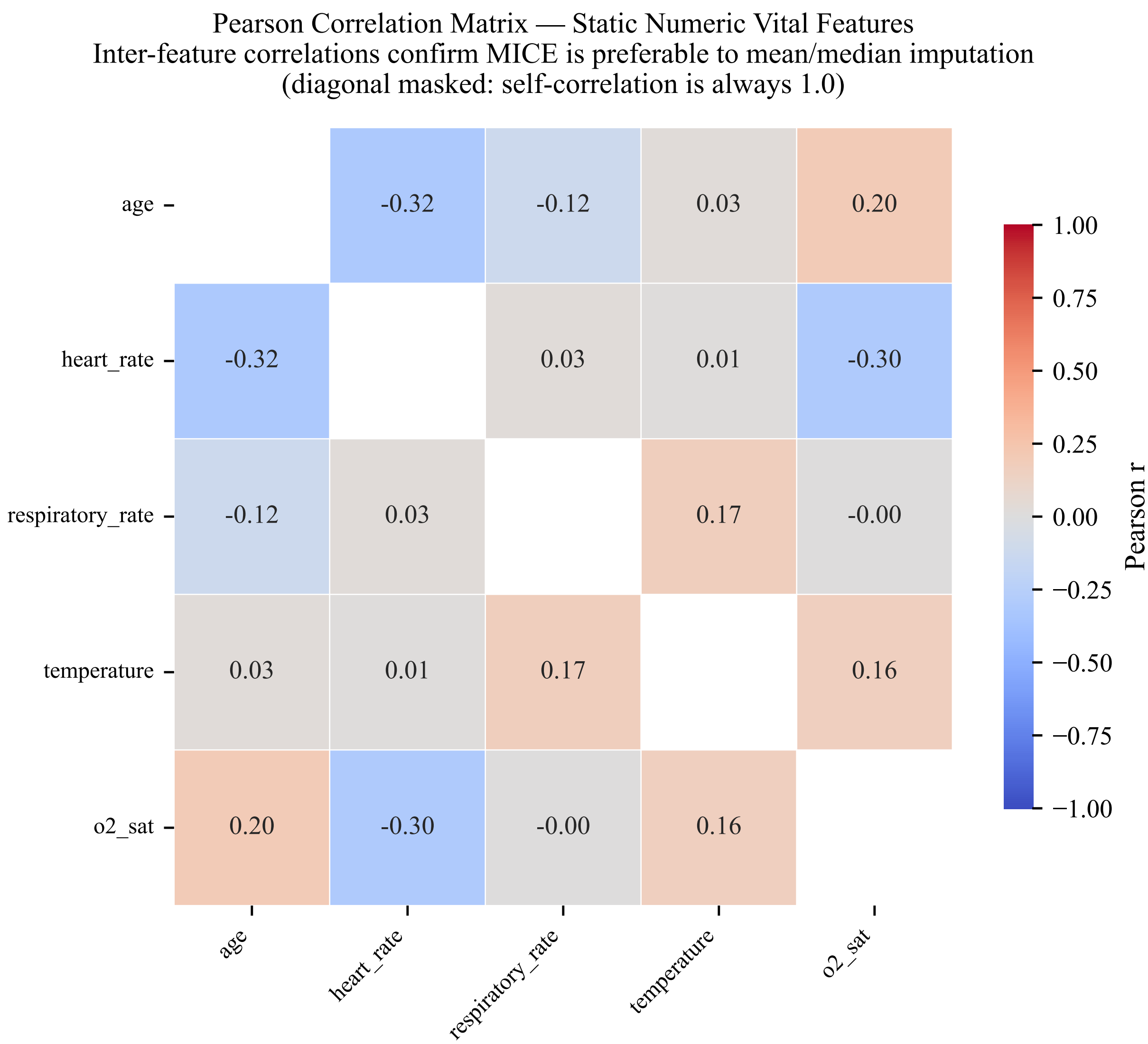
Version 2 (v2) Random Forest Per-Month Performance (M0–M12)



Figures 11–14 follow v2 performance as monitoring data accumulates from M0 through M12, across the Hybrid Bi-LSTM, XGBoost, LightGBM, and Random Forest models. We think the month-by-month view earns its place in TB-DOTS decision support, since it lets a reviewer see when each model actually starts discriminating reliably beyond baseline. Tables 12 and 13 give the single-point M12 numbers, which is one moment in the treatment course. The figures show how prediction quality builds up over the course of

Figure 15

Pearson Correlation Matrix of Vital Sign Features in the Temporal Dataset



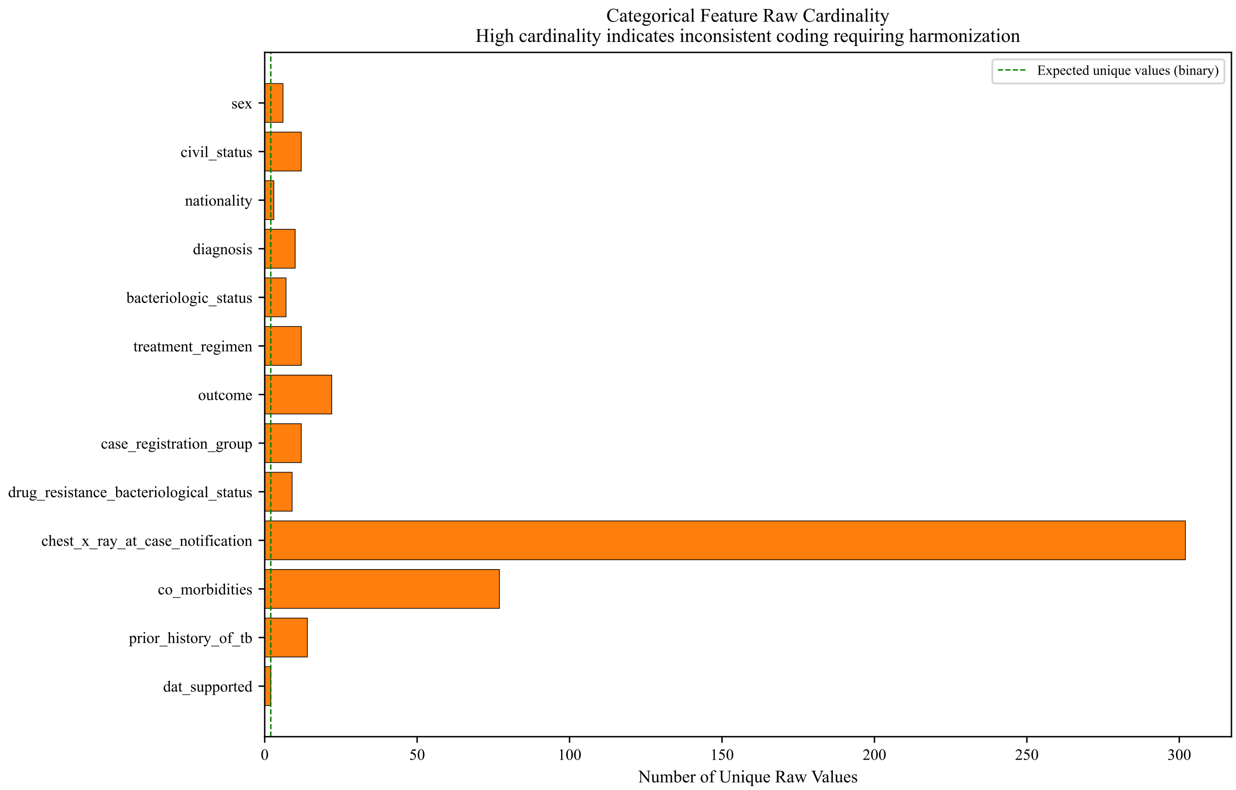
therapy rather than only crystallizing at the end.

Correlation Analysis of Clinical and Temporal Features

Pairwise correlations among the vital signs in the temporal dataset come out moderate, which is the condition MICE needs. The strongest was between age and heart rate ($r = -0.32$), then oxygen saturation and heart rate ($r = -0.30$), and systolic with diastolic blood pressure ($r = 0.24$). The magnitudes are modest,

Figure 16

Categorical Feature Raw Cardinality: Justification for Stage 4 Harmonization



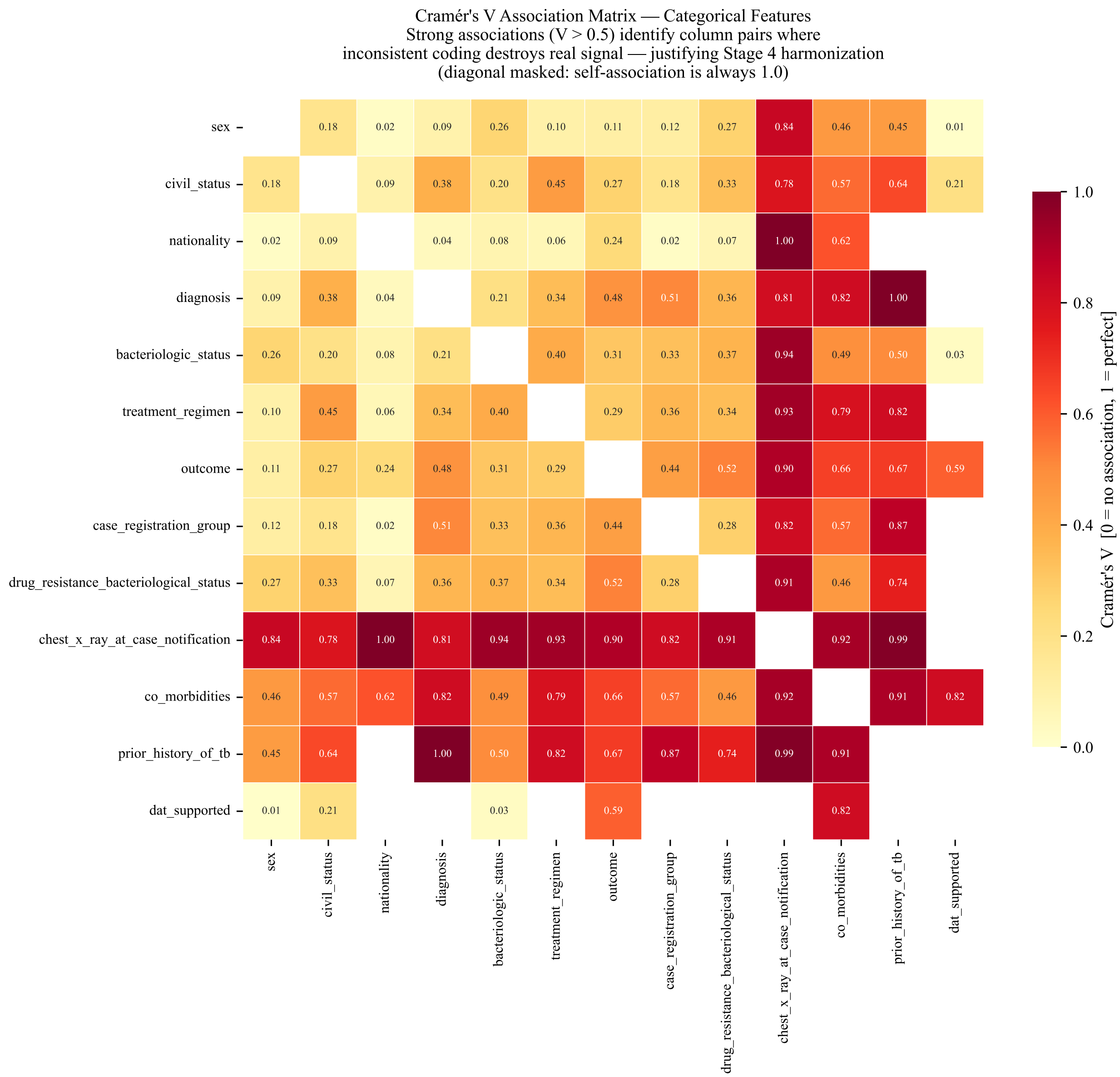
but they are enough for multivariate imputation to borrow shared variance across the vitals and produce more plausible fills than a univariate mean. Respiratory rate and temperature correlated uniformly low with the other vitals ($|r| < 0.10$), so their missingness is better left to the missing indicator with fill than to MICE.

Figure 16 makes a fairly blunt point. The cardinality audit on the raw categorical features showed that data entry inconsistency was widespread in the temporal dataset, with several columns carrying far more unique values than any standardized coding scheme would predict. `chest_x_ray_at_case_notification` held over 300 unique free-text entries; `co_morbidities` broke past 150 variants; `prior_history_of_tb` and `drug_resistance_status` each had more than 20 distinct raw representations. Even fields that are nominally binary, like `sex` and `outcome`, showed 5 to 12 unique raw values, mostly because abbreviation conventions varied from facility to facility.

That kind of heterogeneity is what drove the Stage 4 categorical harmonization step, which maps free-text and inconsistently coded entries onto standardized categories before any analysis touches them.

Figure 17

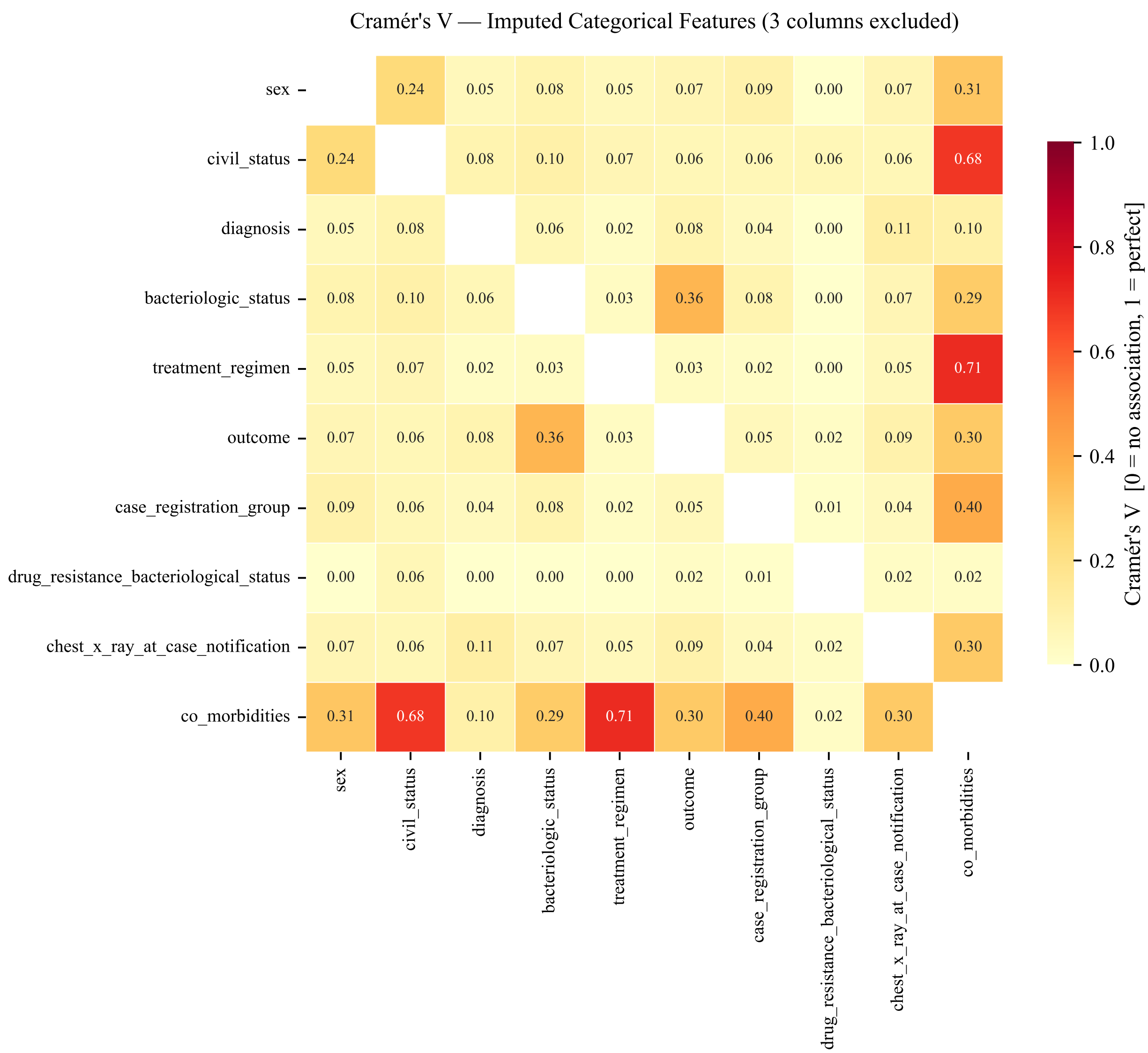
Cramér's V Correlation Matrix of Raw Categorical Variables in the Temporal Dataset



When we look at Cramér's V associations among the raw categorical variables in the temporal dataset, the matrix shows widespread strong associations, with quite a few pairwise values exceeding 0.80. Most of those inflated coefficients are not picking up real clinical associations. They are being driven by `chest_x_ray_at_case_notification`, which holds over 300 unique free-text entries in the raw data, and by `co_morbidities`, where inconsistent coding across facilities artificially boosts the appearance of association. We treat this as a measurement-side artifact rather than a finding about patients, and it is exactly

Figure 18

Cramér's V Correlation Matrix of Imputed Categorical Variables



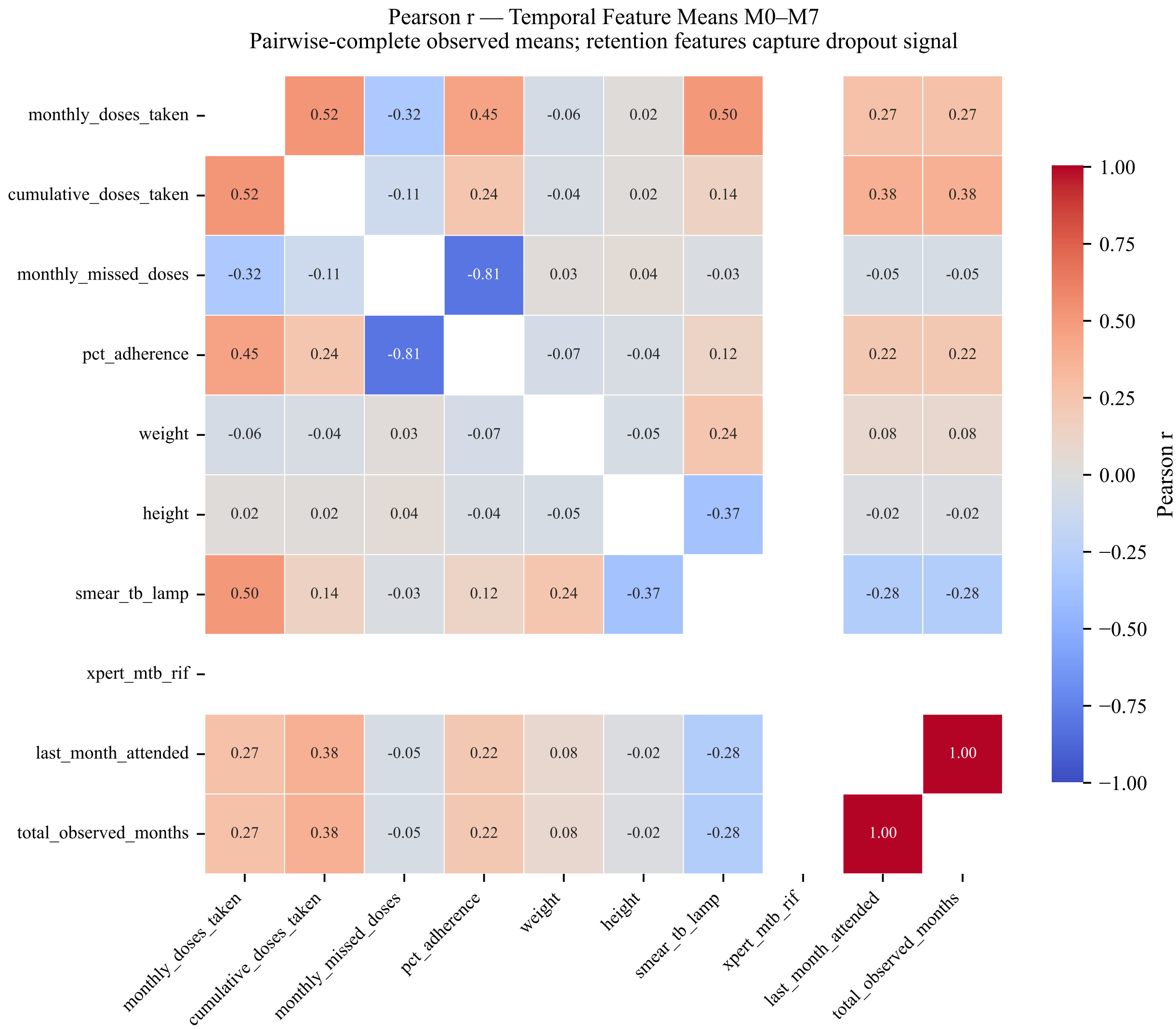
	sex	civil_status	diagnosis	bacteriologic_status	treatment_regimen	outcome	case_registration_group	drug_resistance_bacteriological_status	chest_x_ray_at_case_notification	co_morbidities
sex		0.24	0.05	0.08	0.05	0.07	0.09	0.00	0.07	0.31
civil_status	0.24		0.08	0.10	0.07	0.06	0.06	0.06	0.06	0.68
diagnosis	0.05	0.08		0.06	0.02	0.08	0.04	0.00	0.11	0.10
bacteriologic_status	0.08	0.10	0.06		0.03	0.36	0.08	0.00	0.07	0.29
treatment_regimen	0.05	0.07	0.02	0.03		0.03	0.02	0.00	0.05	0.71
outcome	0.07	0.06	0.08	0.36	0.03		0.05	0.02	0.09	0.30
case_registration_group	0.09	0.06	0.04	0.08	0.02	0.05		0.01	0.04	0.40
drug_resistance_bacteriological_status	0.00	0.06	0.00	0.00	0.00	0.02	0.01		0.02	0.02
chest_x_ray_at_case_notification	0.07	0.06	0.11	0.07	0.05	0.09	0.04	0.02		0.30
co_morbidities	0.31	0.68	0.10	0.29	0.71	0.30	0.40	0.02	0.30	

what the Stage 4 categorical harmonization step is designed to fix, by collapsing the free-text entries into clinically meaningful groupings before the analysis ever runs.

After preprocessing, the same Cramér's V matrix (Figure 18) looks substantively different. Median off-diagonal V drops from roughly 0.46 to 0.07, which is the result we wanted, since it means most of those earlier inflated associations were artifacts of data entry rather than properties of the patients. Three columns that were present in the raw matrix (`prior_history_of_tb`, `dat_supported`, and `nationality`) drop out here because there was not enough variance or enough non-missing pairs in the cleaned data to compute

Figure 19

Temporal Feature Correlation Matrix of Treatment Variables

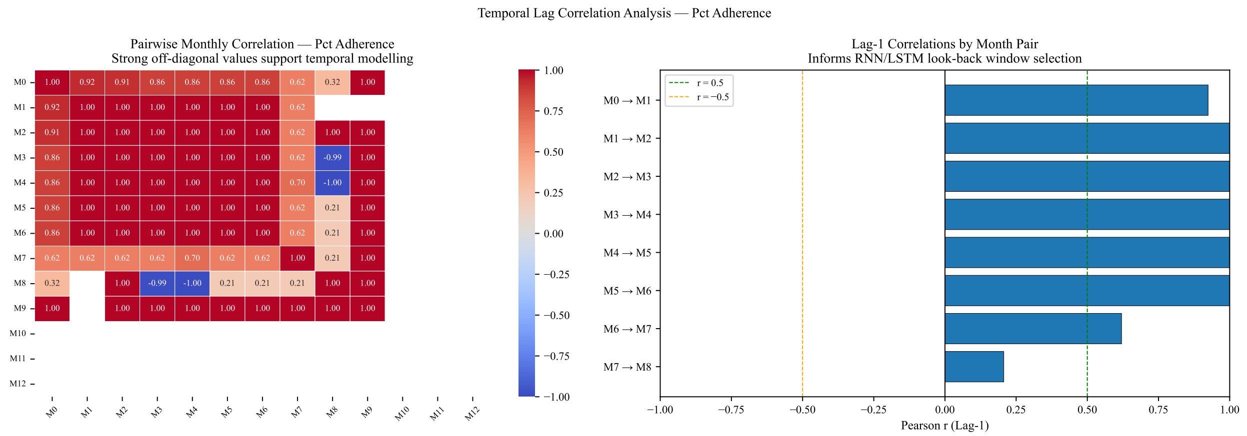


associations reliably.

The temporal feature correlation matrix (Figure 19) maps how the main treatment variables relate over the M0–M7 window. The adherence metrics (`percentage_adherence`, `doses_taken_per_month`, and `cumulative_doses_taken`) form a tight, strongly inter-correlated cluster ($r > 0.85$), which is what internal measurement consistency looks like. Anthropometric features (`weight`, `height`) are orthogonal to every adherence measure ($r \sim 0$), so body size varies independently of treatment compliance in this cohort. Retention features (`last_month_attended`, `total_observed_months`) associate only weakly and

Figure 20

Temporal Lag Correlation of Percentage Adherence Across Consecutive Treatment Months



positively with adherence ($r \approx 0.2$), so visit persistence and dose compliance pick up distinct sides of patient engagement. Diagnostic variables such as `smear_microscopy_result` sit near zero against adherence and retention, in line with their role as outcome indicators rather than behavioral predictors. Xpert MTB/RIF is absent from the matrix because every observed value in the temporal subset was identical (zero variance), which leaves the Pearson coefficient undefined.

Table 16

Lag-1 Temporal Correlation of Pct Adherence Across Consecutive Treatment Months (Justifying Time-Series Model Architecture)

Lag Pair	Pearson r	p-value	Significant?	N (pairs)
M0 → M1	0.9242	0.0000	Yes	300
M1 → M2	1.0	0.0000	Yes	285
M2 → M3	1.0	0.0000	Yes	275
M3 → M4	1.0	0.0000	Yes	271
M4 → M5	0.9999	0.0000	Yes	254
M5 → M6	1.0	0.0000	Yes	197
M6 → M7	0.6202	0.0000	Yes	80
M7 → M8	0.2066	0.6945	No	6

Adherence is stable month to month in the temporal dataset, and that is the empirical case for the temporal architecture. Early months were almost perfectly lag-1 correlated (M0–M1: $r = 0.92$, $p < 0.001$; M1–M2: $r = 1.00$, $p < 0.001$), so adherence is strongly autocorrelated at the start of treatment. The correlation stayed above $r = 0.99$ through M5–M6, dropped at M6–M7 ($r = 0.62$, $p < 0.001$), and went non-significant at M7–M8 ($r = 0.21$, $p = 0.69$), where only six patient pairs were left. Autocorrelation that persistent means adherence carries real sequential structure, which is the reason to use recurrent architectures (RNN, LSTM) that model temporal dependencies (Nayebi et al., 2022). The drop after M6 also matches the retention cascade: patients heading for dropout tend to adhere more erratically before they are lost to follow-up.

Clinical Decision Support System (CDSS) Implementation

The validated models run inside a Clinical Decision Support System (CDSS) meant for use at the point of care. The system takes a patient's data and produces four clinical artifacts that connect the model output to day-to-day TB-DOTS practice in the CAR.

Automated Patient Profiling and Risk Stratification

As soon as a patient's data goes in, the system assembles a single view of where that patient currently stands. The *Patient Overview* screen (Figure 21) brings the demographic data and the baseline health indicators together into one dashboard. From there a *Risk Analysis* (Figure 22) follows, which surfaces the model's estimated probability of treatment failure for that particular patient. The point of sorting patients into risk tiers is operational rather than statistical: it gives health administrators a basis for putting limited resources onto the high-risk patients before treatment is over and the outcome is already decided.

Figure 21

CDSS Patient Overview Dashboard Interface

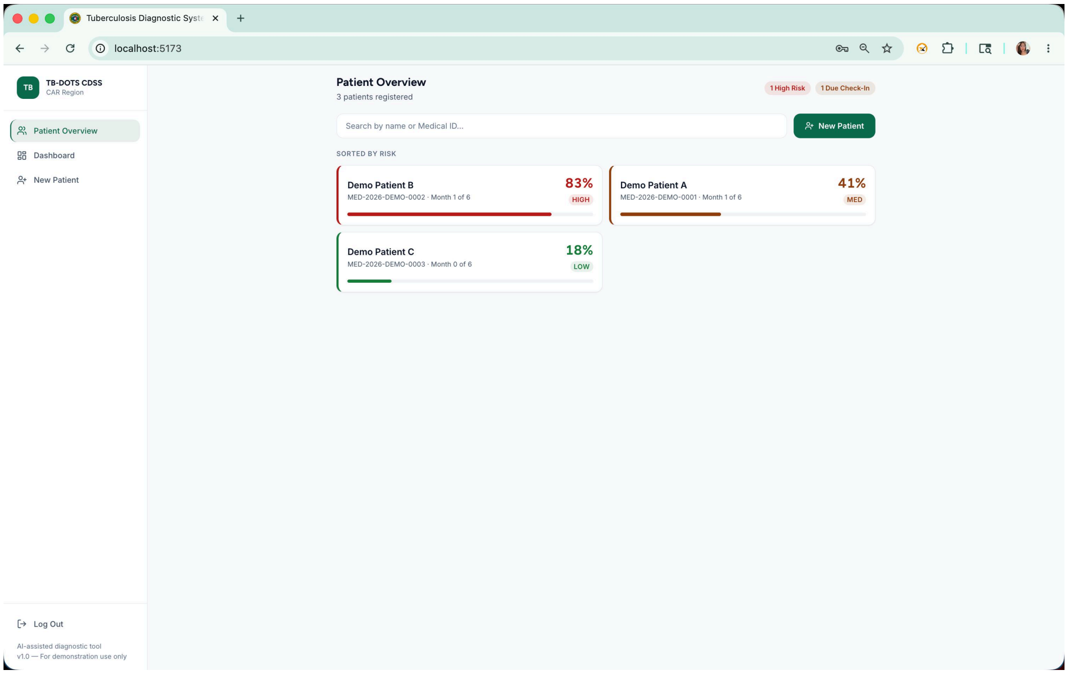
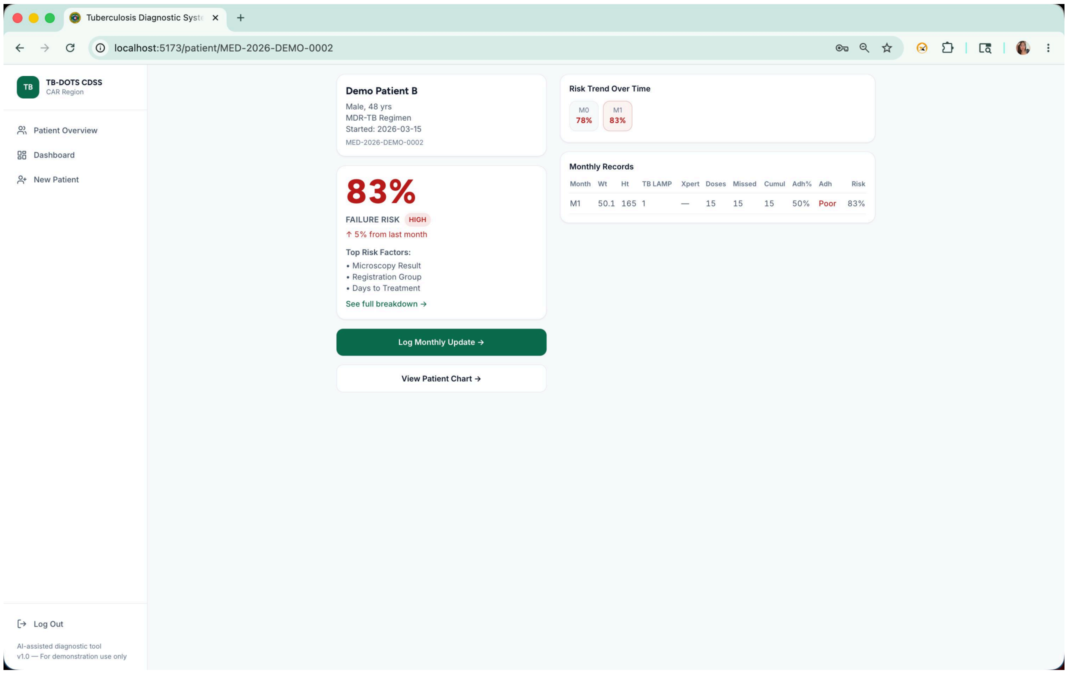


Figure 22

Automated Risk Analysis and Failure Probability Projection



Explainable AI and Intervention Planning

The *Full Breakdown* report (Figure 23) is where SHAP does its work. It shows why a given risk score came out the way it did, and that visibility is what gives a clinician something to verify. The report names the specific variables behind the prediction, things like low BMI or a documented history of TB, rather than leaving the score floating without context.

From there, the *First Step* protocol (Figure 24) translates a prediction into a next move for the clinician. Its Risk Factor Analysis pairs the risk factors the model flagged for that patient with a standard set of initial clinical actions, so the system's output is not just a number but something that can plausibly turn into care.

Figure 23

SHAP-based Predictive Factor Breakdown for Individual Patients

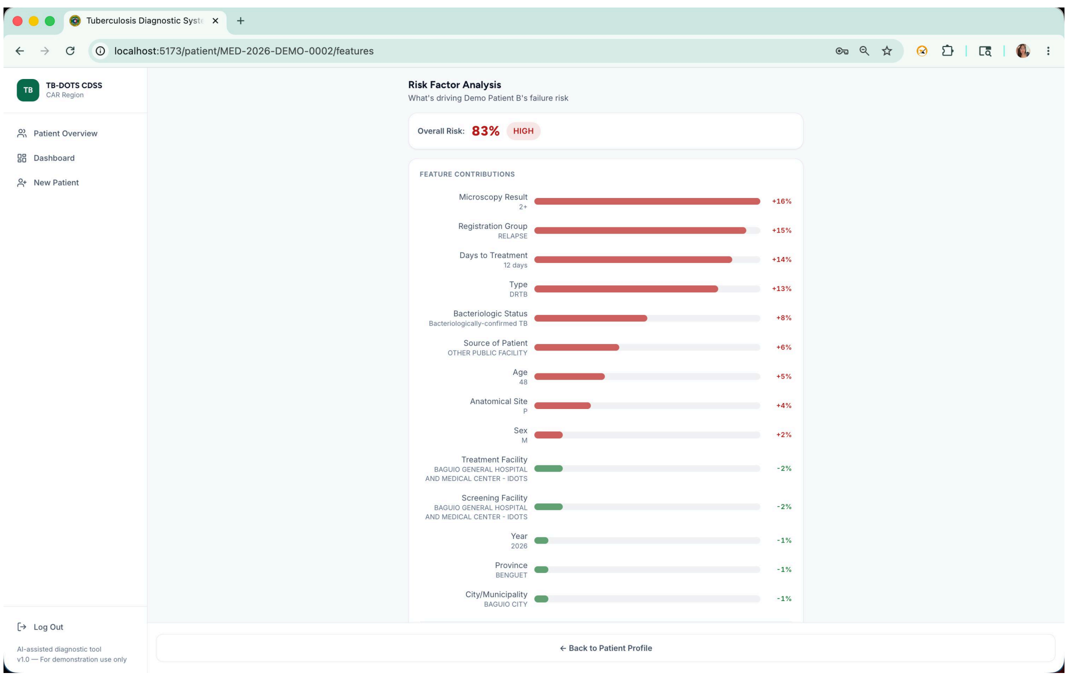
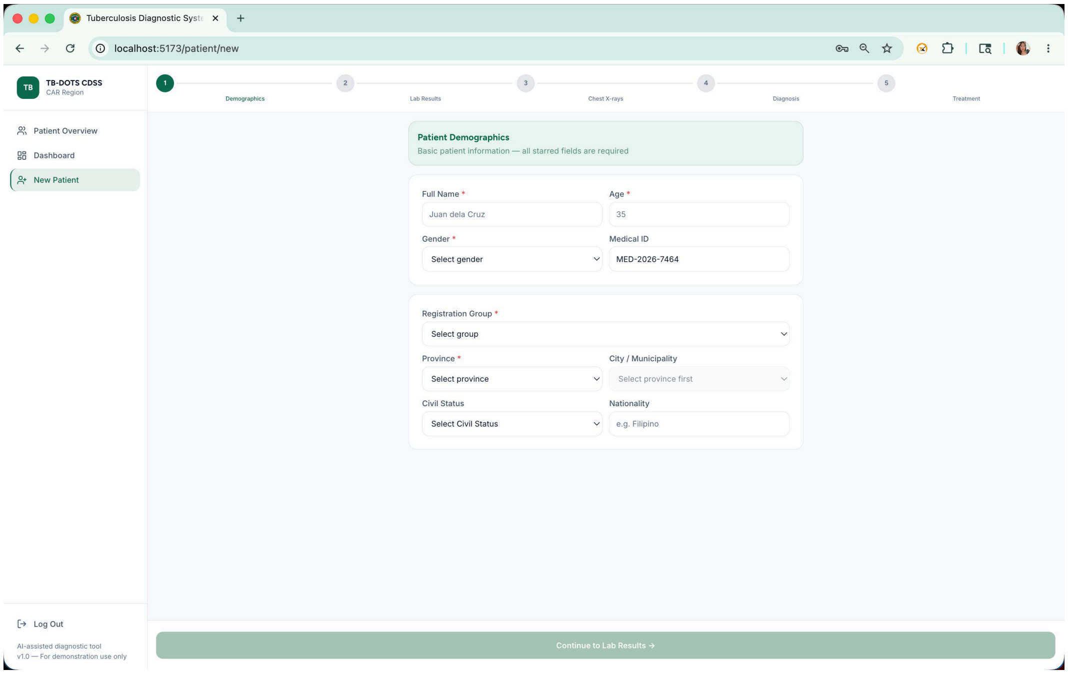


Figure 24

Initial Clinical Intervention and Management Protocol (First Step)



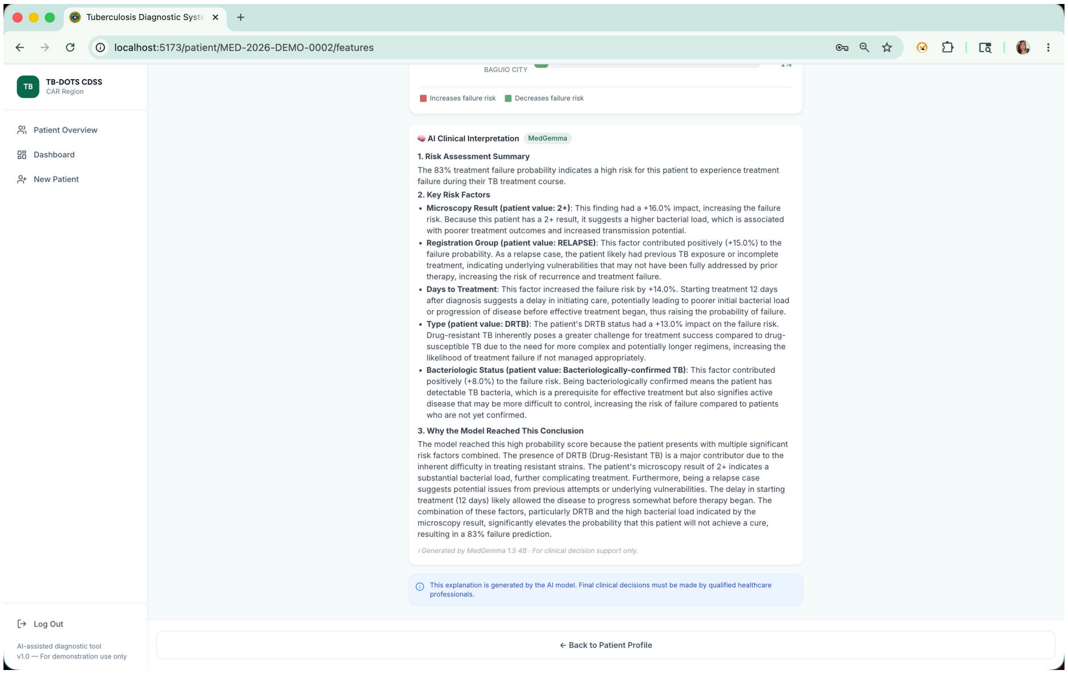
Large Language Model (LLM) Integration for Clinical Context

SHAP handles the quantitative side of feature importance. On top of that, *Med-Gemma*, a fine-tuned medical large language model, adds a short written account of the same risk factors. As Figure 25 shows, Med-Gemma lives inside the Risk Factor Analysis interface and reads the statistical weights back as plain clinical language.

Med-Gemma writes from the specific patient profile in front of it, things like age, co-morbidities, and sputum conversion, and produces a short summary that puts the SHAP factor analysis into clinical terms. Rather than leaving the feature weights to stand on their own, it explains why the variables SHAP picked out matter for this particular patient. The net effect is that the CDSS no longer reports only a risk percentage. It also spells out the reasoning the model leaned on, which makes the output easier for providers to actually read and challenge.

Figure 25

Med-Gemma Clinical Narrative and Risk Factor Explanation



Discussion

What we set out to test was whether machine learning could usefully predict treatment outcomes (Success or Failure) for PTB patients in the DOH-CAR TB-DOTS program, drawing on both monthly monitoring data and baseline clinical features. The temporal half of that work went through two iterations before we trusted it. Version 1 was instructive mainly as a cautionary tale. It showed how easy it is to walk straight into the imbalanced-prediction trap: a model that posts a perfectly respectable accuracy number by siding with the majority Success class every time and quietly missing every real Failure. Specificity was 0.0000 on the held-out test split. Failure identification was simply not happening under that first workflow, and pushing the v1 results forward would have been irresponsible.

Version 2 fixed the underlying issues: leakage risk, the way outcome labels had been handled, and the temporal feature construction. After that rebuild, the SMOTE-ENN tree models landed on the same confusion matrix on the held-out test split, catching 5 of 6 Failure cases and correctly classifying 36 of 38 Success cases. Among those tree models, Random Forest had the strongest held-out ranking (ROC-AUC

0.9693, PR-AUC 0.9950), while LightGBM showed the strongest cross-validated balanced accuracy in the saved artifacts. The Hybrid Bi-LSTM baseline went the other direction. It detected every Failure case on the held-out test split (specificity 1.0000) but raised more false Failure alerts among the true Success cases, which dragged its overall accuracy down. That tradeoff is not necessarily wrong for a screening CDSS, where missing a high-risk patient costs more than an extra clinical review, but it does need to be managed, otherwise the alerts lose their meaning.

A few things came out of all this that feel worth naming. The first is that pairing monthly monitoring data with the static baseline features really did predict better than the baseline data alone, so the longitudinal half was not just expensive ornament. The second is that SMOTE-ENN, alongside the Bi-LSTM data augmentation, did help the models catch high-risk patients under the extreme imbalance of TB outcomes, which had been the main obstacle the v1 workflow could not get past. The third is harder to compress into a contribution bullet but is probably the most important: prediction on its own is not enough in a clinical setting. A risk score the clinician cannot inspect is a risk score the clinician will not trust, and SHAP explanations together with the bounded SLM narrative layer are how this study tries to make the output inspectable. The point is to support clinician review, not replace it.

Taken together, the v2 temporal outputs look reasonable as a starting point for further evaluation inside real TB monitoring workflows, particularly for catching earlier the patients who need closer review. If deployment ever comes up, the v2 tree models, Random Forest and LightGBM with SMOTE-ENN in particular, hit a useful middle between Failure detection and overall correctness, while the Hybrid Bi-LSTM baseline fits a more cautious screening stance where missing nothing matters more than minimizing false alerts. None of these models should reach a patient without external validation on independent data, prospective testing in real workflows, and review by healthcare professionals for calibration, safety, and clinical fit. Future work should also push toward larger and multi-region datasets to improve generalization, track temporal calibration over time, and explore hybrids that combine deep temporal representations with tree-based learners. Adding interpretability over the temporal dynamics themselves, attention across treatment months for instance, would probably help clinician trust as much as anything else we could add.

Beyond the modeling, the CDSS artifacts shift the program in a particular direction. NTP protocols, by design, register treatment failure mostly after it has already happened. The predictive outputs instead create room for intervention during the early months of therapy, when adjustments still have a chance to change the outcome. The *Full Breakdown* report works against the black-box reputation of ensemble models by letting health workers compare the model's logic against their own clinical experience. The *First Step* document then carries that into the care pathway, which can cut down the MDR-TB that unrecognized non-adherence would otherwise generate. Pairing SHAP with Med-Gemma is also a deliberate response to the skepticism providers often bring to AI in healthcare: with both a statistical breakdown and a plain-language narrative on the same screen, the CDSS keeps a human in the loop and gives DOH-CAR health workers something they can actually argue with before acting.

Recommendations

Our concrete suggestion to healthcare facilities and policymakers is to consider adding the v2 tree-based models (Random Forest and LightGBM with SMOTE-ENN especially) into the existing DOH-CAR TB-DOTS monitoring stack as a decision-support layer for early risk stratification. On the held-out test set, these two detected treatment failure in a balanced way while keeping overall accuracy high, which is what makes them a practical way to flag patients who may need closer clinical review. Before any deployment, though, the models still need external validation on independent multi-facility data, prospective testing under real workflow conditions, and review by clinicians for calibration, safety, and fit with current practice. None of that work has happened yet.

The explanation layer needs its own evaluation before it goes anywhere near a patient. We ran a preliminary internal faithfulness check on the 44-patient test cohort and the finding was unambiguous. Stripped of the attribution signal, the narrative layer almost never reproduced the model's SHAP attribution, and not one of the 44 cases passed the faithfulness criterion under that blind condition (0 of 44), regardless of which SLM backend we tried. Hand the SLM the SHAP contribution signals in the prompt and faithfulness rises sharply, which is exactly what we expected, since that confirms the layer for what it is: a constrained verbalizer of the model's reasoning, not a reasoner of its own. What follows from this is straightforward. The faithfulness evaluation needs to be widened to a larger multi-facility cohort, run with several SLM backends, and paired with clinician-rated readability, trust, and actionability as outcome measures. Without that evidence base, deploying the explanation component would be premature.

Safety Scope and Interpretation Limits

The explanation layer is there to make the model's reasoning readable, turning contribution signals into clinical text someone can follow. But its output has to sit inside fairly firm safety bounds. What SHAP actually says is how much a given feature pushed the predicted risk up or down. It does not establish a causal mechanism of disease progression, and nothing about a SHAP value promises that changing a highlighted factor will change the patient's outcome. That distinction earns its keep in high-stakes settings, where a generative output is very easy to over-read past what actually supports it (Vasey et al., 2022; World Health Organization, 2021).

We have therefore positioned the system as decision support and nothing more. It flags risk, surfaces the factors that contributed to that risk, and summarizes how the model behaved. Diagnosis and treatment decisions stay with the clinician. That framing lines up with current reporting and governance guidance for AI-assisted clinical prediction, where transparency, bounded use, and human oversight are still the core requirements (Collins et al., 2024).

The preliminary faithfulness check described above feeds back into this same framing. Because the narrative layer turned out to depend on SHAP contribution signals rather than reconstructing the model's attribution from patient context alone, the operational design keeps SHAP as the primary explanation surface and treats the narrative output as a readability aid, not a substitute.

Conclusion

TB treatment in the Cordillera Administrative Region looks the way most clinical realities look up close: heterogeneous patients, uneven record-keeping, and external shocks that arrive on their own schedule. Single-variable analyses cannot really see across that, so the study began with something less glamorous than a model, namely a repeatable cleaning pipeline for two related datasets, the 7,389-patient DOH-CAR registry and the 599-patient month-by-month monitoring set the Pharmacy Co-authors transcribed by hand. That pipeline standardized formats, stripped patient identifiers, and decided each missing value by first asking why it was missing (MCAR, MAR, or MNAR) before routing it to one of four matching strategies. Once we trusted the cleaned data, rather than just having filled it, the descriptive and exploratory work could surface what actually mattered: outcome trends across the decade, the rise in deaths and lost follow-ups through the COVID-19 period, and the patient-level, time-based, and operational patterns that ended up shaping the models.

Those signals carried straight through into model performance. On the static dataset, XGBoost with SMOTE-ENN became our strongest screening configuration (composite score $S = 0.699$; failure recall 72.9%), while XGBoost without resampling produced the best-calibrated risk ranking (test AUC 0.712, MCC 0.258), so we ended up with two setups serving two different clinical uses. The longitudinal models went further still. After v2 cleared up the leakage and label-handling problems that had crippled v1, the SMOTE-ENN tree models (Random Forest, LightGBM, XGBoost) reached 0.9318 accuracy and 0.8333 specificity, catching five of six failure cases on the held-out test set, while the Hybrid Bi-LSTM caught all six but raised more false alarms. Sitting on top of those predictions are SHAP feature explanations and a bounded SLM narrative layer, all wrapped into a Clinical Decision Support System that displays each patient's risk, the factors behind it, and a refreshed estimate as new monthly data comes in.

We think the results are promising clinically, but we do not think they are ready for actual clinical use, and we want to say that clearly. The test cases came from a single regional registry. The narrative layer's faithfulness was checked on only 44 patients. So before any of this becomes part of clinical practice, the system still has to clear external validation on data from other facilities, prospective testing under real workflow conditions, and a careful review by healthcare professionals for safety, calibration, and fit. The narrative layer in particular needs a much larger faithfulness study before it goes anywhere near a clinical decision. Once those conditions are met, the framework, together with the published pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset, could support regional TB control and further work on explainable clinical prediction. Not before.

Acknowledgments

This study was carried out at Saint Louis University, and we owe much of it to people who made room for the work to happen. The Department of Computing and Information Studies, under SAMCIS, gave us the lab time, the access, and the patience to figure things out as we went. Dr. Randy Domantay opened doors we would not have known to knock on. Our adviser, Dr. Josephine S. Dela Cruz, read drafts that were not always easy reading, pushed back where it counted, and stayed with the project through the long stretches when nothing was working. Our partner researchers in the SONAHBS Department of Pharmacy did the part of this work that no one else could have done: they went to the TB-DOTS facilities in person, sat with paper Treatment Cards, and transcribed the 599-patient monitoring set by hand from records that had never been digitized. Without their fieldwork, the temporal half of this thesis simply would not exist.

References

Allgaier, J., Mulansky, L., Draelos, R. L., & Pryss, R. (2023). How does the model make predictions? A systematic literature review on the explainability power of machine learning in healthcare. *Artificial Intelligence in Medicine*, *143*, 102616. https://doi.org/10.1016/j.artmed.2023.102616

Alruhaymi, A., & Kim, C. (2021). Why can multiple imputations and how (mice) algorithm work? *Open Journal of Statistics*, *11*(5), 759–777. https://doi.org/10.4236/ojs.2021.115045

Amante, T. D., & Abdosh, T. (2015). Risk factors for unsuccessful tuberculosis treatment outcome (failure, default and death) in public health institutions, eastern ethiopia. *Pan African Medical Journal*, *20*. https://doi.org/10.11604/pamj.2015.20.247.3345

Aracri, F., Bianco, M. G., Quattrone, A., & Sarica, A. (2025). Bridging the gap: Missing data imputation methods and their effect on dementia classification performance. *Brain Sciences*, *15*(6), 639. https://doi.org/10.3390/brainsci15060639

Austin, P. C., White, I. R., Lee, D. S., & van Buuren, S. (2021). Missing data in clinical research: A tutorial on multiple imputation. *Canadian Journal of Cardiology*, *37*(9), 1322–1331. https://doi.org/10.1016/j.cjca.2020.11.010

Azeez, A., Ndege, J., & Mutambayi, R. (2018). Associated factors with unsuccessful tuberculosis treatment outcomes among tuberculosis/hiv coinfected patients with drug-resistant tuberculosis. https://journals.lww.com/ijmy/fulltext/2018/07040/Associated_Factors_with_Unsuccessful_Tuberculosis.8.aspx

Baron, G. (2025). Many tb patients left untreated. https://tribune.net.ph/2025/03/29/many-tb-patients-left-untreated

Batista, G. E. A. P. A., Prati, R. C., & Monard, M. C. (2004). A study of the behavior of several methods for balancing machine learning training data. *ACM SIGKDD Explorations Newsletter*, *6*(1), 20–29. https://doi.org/10.1145/1007730.1007735

Bendre, A. D., Peters, P. J., & Kumar, J. (2021). Tuberculosis: Past, present and future of the treatment and drug discovery research. *Current Research in Pharmacology and Drug Discovery*, *2*, 100037. https://doi.org/10.1016/j.crphar.2021.100037

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357. https://doi.org/10.1613/jair.953

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Chicco, D., & Jurman, G. (2020). The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. *BMC Genomics*, *21*(1), 6. https://doi.org/10.1186/s12864-019-6413-7

Collins, G. S., Moons, K. G. M., Dhiman, P., Riley, R. D., Beam, A. L., Calster, B. V., Ghassemi, M., Liu, X., Reitsma, J. B., van Smeden, M., Boulesteix, A.-L., Camaradou, J., Celi, L. A., Denaxas, S., Denniston, A. K., Glocker, B., Golub, R. M., Harvey, H., Heinze, G., et al. (2024). TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, *385*, e078378. https://doi.org/10.1136/bmj-2023-078378

Davis, J., & Goadrich, M. (2006). The relationship between precision-recall and roc curves. *Proceedings of the 23rd International Conference on Machine Learning*, 233–240. https://doi.org/10.1145/1143844.1143874

Department of Health [DOH]. (2014). *National tb control program manual of procedures (5th ed.)* Department of Health. https://www.philhealth.gov.ph/partners/providers/pdf/NTCP_MoP2014.pdf

Department of Health - Republic of the Philippines. (2022). Doh-car. doh-cordillera, usaid, recalibrate plan to detect and mitigate tuberculosis hot spots. https://caro.doh.gov.ph/

Dong, Y., & Peng, C.-Y. J. (2013). Principled missing data methods for researchers. *SpringerPlus*, *2*(1), 222. https://doi.org/10.1186/2193-1801-2-222

Emmanuel, T., Maupong, T., Mpoeleng, D., Semong, T., Mphago, B., & Tabona, O. (2021). A survey on missing data in machine learning. *Journal of Big Data*, *8*(1), 140. https://doi.org/10.1186/s40537-021-00516-9

Florentino, J. L., Arao, R. M. L., Garfin, A. M. C., Gaviola, D. M. G., Tan, C. R., Yadav, R. P., Hiatt, T., Morishita, F., Siroka, A., Yamanaka, T., & Nishikiori, N. (2022). Expansion of social protection is necessary towards zero catastrophic costs due to tb: The first national tb patient cost survey in the philippines. *PLOS One*, *17*(2), e0264689. https://doi.org/10.1371/journal.pone.0264689

Groenwold, R. H. H., White, I. R., Donders, A. R. T., Carpenter, J. R., Altman, D. G., & Moons, K. G. M. (2012). Missing covariate data in clinical research: When and when not to use the missing-indicator method for analysis. *Canadian Medical Association Journal*, *184*(11), 1265–1269. https://doi.org/10.1503/cmaj.110977

Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd). Springer. https://doi.org/10.1007/978-0-387-84858-7

Hinay, A. A., Mamalintaw, M. A., Damasin, M. L., Shamera, J., Adrian, B., Joy, C., Evan, L., Insular, L. C., Joy, A., Cadotdot, M. T., Elipio, M. R., Ashley, J., & Dayaganon, J. B. (2025). Sociodemographic and clinical predictors of tuberculosis and unsuccessful treatment outcomes in davao city, philippines: A retrospective cohort study. *International Journal of Environmental Research and Public Health*, *22*(7), 1154–1154. https://doi.org/10.3390/ijerph22071154

Junaid, K. P., Kiran, T., Gupta, M., Kishore, K., & Siwatch, S. (2025). How much missing data is too much to impute for longitudinal health indicators? A preliminary guideline for the choice of the extent of missing proportion to impute with multiple imputation by chained equations. *Population Health Metrics*, *23*(1), 2. https://doi.org/10.1186/s12963-025-00364-2

Kaur, R., & Sharma, A. (2021). An accurate integrated system to detect pulmonary and extra pulmonary tuberculosis using machine learning algorithms. *Inteligencia Artificial*, *24*(68), 104–122. https://doi.org/10.4114/intartif.vol24iss68pp104-122

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, *30*, 3146–3154.

Khalid, T. N. F. T., Mohammad, W. M. Z. W., Samat, R. A., & Husain, N. R. N. (2022). Predictors of tuberculosis disease in smokers. *PeerJ*, *10*, e13984. https://doi.org/10.7717/peerj.13984

Kossakov, M., Kulichenko, D., Butusov, D., & Popov, A. (2024). Quantitative comparison of machine-learning clustering methods for tuberculosis data analysis. *Engineering Proceedings*, *20*(1), 60. https://doi.org/10.3390/engproc2024060020

Laine, S., & Aila, T. (2016). Temporal ensembling for semi-supervised learning. https://arxiv.org/abs/1610.02242

Lamsal, D. K., Lewis, O. D., Smith, S., & Jha, N. (2009). Factors related to defaulters and treatment failure of tuberculosis patients in the DOTS program in the Sunsari district of Eastern Nepal. *SAARC Journal of Tuberculosis, Lung Diseases and HIV/AIDS*, *6*(1), 25–30.

Lu, J. Y., Zhu, J., Zhu, J., & Duong, T. Q. (2022). Long-short-term memory machine learning of longitudinal clinical data accurately predicts acute kidney injury onset in covid-19. *International Journal of Infectious Diseases*, *122*, 802–810. https://doi.org/10.1016/j.ijid.2022.07.034

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, *2*, 56–67. https://doi.org/10.1038/s42256-019-0138-9

Lundberg, S. M., & Lee, S.-I. (2017a). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, *30*, 4765–4774.

Lundberg, S. M., & Lee, S.-I. (2017b). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, *30*, 4765–4774. https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions

Nayebi, A., Tipirneni, S., Foreman, B., Ratcliff, J., Reddy, C. K., & Subbian, V. (2022). Recurrent neural network based time-series modeling for long-term prognosis following acute traumatic brain injury. https://pmc.ncbi.nlm.nih.gov/articles/PMC8861707/

Ndambuki, J., Nzomo, J., Muregi, L., Mutuku, C., Makokha, F., Nthusi, J., Ambale, C., Lynen, L., & Decroo, T. (2020). Comparison of first-line tuberculosis treatment outcomes between previously treated and new patients: A retrospective study in machakos subcountry, kenya. *International Health*. https://academic.oup.com/inthealth/article/13/3/272/5898744?login=false

Nishat, M. M., Faisal, F., Ratul, I. J., Al-Monsur, A., Ar-Rafi, A. M., Nasrullah, S. M., & Khan, M. R. H. (2022). A comprehensive investigation of the performances of different machine learning classifiers with SMOTE-ENN oversampling technique and hyperparameter optimization for imbalanced heart failure dataset. *Scientific Programming*, *2022*, 3649406. https://doi.org/10.1155/2022/3649406

Noubiap, J. J., Nansseu, J. R., Nyaga, U. F., Nkeck, J. R., Endomba, F. T., Kaze, A. D., Agbor, V. N., & Bigna, J. J. (2019). Global prevalence of diabetes in active tuberculosis: A systematic review and meta-analysis of data from 2.3 million patients with tuberculosis. *The Lancet Global Health*, *7*(4), e448–e460. https://doi.org/10.1016/S2214-109X(18)30487-X

Peng, A., Kong, X.-H., Liu, S., Zhang, H., Xie, L., Ma, L., Zhang, Q., & Chen, Y. (2024). Explainable machine learning for early predicting treatment failure risk among patients with tb-diabetes comorbidity. *Scientific Reports*, *14*(1), 7567. https://doi.org/10.1038/s41598-024-57446-8

Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). Ai in health and medicine. *Nature Medicine*, *28*(1), 31–38. https://doi.org/10.1038/s41591-021-01614-0

Salmi, M., Atif, D., Oliva, D., Abraham, A., & Ventura, S. (2024). Handling imbalanced medical datasets: Review of a decade of research. *Artificial Intelligence Review*, *57*(10), 273. https://doi.org/10.1007/s10462-024-10884-2

Sellergren, A., Kazemzadeh, S., Jaroensri, T., Kiraly, A., Traverse, M., Kohlberger, T., Xu, S., Jamil, F., Hughes, C., Lau, C., et al. (2025). Medgemma technical report. https://arxiv.org/abs/2507.05201

Shickel, B., Tighe, P. J., Bihorac, A., & Rashidi, P. (2018). Deep EHR: A survey of recent advances in deep learning techniques for electronic health record (EHR) analysis. *IEEE Journal of Biomedical and Health Informatics*, *22*(5), 1589–1604. https://doi.org/10.1109/JBHI.2017.2767063

Sibanda, S., & Ndlovu, B. (2026). Explainable transformer and machine learning models in predicting tuberculosis treatment outcomes: A systematic review. *Journal of Applied Informatics and Computing*, *10*(1), 150–164. https://doi.org/10.30871/jaic.v10i1.11846

Siddiqui, A. N., Khayyam, K. U., & Sharma, M. (2016). Effect of diabetes mellitus on tuberculosis treatment outcome and adverse reactions in patients receiving directly observed treatment strategy in india. *BioMed Research International*, 1–11. https://doi.org/10.1155/2016/7273935

Sisk, R., Sperrin, M., Peek, N., van Smeden, M., & Martin, G. P. (2023). Imputation and missing indicators for handling missing data in the development and deployment of clinical prediction models: A simulation study. *Statistical Methods in Medical Research*, *32*(8), 1461–1477. https://doi.org/10.1177/09622802231165001

Smith, I. D. M., Jones, G. T., Simpson, C. R., Smith, B. H., & Lee, R. J. (2021). Ensemble machine learning models for predicting chronic disease outcomes in electronic health records. *Medical Decision Making*, *41*(2), 130–141. https://doi.org/10.1177/0272989X20982548

Stop TB Partnership. (2021). Tb and covid: One year on - media brief. https://stoptb.org/webadmin/cms/docs/20210316_TB%20and%20COVID_One%20Year%20on_Media%20Brief_FINAL.pdf

The Borgen Project. (2025). Tb in the philippines: A persistent public health crisis. https://borgenproject.org/tb-in-the-philippines/

Ting, K. M. (2011). Confusion matrix. In C. Sammut & G. I. Webb (Eds.), *Encyclopedia of machine learning* (pp. 209–209). Springer. https://doi.org/10.1007/978-0-387-30164-8_157

Tobin, E. H., & Tristram, D. (2024). Tuberculosis overview. In *Statpearls*. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK441916/

Tok, P. S. K., Liew, S. M., Wong, L. P., Razali, A., Loganathan, T., Chinna, K., Ismail, N., & Kadir, N. A. (2020). Determinants of unsuccessful treatment outcomes and mortality among tuberculosis patients in malaysia. *PLOS One*. https://doi.org/10.1371/journal.pone.0231986

Vasey, B., Nagendran, M., Campbell, B., Clifton, D. A., Collins, G. S., Denaxas, S., Denniston, A. K., Faes, L., Geerts, B., Ibrahim, M., Liu, X., Mateen, B. A., McCradden, M. D., Morgan, L., Ordish, J., Rogers, C., Saria, S., Ting, D. S. W., Watkinson, P., . . . the DECIDE-AI Expert Group. (2022). Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI. *Nature Medicine*, *28*, 924–933. https://doi.org/10.1038/s41591-022-01772-9

von Hippel, P. T. (2018). How many imputations do you need? A two-stage calculation using a quadratic rule. *Sociological Methods & Research*, *49*(3), 699–718. https://doi.org/10.1177/0049124117747303

Wang, Z., Guo, Z., Wang, W., Zhang, Q., Song, S., Xue, Y., Zhang, Z., & Wang, J. (2025). Prediction of tuberculosis treatment outcomes using biochemical makers with machine learning. *BMC Infectious Diseases*, *25*(1), 229. https://doi.org/10.1186/s12879-025-10609-y

World Health Organization. (2021). *Ethics and governance of artificial intelligence for health: WHO guidance*. https://www.who.int/publications/i/item/9789240029200

World Health Organization. (2022). *Global tuberculosis report 2021*. https://www.who.int/publications/i/item/9789240037021

World Health Organization. (2023). *Global tuberculosis report 2023* (tech. rep.). World Health Organization. Geneva. https://www.who.int/publications/i/item/9789240083851

Xu, G., Hu, X., Lian, Y., & Li, X. (2023). Diabetes mellitus affects the treatment outcomes of drug-resistant tuberculosis. https://doi.org/10.1186/s12879-023-08765-0

Zeng, X., & Zhu, K. (2024). Enhancing the interpretability of shap values using large language models. https://arxiv.org/abs/2409.00079

Appendix A: CDSS Web Application Walkthrough

This appendix presents a sequential walkthrough of the deployed CDSS web application, following the order in which a clinical user would traverse the system: authentication, patient intake, profile review, longitudinal monitoring, and risk explanation. Several interfaces shown here also appear in the body of the paper; they are reproduced in this appendix for continuity of the walkthrough.

A.1 Authentication and Dashboard

Figure 26

CDSS login screen.

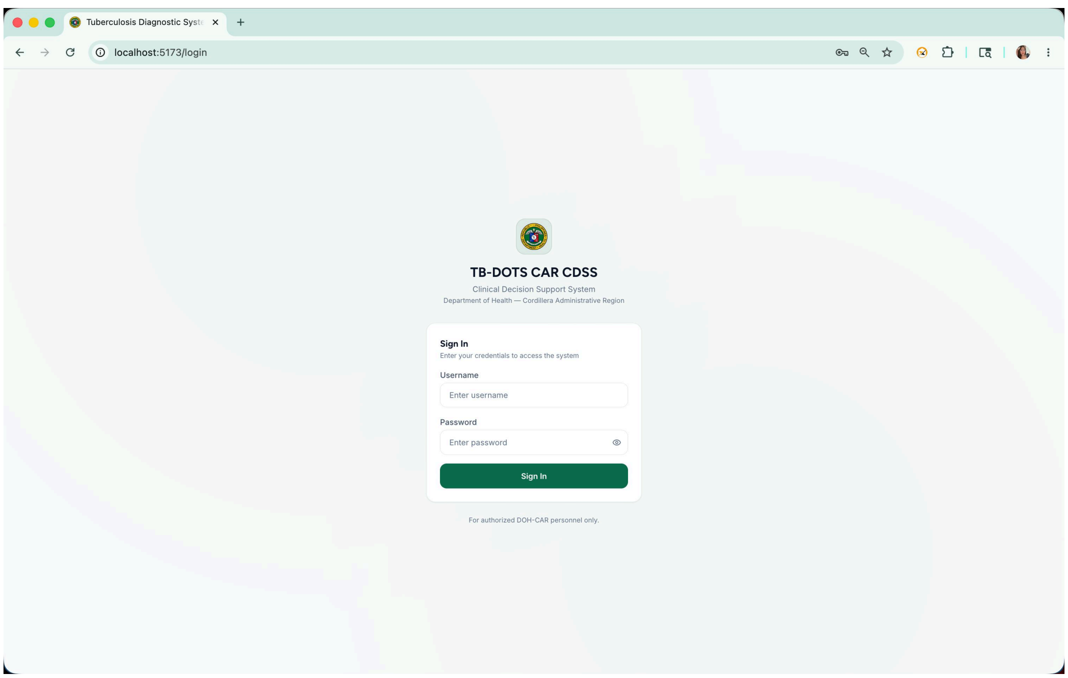
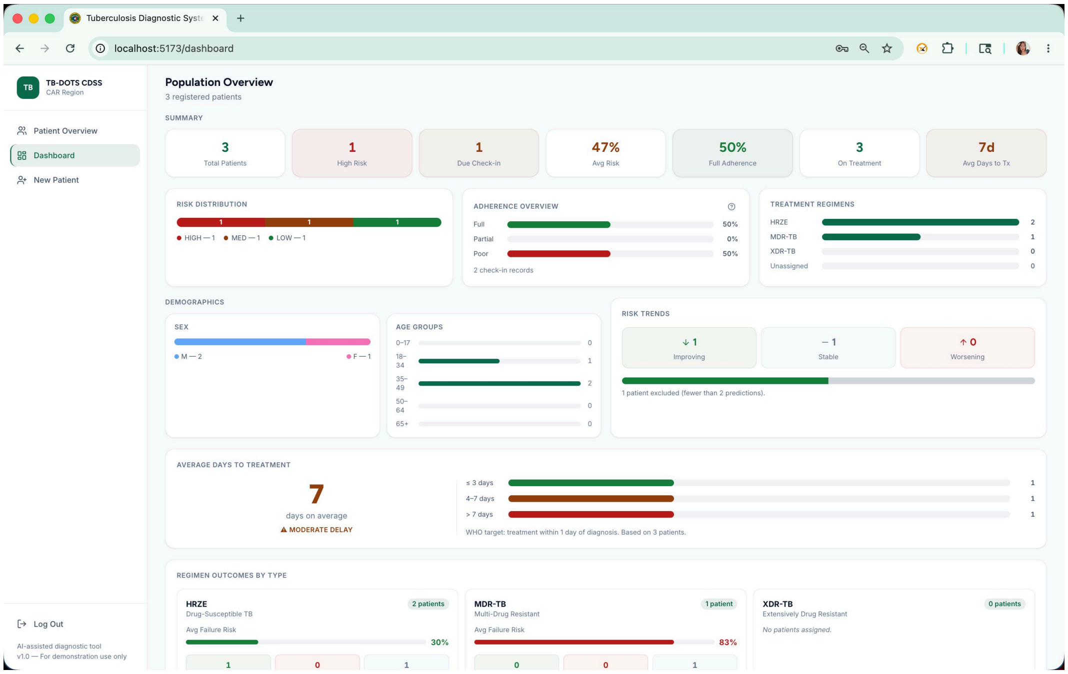


Figure 27

Main dashboard listing enrolled patients.



A.2 Patient Intake Workflow

The intake workflow consists of five sequential steps. The Lab Results step (Step 2) is omitted here as it was not captured in the available screenshot set.

Figure 28

Patient intake Step 1: Demographics.

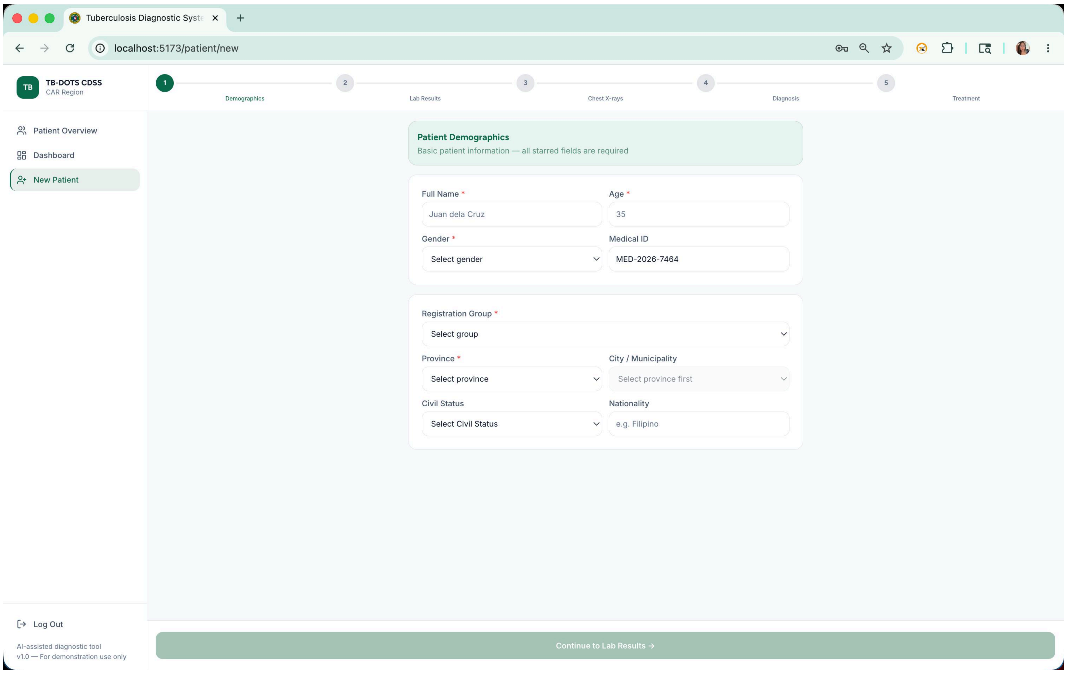


Figure 29

Patient intake Step 3: Chest X-ray upload.

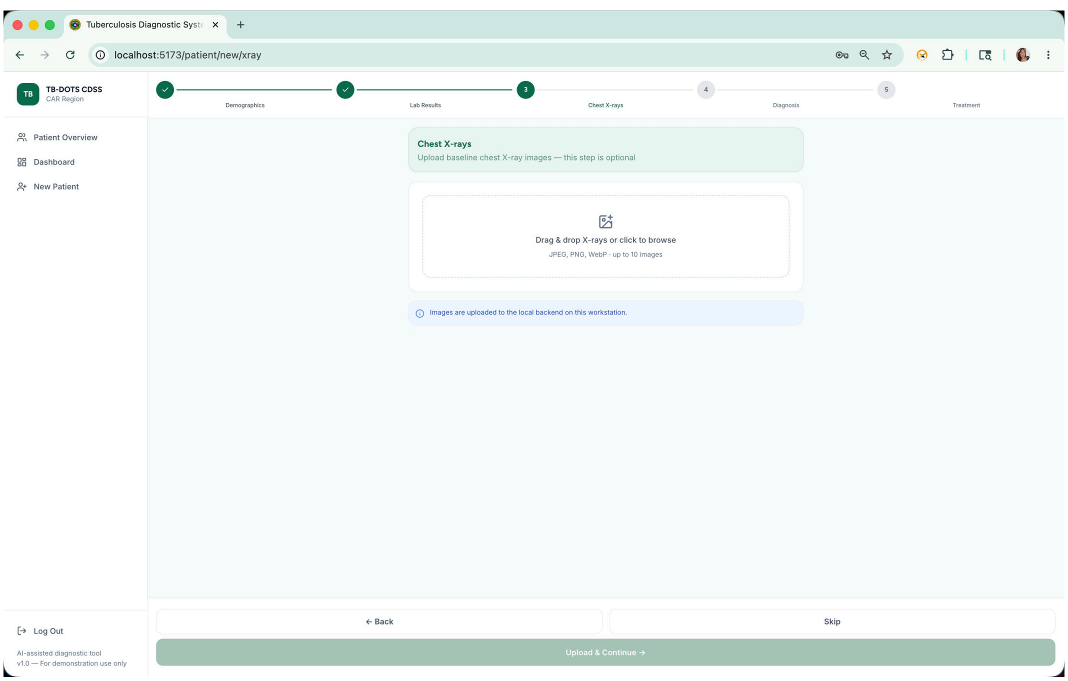


Figure 30

Patient intake Step 4: Diagnostic result.

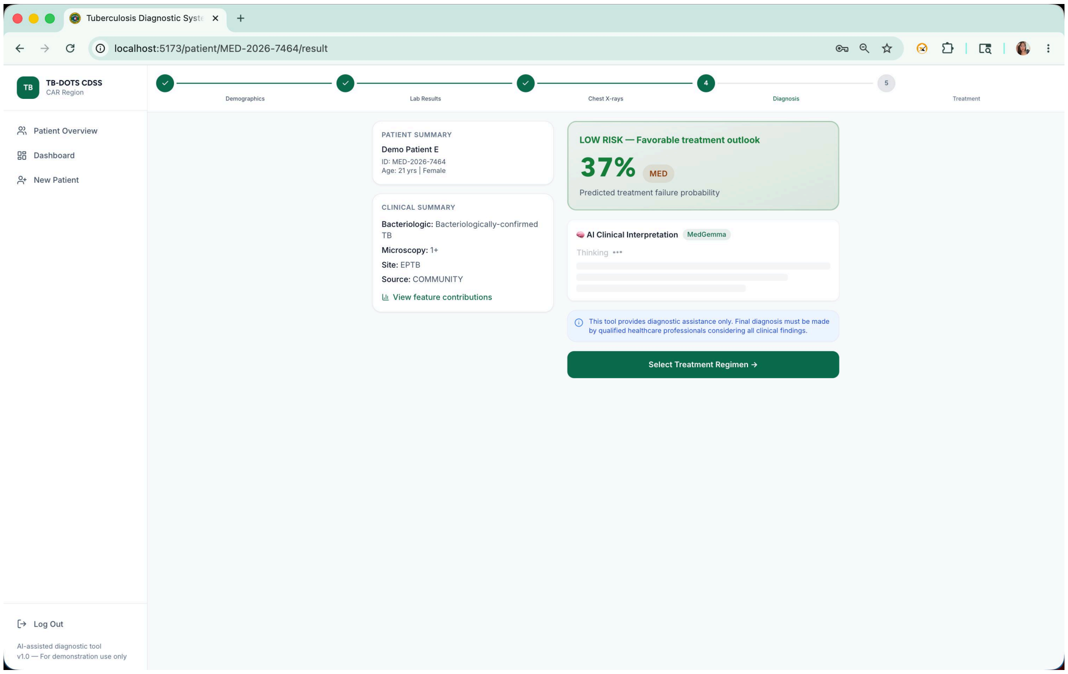
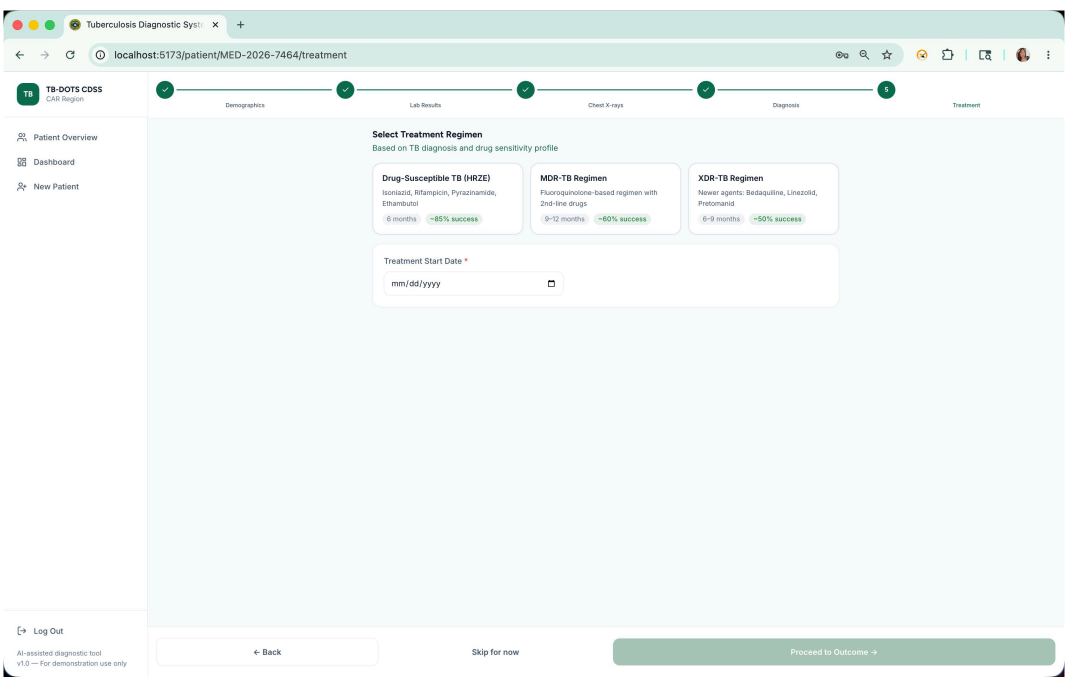


Figure 31

Patient intake Step 5: Treatment regimen selection.



A.3 Patient Profile and Records

Figure 32

Patient profile overview tab.

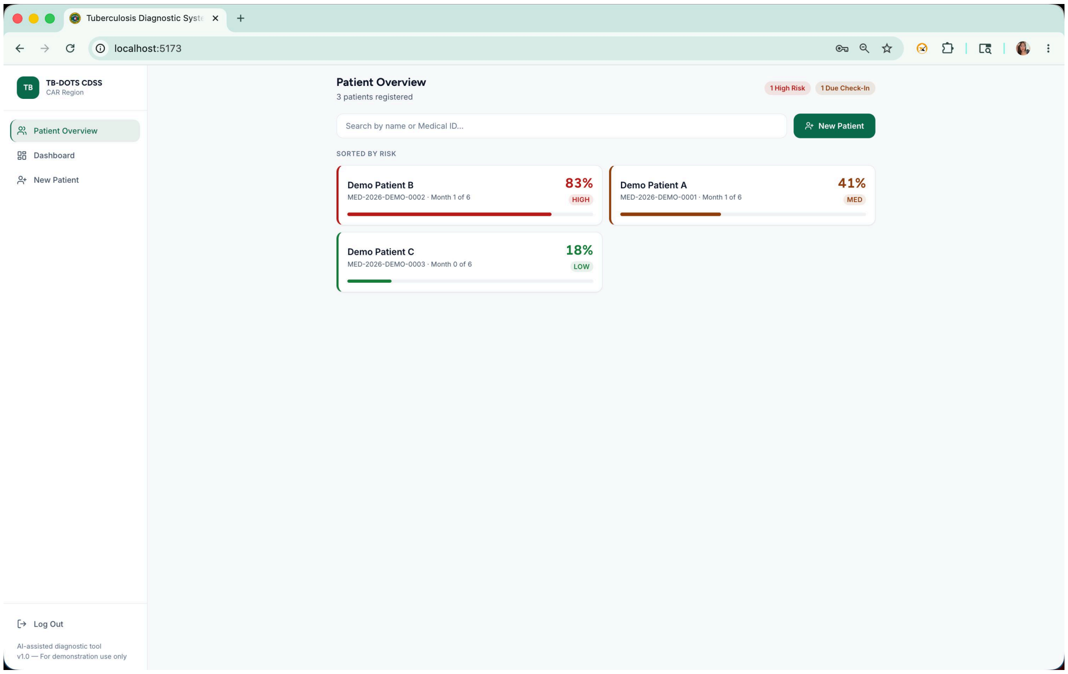
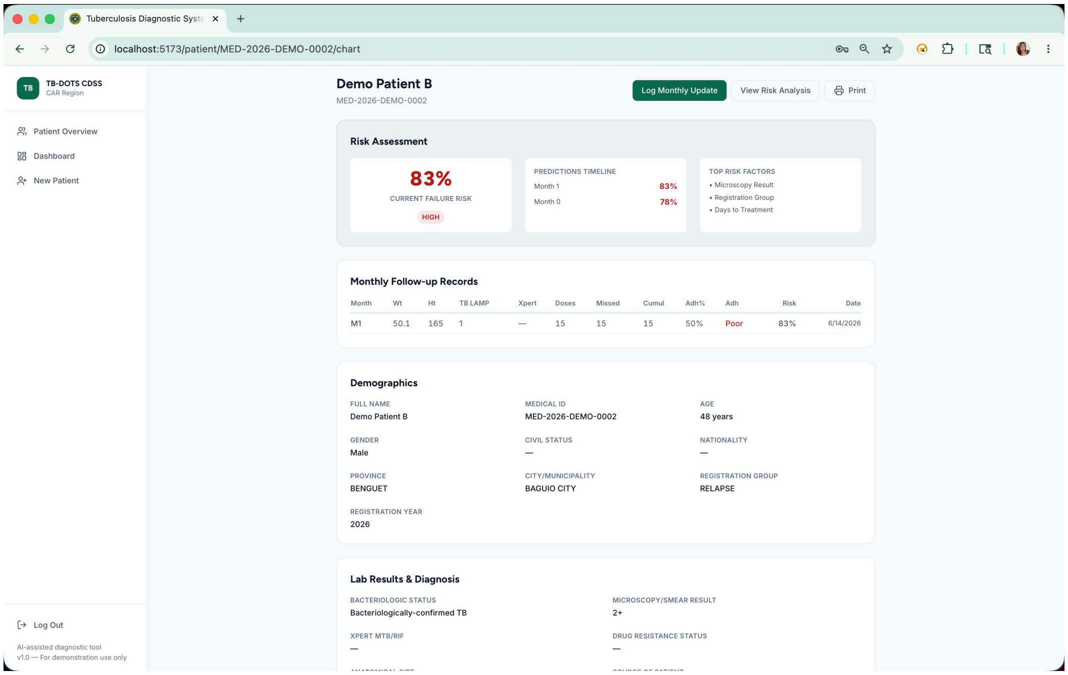


Figure 33

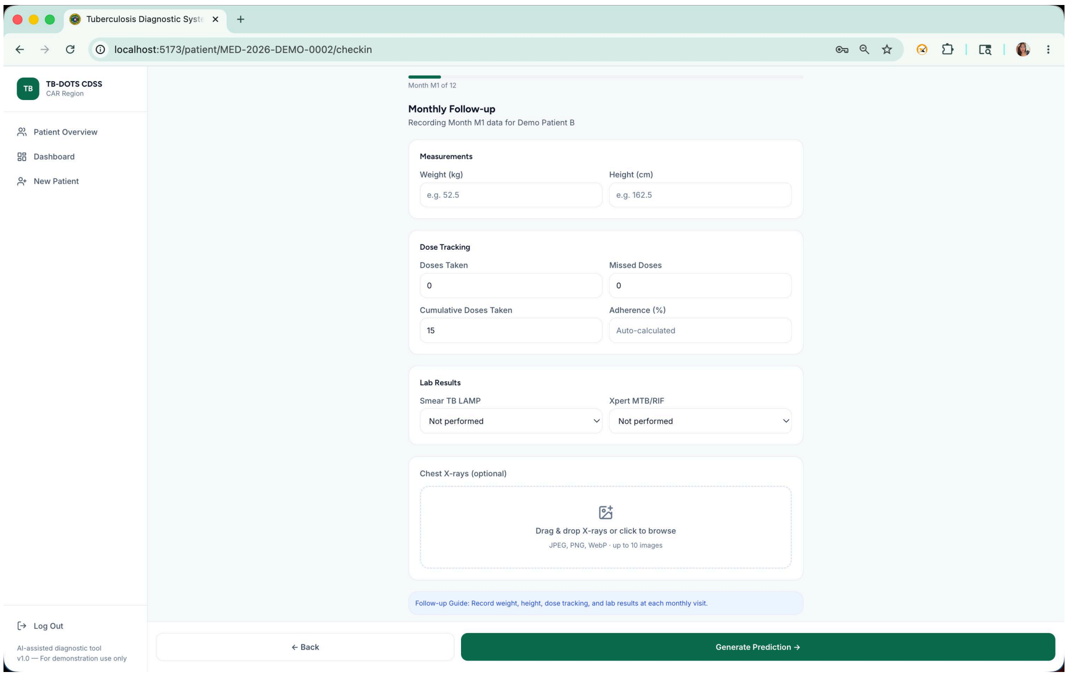
Complete patient chart with longitudinal record.



A.4 Longitudinal Monitoring

Figure 34

Monthly check-in log for adherence and follow-up imaging.



A.5 Risk Prediction and Explainability

Figure 35

Updated risk analysis after monthly follow-up.

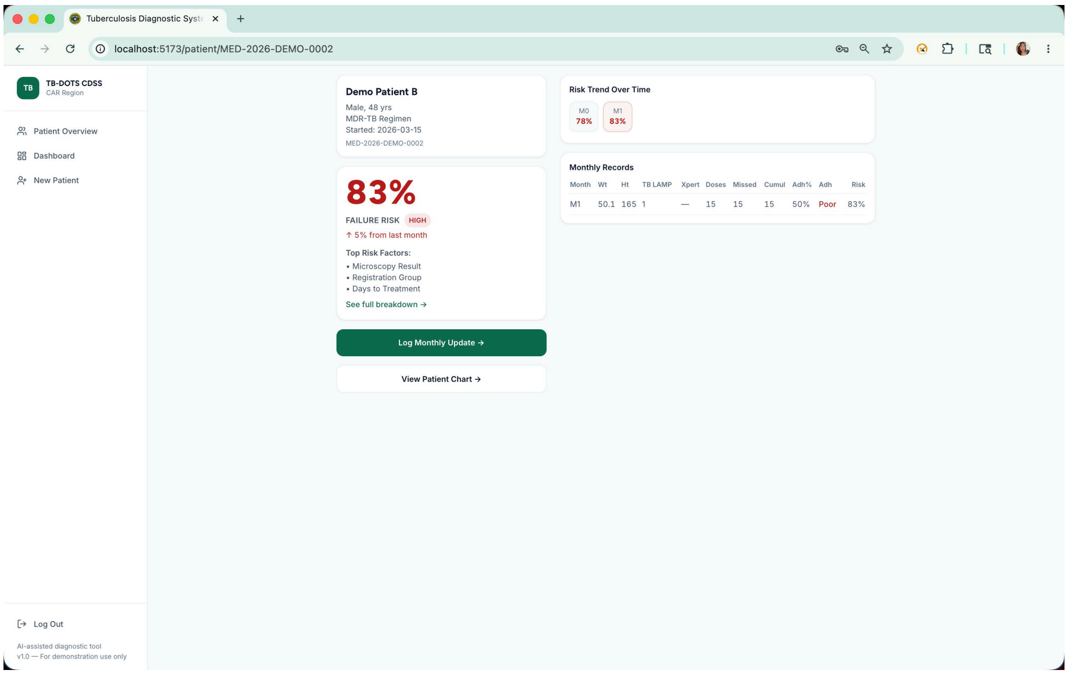


Figure 36

SHAP-based full breakdown of contributing factors.

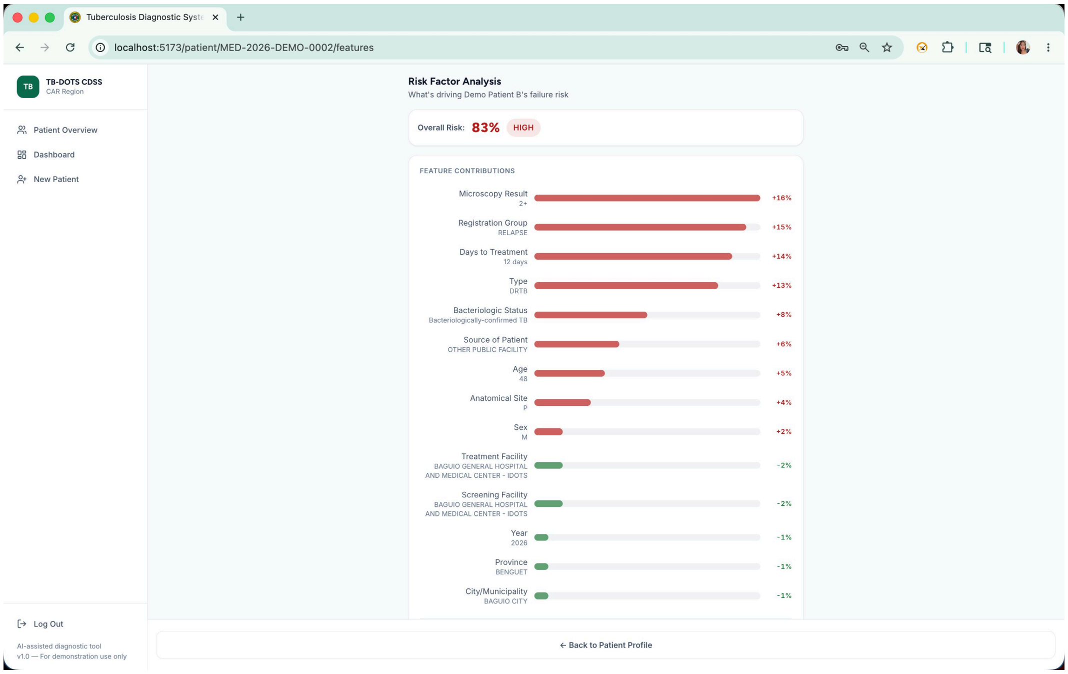
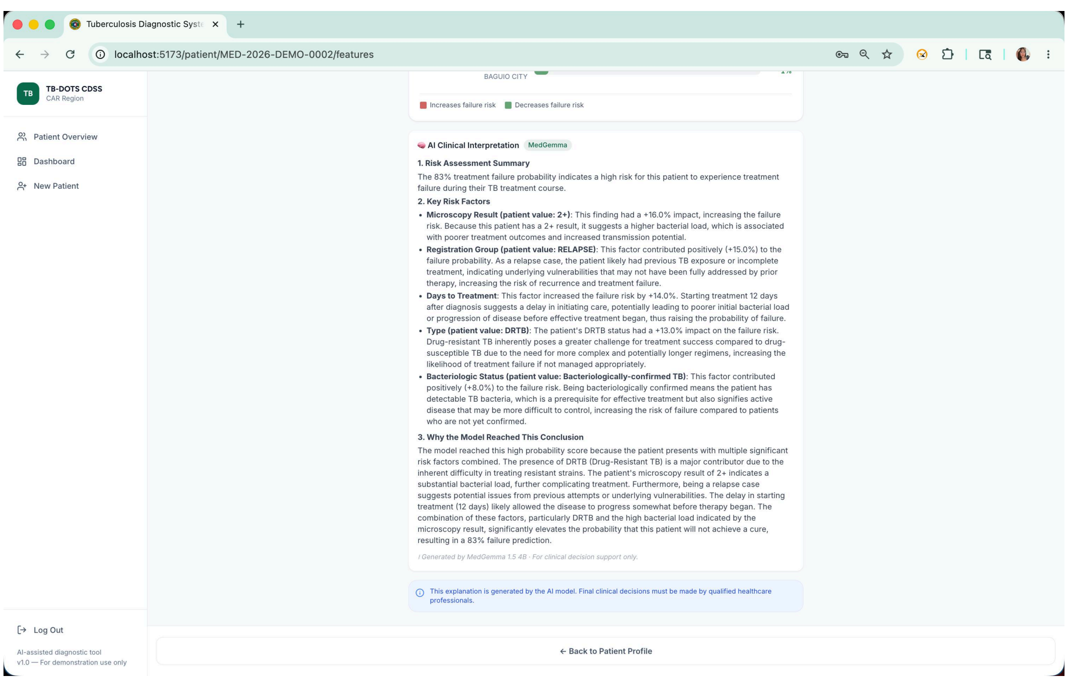


Figure 37

Med-Gemma clinical narrative explaining risk factors.



Appendix B: iCreate Supplementary Material

The following pages reproduce the original iCreate 2026 supplementary material in its source format for archival completeness.

Treatment Success and Failure Prediction using Machine Learning Models for Pulmonary Tuberculosis Patients under the DOH-CAR TB-DOTS Program

Author Names
Aaron Kyle M. Aguilar[1], Mark Lestat C. Agustin[2], Keanu Sonn M, Fortaleza[3], Benny Gil A. Lactaotao[4],
Marven Joffre G. Luis[5], Georcelle Leisley U. Nuarin[6], Sean Drei A. Octavo[7], Gebreyl Isaac P. Rabang[8]

Affiliations
[1 2 3 4 5 6 7 8] Department of Computing and Information Studies, Saint Louis University, Baguio City, Philippines

Corresponding Author:
Josephine S. Dela Cruz, DIT – jdelacruz@slu.edu.ph

Abstract
This study proposes a multi-layer analytical framework using machine learning to predict pulmonary tuberculosis (PTB) treatment success or failure in the Cordillera Administrative Region (CAR), Philippines. Utilizing demographic, clinical, and diagnostic data from 2015 to 2025, supervised models (Logistic Regression, Support Vector Machines, and Extreme Gradient Boosting) and temporal models (Recurrent Neural Networks and Long Short-Term Memory) are trained to identify risk factors. Shapley Additive Explanations (SHAP) ensure interpretability. The models will be integrated into a clinical decision support tool for early intervention. Aligning with the Sustainable Development Goals for good health and well-being, this predictive tool aims to enhance TB control efforts, optimize resource allocation, and improve patient management.

Keywords
Clinical decision support, machine learning, predictive modeling, pulmonary tuberculosis, treatment outcomes

1. Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). In the Philippines, TB remains a major public health concern, with the country being one of the top contributors to the global TB burden (World Health Organization, 2022). In the Cordillera Administrative Region (CAR), the Department of Health (DOH) reported 437 confirmed TB cases in Baguio City alone in 2022, alongside an estimated 4,405 missing or unreported cases (Department of Health Republic of the Philippines, n.d.).

While multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, findings remain fragmented as traditional statistical analyses often assume linear relationships and examine only a few factors simultaneously (Amante &

Abdosh, 2015). To address this gap, this study adopts an integrative machine learning framework to combine multiple dimensions of data in analyzing interrelated variables.

The specific research objectives of this study are as follows:

1. **Data Pre-processing & Feature Engineering**: To clean, pre-process, and digitalize PTB patient records, creating engineered features that ensure the dataset is optimized for statistical analysis and machine learning applications.
2. **Statistical Analysis & Profiling**: To perform descriptive and inferential statistics to profile the demographic, clinical, and treatment characteristics of the study population, identifying baseline variables that significantly influence treatment outcomes.
3. **AI Modeling & Comparative Analysis**: To develop, validate, and evaluate both non-temporal and temporal machine learning models for predicting PTB treatment outcomes, comparing the predictive performance of static baseline data against dynamic, time-dependent patient patterns throughout the course of therapy.
4. **Data Visualization & Clinical Interpretation**: To visualize patient risk levels, statistical trends, and predictive outputs from the AI models, providing clear insights to support clinical interpretation and the early identification of individuals at risk of poor outcomes.

2. Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering the period from 2015 to 2025, consolidated by the DOH-CAR. Data undergoes a standardized preprocessing pipeline, including schema harmonization, error correction, and missing value imputation via Multivariate Imputation by Chained Equations, or MICE (Alruhaymi & Kim, 2021).

To predict treatment outcomes defined by the National Tuberculosis Program criteria, supervised models including Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) are utilized. Furthermore, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are employed to capture longitudinal treatment dynamics (Lu et al., 2022; Nayebi et al., 2022). Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are applied to ensure clinical interpretability (Peng et al., 2024).

3. Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

3.1 Dataset Characteristics and Class Distribution

The final dataset exhibited significant class imbalance, with 86.2% of cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, including four newly introduced geographic and facility-level variables and one derived clinical delay variable (Diagnosis_to_Treatment_days). Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio. This approach mitigated class bias while preserving realistic evaluation conditions in the test and validation sets.

3.2 Validation Set Performance and Generalization

Table 1: Test Set Model Performance

Model	**AUC (Area Under the Curve)**	**Recall (Failure)**	**F1 (Failure)**
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy (~51%). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

3.3 Validation Set Performance and Generalization

Table 2: Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 1 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

4. Conclusions and Recommendations

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes. By accurately identifying patients at risk of treatment failure, these predictive models have the potential to reduce TB-related morbidity and mortality, inform targeted public health interventions, and contribute to more efficient and effective TB care within the Cordillera Administrative Region.

Acknowledgments

The authors would like to express their sincere gratitude to Saint Louis University for providing the academic environment and resources that made this research possible. We extend our appreciation to the School of Accountancy, Management, Computing, and Information Studies (SAMCIS), specifically the Department of Computing and Information Studies, for their continuous support and guidance throughout the study. Our deepest thanks go to Dr. Randy Domantay, whose leadership and encouragement have been invaluable. We are especially grateful to our research adviser, Dr. Josephine S. Dela Cruz, for their expert guidance, insightful feedback, and unwavering support at every stage of the research. Lastly, we acknowledge the collaboration and contributions of our partner researchers from the SONAHBS Department of Pharmacy, whose expertise and assistance significantly enriched this study.

References

Alruhaymi, A., & Kim, C. (2021). Why can multiple imputations and how (MICE) algorithm work? Open Journal of Statistics, 11(5), 759–777. https://doi.org/10.4236/ojs.2021.115045

Amante, T. D., & Abdosh, T. (2015). Risk factors for unsuccessful tuberculosis treatment outcome (failure, default and death) in public health institutions, Eastern Ethiopia. Pan African Medical Journal, 20. https://doi.org/10.11604/pamj.2015.20.247.3345

Department of Health Republic of the Philippines. (n.d.). DOH-CAR. DOH-Cordillera, USAID, recalibrate plan to detect and mitigate tuberculosis hot spots. https://caro.doh.gov.ph/%F0%9D%90%83%F0%9D%90%8E%F0%9D%90%87-%F0%9D%90%82%F0%9D%90%A8%F0%9D%90%AB%F0%9D%90%9D%F0%9D%90%A2%F0%9D%90%A5%F0%9D%90%A5%F0%9D%90%9E%F0%9D%90%AB%F0%9D%90%9A-%F0%9D%90%94%F0%9D%90%92%F0%9D%90%80/

Lu, J. Y., Zhu, J., Zhu, J., & Duong, T. Q. (2022). Long-short-term memory machine learning of longitudinal clinical data accurately predicts acute kidney injury onset in COVID-19: a two-center study. International Journal of Infectious Diseases, 122, 802–810. https://doi.org/10.1016/j.ijid.2022.07.034

Nayebi, A., Tipirneni, S., Foreman, B., Ratcliff, J., Reddy, C. K., & Subbian, V. (2022). Recurrent Neural Network based Time-Series Modeling for Long-term Prognosis Following Acute Traumatic Brain Injury. https://pmc.ncbi.nlm.nih.gov/articles/PMC8861707/

Peng, A., Kong, X.-H., Liu, S., Zhang, H., Xie, L., Ma, L., Zhang, Q., & Chen, Y. (2024). Explainable machine learning for early predicting treatment failure risk among patients with TB-diabetes comorbidity. Scientific Reports, 14(1), 7567. https://doi.org/10.1038/s41598-024-57446-8

Tobin, E. H., & Tristram, D. (2024). Tuberculosis overview. In StatPearls. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK441916/

World Health Organization. (2022). Global tuberculosis report 2021: supplementary material. World Health Organization. https://www.who.int/publications/i/item/9789240037021

Appendix C: Publication Paper for MSJI

The following pages reproduce the publication paper submitted to the Mapua Student Journal of Information (MSJI) in its source format for archival completeness.

Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the DOH-CAR TB-DOTS Program

Aaron Kyle M. Aguilar, Mark Lestat C. Agustin, Keanu Sonn M. Fortaleza, Benny Gil A. Lactaotao, Marven Joffre G. Luis, Georcelle Leisley U. Nuarin, Sean Drei A. Octavo, and Gebreyl Isaac P. Rabang

Department of Computing and Information Studies

Saint Louis University, Baguio City, Philippines

Abstract

Pulmonary Tuberculosis (PTB) remains a critical public health challenge in the Cordillera Administrative Region (CAR), Philippines, where complex patient profiles and programmatic gaps contribute to persistent treatment failure. This study develops machine learning models to predict treatment success or failure among PTB patients in DOH-CAR TB-DOTS facilities covering 2015–2025. A standardized preprocessing pipeline performs schema harmonization, pseudonymization, and diagnostic-first missing data handling classifying each variable's missingness mechanism. Supervised ensemble tree-based classifiers—XGBoost, LightGBM, and Random Forest—were trained alongside Recurrent Neural Network and Long Short-Term Memory temporal architectures to capture longitudinal treatment dynamics. Shapley Additive Explanations (SHAP) were applied to quantify the contribution of each variable to model predictions. The validated models are embedded in a Clinical Decision Support System (CDSS) providing dynamic risk estimates and narrative explanations to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by Mycobacterium tuberculosis, a slow-growing and acid-fast bacillus that continues to challenge public health. Despite effective therapies, PTB remains widespread due to its strong association with poverty, undernutrition, and comorbidities such as diabetes and HIV, which weaken immune defenses and increase susceptibility. These factors delay diagnosis, complicate treatment, and contribute to disease progression. Globally, TB remains one of the leading causes of death, with 1.5 million deaths and 10 million new cases recorded in 2018, including half a million with rifampicin resistance.

In the Philippines, TB continues to be a major public health concern. The World Health Organization classifies the country as one of the top contributors to the global TB burden, accounting for approximately 6.8 percent of cases. In 2023, around 37,000 Filipinos died from TB out of 739,000 diagnosed cases. In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with over 4,405 missing or unreported cases, making the area a priority for surveillance and treatment improvement.

Contributing factors include high population density, internal migration, the city's cool and humid highland climate conducive to transmission, and its role as a tertiary referral center. Patient comorbidities such as diabetes mellitus, HIV infection, and malnutrition impair immune response and reduce treatment completion rates, while poverty-related barriers disrupt adherence to directly observed treatment short-course (DOTS).

Risk Factors for Treatment Failure

TB patients who acquired infection through close contact were shown to be at higher risk of delayed diagnosis and non-adherence. Patients who missed follow-ups had higher risk of reactivation of infection and mortality. HIV co-infection was determined to be a substantial risk factor for treatment failure. A study in the Philippines revealed that 58.82% of 51 patients who experienced treatment failure had at least one comorbidity. Diabetes Mellitus (DM) further increases the risk of treatment failure, relapse, and death compared to non-diabetic patients.

Machine Learning for TB Outcome Prediction

Machine learning offers distinct advantages in modeling complex and non-linear interactions among demographic, clinical, diagnostic, and treatment-related variables. By leveraging algorithms such as logistic regression, Support Vector Machines, and ensemble-based methods including Extreme Gradient Boosting (XGBoost), machine learning can uncover subtle dependencies that contribute to treatment success or failure. Shapley Additive Explanations (SHAP) quantify the contribution of each variable to a model prediction, summarizing both patient-level explanations and overall feature importance.

The limitations of fragmented risk factor studies and capabilities of interpretable machine learning highlight the need for an integrative approach combining multiple dimensions of data. This study adopts a holistic analytical framework integrating descriptive analysis, supervised predictive modeling, and explanation methods to help healthcare providers identify patients at higher risk of treatment failure, understand factors influencing model's risk estimate, and use that information to support early intervention, tailored monitoring, and improved patient management.

Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering 2015 to 2025, consolidated by DOH-CAR. After ethical approval, records are pseudonymized and processed using a standardized preprocessing pipeline, including schema harmonization, error correction, and feature scaling, followed by a diagnostic-first missing data handling protocol.

Data Acquisition and Preprocessing

The study draws on two complementary datasets from the CAR: (1) a non-temporal consolidated registry maintained by DOH-CAR comprising 7,389 patient records with cross-sectional baseline features including demographic variables, clinical indicators, and diagnostic results, used to train static tree-based classifiers; and (2) a temporal monitoring dataset of 599 patients with complete month-by-month treatment monitoring records, manually compiled by the study's Pharmacy Co-authors from facility treatment monitoring forms and used to train longitudinal deep learning and temporal tree-based models.

The non-temporal preprocessing pipeline implements schema harmonization to standardize variable formats and units and pseudonymization to enforce patient privacy. Outcome annotation defines the binary target variable: patients classified as 'Cured' or 'Treatment Completed' are annotated as 'Success,' whereas those marked as 'Failed,' 'Died,' or 'Lost to Follow-up' are annotated as 'Failure.' To prevent data leakage, the dataset is partitioned into training and testing sets prior to any data transformation steps.

Missing Data Handling

Clinical TB registries exhibit heterogeneous missingness arising from inconsistent documentation and varying facility capacities. This study implements a diagnostic-first missing data pipeline that classifies each variable's missingness mechanism—Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR)—and routes the variable to one of four handling strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multivariate Imputation by Chained Equations (MICE). Variables exceeding 80% missingness are excluded entirely. Four clinical vital signs confirmed MAR are imputed using stochastic MICE with an ExtraTreesRegressor as the conditional model.

Supervised Learning Model Development

The study leverages ensemble tree-based algorithms—XGBoost, LightGBM, and Random Forest. Two sampling strategies are compared for each classifier: no resampling with native class-weighting, and SMOTE combined with Edited Nearest Neighbors (SMOTE-ENN), which increases the training-set failure class by 433%. Three progressive feature set versions are evaluated, ranging from 11 core demographic and diagnostic variables to an extended set of 21 features incorporating facility-level geographic and temporal delay variables.

For temporal modeling, a Hybrid Bidirectional LSTM architecture was adopted to model long-term dependencies, with tree-based temporal baselines including XGBoost and LightGBM, all trained on monthly patient monitoring data from Month 0 through Month 7. The dataset is partitioned using stratified 70/20/10 patient-level split into training, test, and validation sets. Model selection employs a composite scoring metric: score = 0.6 × AUC + 0.4 × Recall(Failure), prioritizing discrimination ability while rewarding sensitivity to the minority failure class.

Explainability and Clinical Decision Support

Shapley Additive Explanations (SHAP) are applied to quantify the marginal contribution of each variable to individual model predictions, surfacing the patient-level factors that drive each risk estimate for clinician review. The validated models are embedded within a Clinical Decision Support System (CDSS) that visualizes patient risk levels, highlights contributing factors through SHAP-based explanations, and dynamically updates risk estimates as new follow-up data become available.

Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static tree-based models under two sampling conditions and temporal models are compared, and feature importance analyses provide interpretable evidence about factors influencing model predictions.

Dataset Characteristics and Class Distribution

The final validated dataset comprised 7,389 complete patient records encompassing demographic, geographic, and clinical features. Overall, the dataset exhibited significant class imbalance, with 86.2% of evaluated cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, integrating spatial identifiers with temporal markers and derived clinical delays. Resampling using SMOTE combined with Edited Nearest Neighbors was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio.

Outcome Distribution and Longitudinal Trends

The distribution of treatment outcomes highlighted both programmatic successes and persistent challenges. The vast majority of the cohort achieved positive outcomes, with 65.1% classified as 'Treatment Completed' and 14.2% as 'Cured.' Conversely, adverse outcomes accounted for a critical minority: patients 'Lost to Follow-up' comprised 8.4%, patients who 'Died' accounted for 3.3%, and treatment failure was 0.9%. Analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts: success rates remained in the 80%–89% range, with 88.92% achieved in 2023. Mortality peaked at 6.63% in 2021, aligning with COVID-19 pandemic disruptions, before declining to 3.66% by 2024. Loss to follow-up rates declined from 11.99% (2020) to 5.56% (2024), suggesting improved patient tracking and adherence interventions.

Figure 1
Distribution of Treatment Outcomes Among PTB Patients

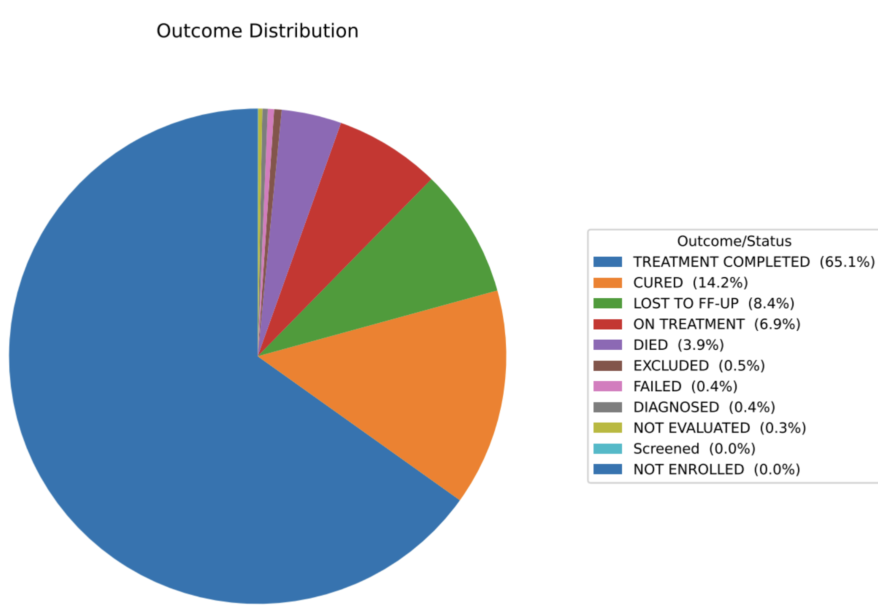


Figure 1 displays the distribution of programmatic treatment outcomes for the entire cohort. The dominance of 'Treatment Completed' and 'Cured' classifications demonstrates the overall effectiveness of the DOH-CAR TB-DOTS program. However, the 8.4% loss to follow-up rate and 3.3% mortality rate underscore the importance of early risk identification and targeted interventions.

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. Treatment Success Rates: The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to a peak of 88.92% in 2023.

2. Mortality and Pandemic Impact: The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.

3. Loss to Follow-up: The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.

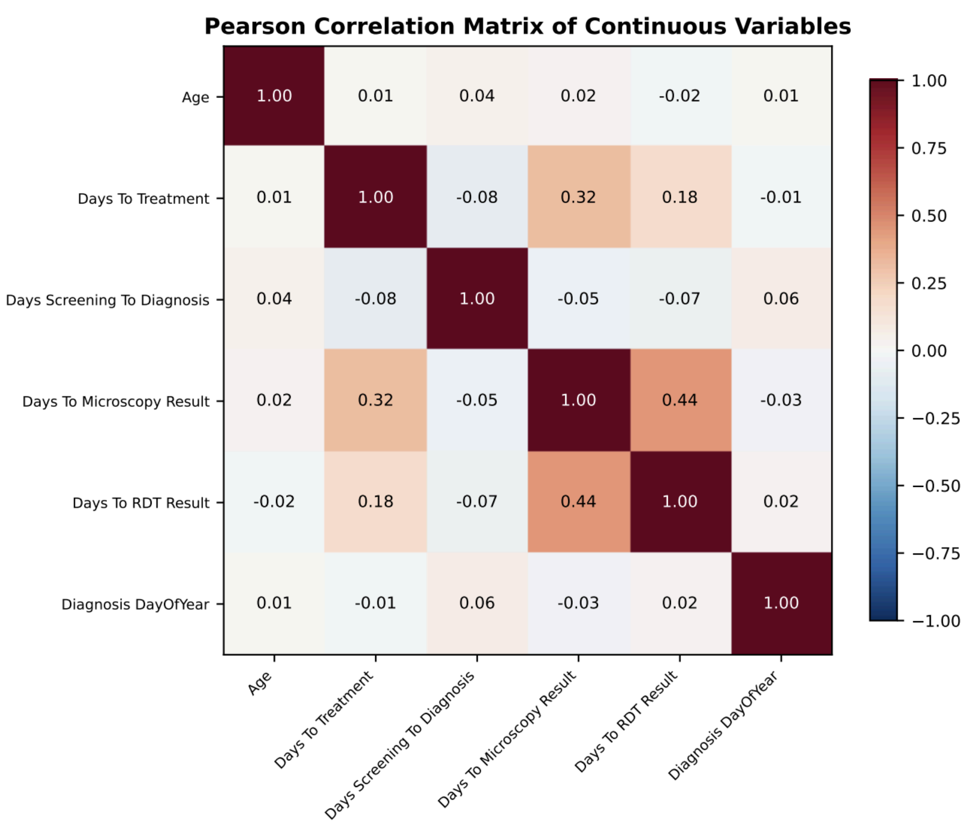
4. 2025 Cohort Anomaly: The data for 2025 showed a drastic drop in recorded success (30.01%) and 52 a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for disease transmission and treatment failure. Median times revealed: (1) screening to diagnosis: 2.0 days (Q3 = 21.09 days); (2) diagnosis to treatment: 2.68 days (Q3 = 13.12 days); and (3) microscopy results: 6.60 days (Q3 = 14.95 days). While median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing algorithms to correlate logistical delays with eventual probability of mortality or loss to follow-up.

Correlation Analysis

Figure 2
Pearson Correlation Matrix of Continuous Variables



The Pearson correlation analysis revealed weak-to-moderate associations among clinical delay variables. Age exhibited weak negative correlations with time intervals ($|r| < 0.10$), indicating older patients experienced marginally shorter delays. Screening-to-diagnosis and diagnosis-to-treatment intervals showed moderate positive correlation ($r = 0.32$), suggesting that delays at one care cascade stage are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming seasonal timing is independent of demographic and operational factors.

Static Model Performance

Among all evaluated static models, Random Forest achieved the highest overall performance on the test set.

Table 1
Test Set Model Performance Metrics of Static Classifiers

Model	AUC	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Note. AUC = Area Under the Receiver Operating Characteristic Curve.

Random Forest achieved an AUC of 0.7154 and an F1-score of 0.3539 for the failure class, with balanced recall of 57.98% without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (approximately 0.71), though with slightly lower recall for treatment failure. On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure, confirming robustness and suitability for deployment within a clinical decision support framework.

Temporal Model Performance

Table 2 presents the performance metrics of temporal models evaluated on the held-out test set.

Table 2
Test Set Performance Metrics of Temporal Models

Metric	Bi-LSTMBaseline	Bi-LSTMAugmented	XGBoostBaseline	XGBoostSMOTE-ENN	LightGBMSMOTE-ENN	RFSMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419

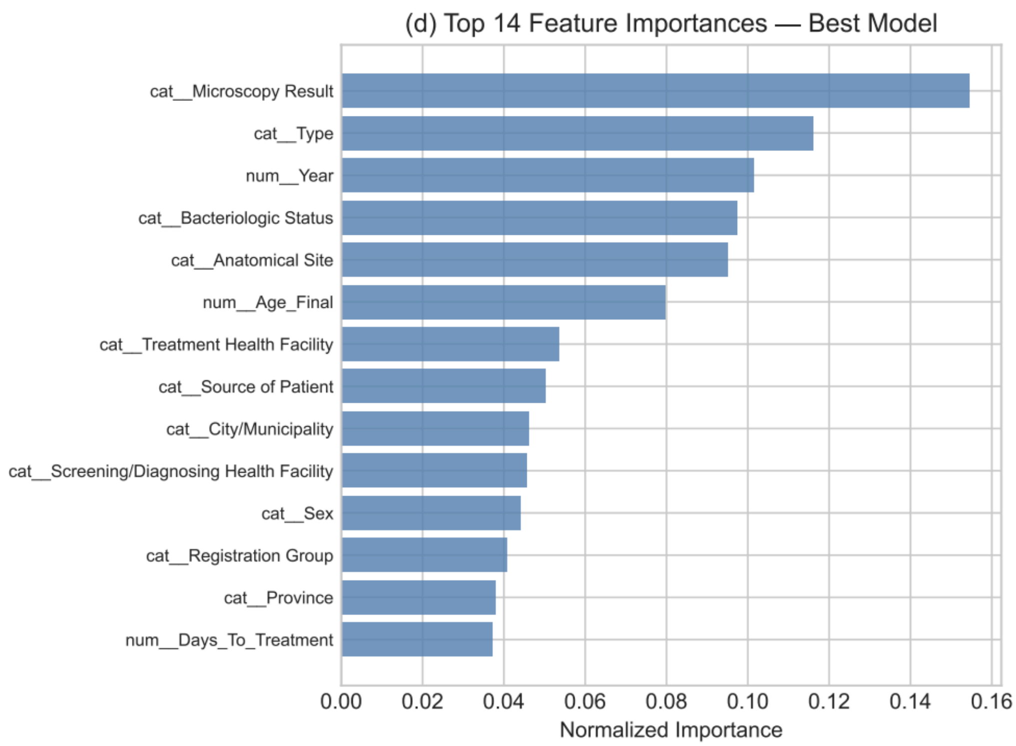
Note. Bi-LSTM = Bidirectional Long Short-Term Memory; RF = Random Forest; ROC-AUC = Area Under Receiver Operating Characteristic Curve; PR-AUC = Area Under Precision-Recall Curve.

Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. However, specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Random Forest with SMOTE-ENN showed more balanced performance, achieving slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. These results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

Feature Importance Analysis and SHAP Explanations

To understand which patient-level factors most strongly influence treatment outcomes, we analyzed feature importance across the ensemble models using both model-native importance metrics and post-hoc SHAP values. Figure 3 presents the top 15 features ranked by mean absolute SHAP values for the Random Forest model, which integrates local explanations from all patients in the validation set.

Figure 3
Top 15 Features Ranked by Mean Absolute SHAP Values (Random Forest Model)



The SHAP analysis reveals that diagnostic and treatment intervals—particularly days from diagnosis to treatment—emerge as the strongest global predictor of treatment outcomes. Monthly adherence metrics (doses taken, percentage adherence) in mid-course months (M2–M4) are also highly influential, indicating that early adherence patterns are predictive of final treatment success or failure. Clinical and demographic factors such as bacteriologic status, anatomical site, age, and comorbidities contribute meaningfully but with lower average impact than operational delays and adherence metrics.

Patient-Level SHAP Explanations and Risk Stratification

Beyond global feature importance, SHAP provides individual patient-level explanations that clinicians can review to understand why the model assigned a particular risk score. Figure 5 illustrates a hypothetical example of how SHAP values are visualized for a single high-risk patient, showing which factors pushed the model's prediction toward treatment failure and by how much.

Clinical Decision Support System Implementation

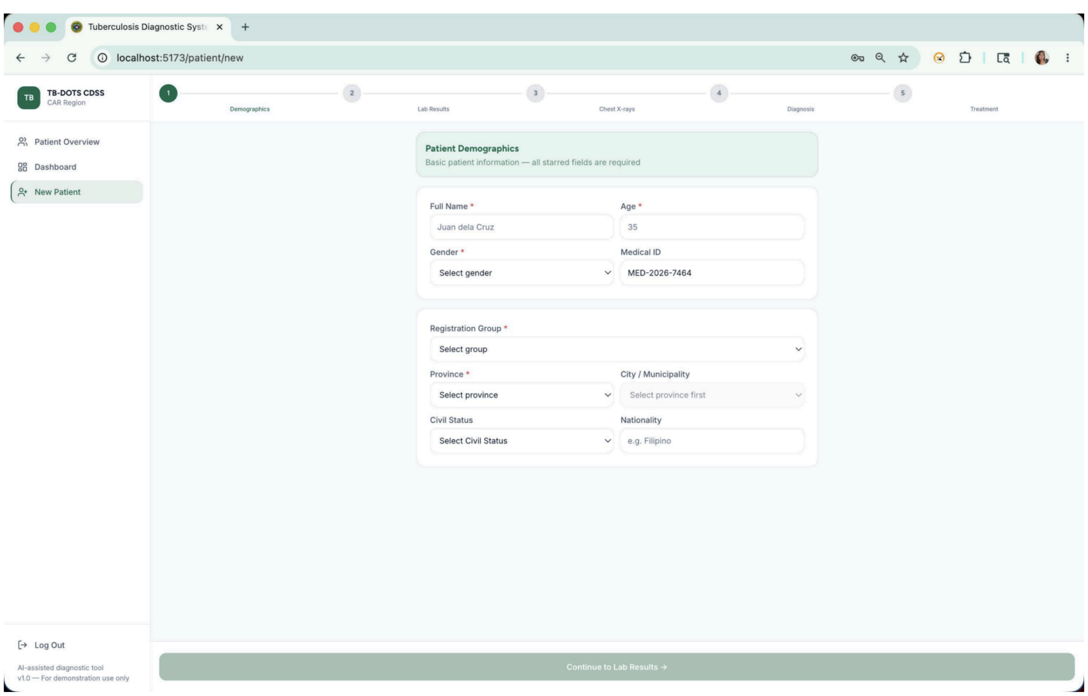
The validated machine learning models are embedded into a Clinical Decision Support System (CDSS) designed for healthcare providers in TB-DOTS facilities. The system is structured around a clear separation of responsibilities: the predictive model estimates the probability of treatment success or failure, the SHAP layer identifies the factors contributing to the model's prediction, the interface visualizes risk and contributing factors, and the clinician interprets these outputs in the context of patient care.

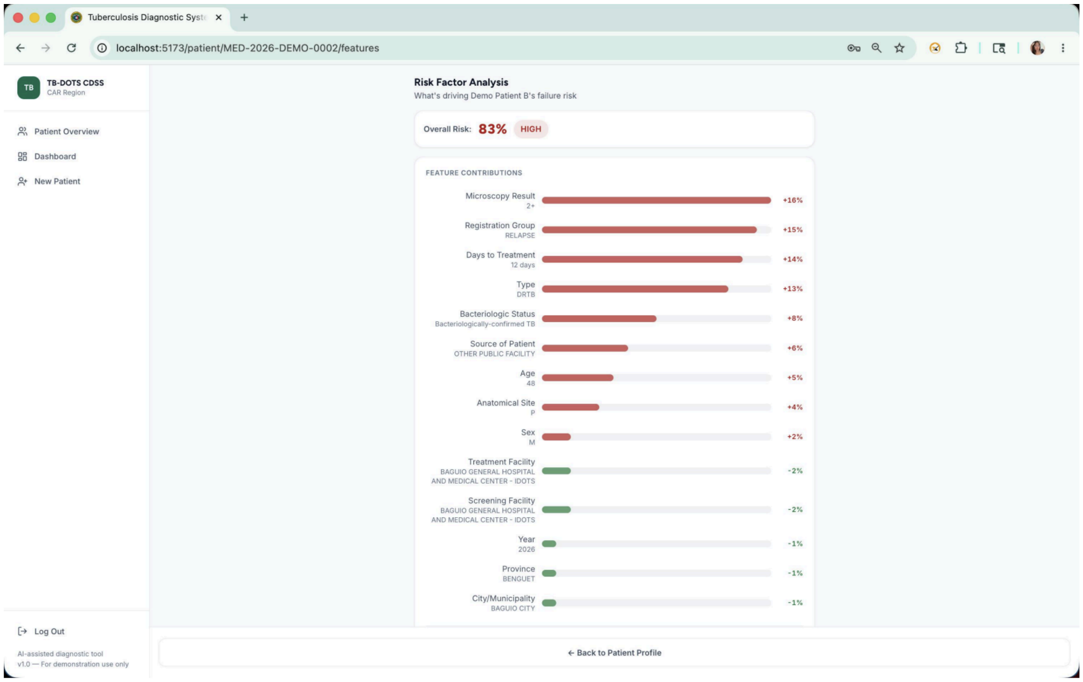
Key Features of the CDSS

The Clinical Decision Support System (CDSS) is designed as a risk stratification and explanation tool to assist healthcare providers rather than act as an autonomous diagnostic or treatment-prescribing system, ensuring that final clinical judgment remains with the treating professional. At its core, the system features a predictive dashboard that displays the probability of treatment failure and success using color-coded risk stratification, where green signifies low risk <20%, yellow indicates medium risk 20% - 50%, and red denotes high risk >50%, allowing providers to easily prioritize high-risk patients for closer monitoring and targeted interventions. To provide transparency, a risk factor visualization layer utilizes SHAP-based feature contribution visualizations to highlight specific variables driving the risk score—such as adherence gaps, laboratory results, or diagnostic delays—enabling clinicians to inspect exactly which patient factors the model identified as concerning. This quantitative data is further contextualized by a narrative explanation layer, where a constrained small language model verbalizes the SHAP-based attributions into readable clinical text, explaining, for instance, how a seven-day delay in treatment initiation or low adherence in a specific month quantitatively elevates failure risk. Finally, the CDSS supports dynamic updates by accepting new follow-up data—including monthly weight, smear results, and adherence percentages—and recomputing risk scores in real time to reflect the patient's current clinical status throughout the entire duration of their treatment.

The CDSS is framed as a risk stratification and explanation tool, not as an autonomous diagnostic or treatment-prescribing system. Final clinical judgment remains with the treating healthcare professional.

Figure 4
Clinical Decision Support System Workflow and User Interface Mockup





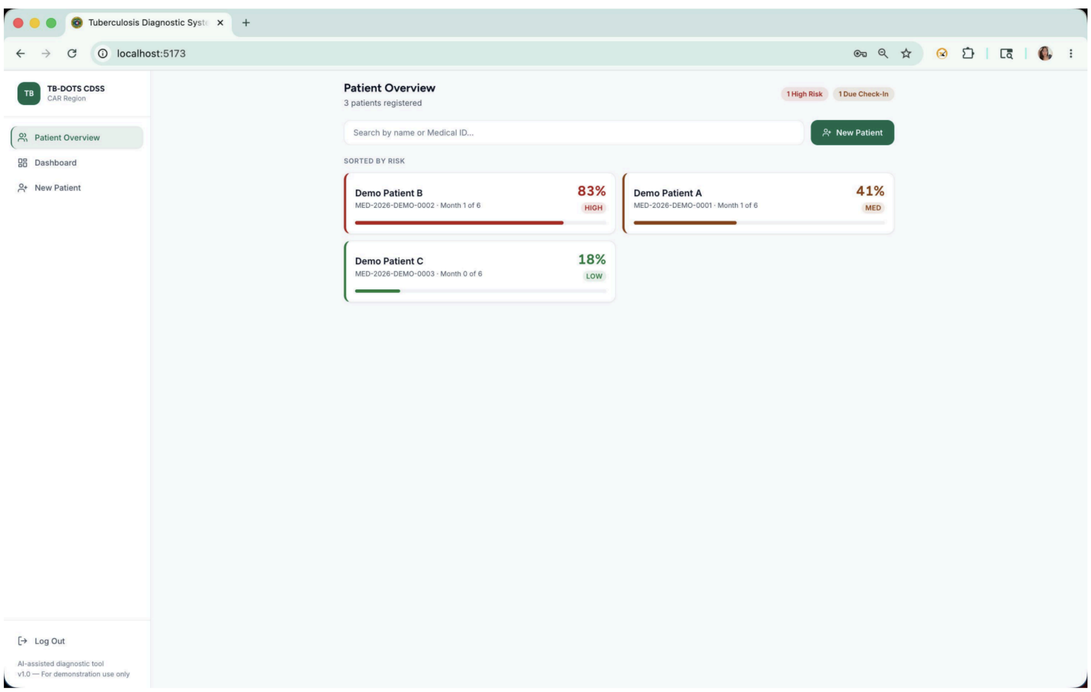


Figure 4 illustrates the integrated workflow of the CDSS. Patient data flows into the predictive model, which generates a treatment failure risk probability. SHAP explanations are automatically computed and visualized, showing the contributing factors. The healthcare provider reviews both the risk score and the contributing factors, and then makes clinical decisions informed by but not dictated by the system.

Model Generalization and Clinical Utility

The validation set performance (AUC = 0.7380, Recall = 52.13% for failure class) demonstrates that Random Forest with SMOTE-ENN generalizes well beyond the training cohort. This level of discriminative performance, combined with interpretable SHAP explanations, provides clinically useful support for identifying high-risk patients. The 52% recall for the failure class indicates that the model captures approximately half of all patients who will experience treatment failure, enabling earlier identification and intervention compared to standard paper-based monitoring.

The moderate AUC (0.7380) reflects the inherent difficulty of predicting treatment failure from baseline clinical and operational data: many patients with similar risk profiles have disparate outcomes due to unmeasured social determinants, medication side effects, or unpredictable treatment adherence. However, the model's ability to identify a substantial minority of high-risk patients, combined with interpretable explanations of why they are flagged, provides meaningful value for TB control programs aiming to concentrate limited resources on the most vulnerable populations.

Conclusions

This study investigated the use of machine learning models to predict treatment outcomes among PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes.

The study demonstrates that integrating temporal monitoring data with static baseline features improves predictive performance over static data alone. The application of SMOTE-ENN effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Shapley Additive Explanations prove invaluable in bridging the gap between predictive performance and clinician interpretability: by surfacing the specific patient-level factors driving each risk prediction, SHAP enables healthcare providers to understand and trust the model's outputs.

Predictive models—particularly Random Forest with SMOTE-ENN—have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly. The embedded CDSS, with its real-time risk updates and narrative explanations, provides a scalable tool for TB program managers to strengthen treatment outcomes. Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes in the Cordillera Administrative Region.

Recommendations

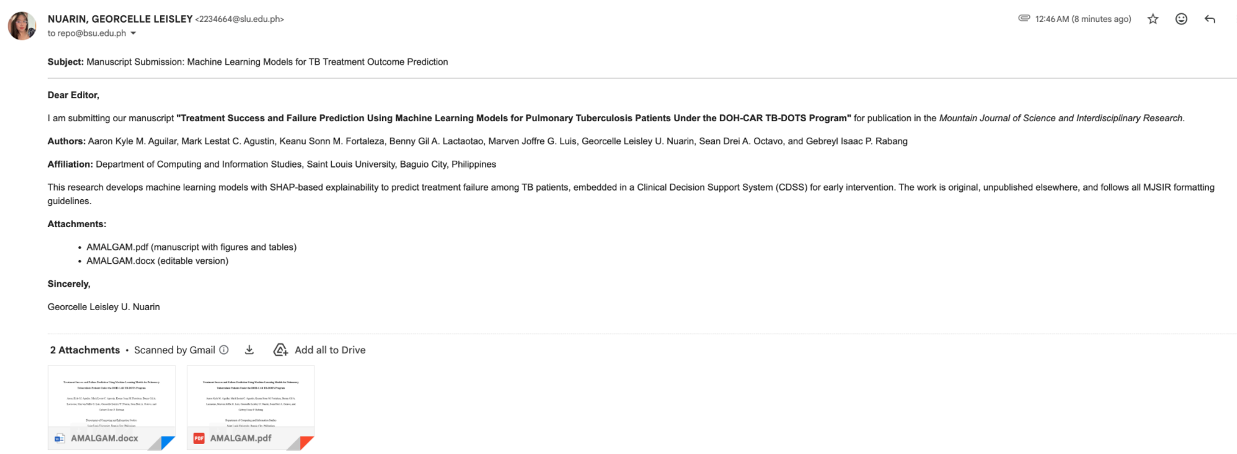
Based on the findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance model generalization and explore hybrid modeling approaches combining deep learning and tree-based methods. Investigations into additional features such as social determinants of health and geographic risk factors are encouraged. Enhancing model interpretability—particularly of Bi-LSTM attention outputs—could improve clinician trust and adoption. Standardizing digital record-keeping practices across TB-DOTS facilities will support development of more comprehensive longitudinal datasets for future studies and enable real-time implementation of the CDSS at scale.

Acknowledgment

The authors would like to express their sincere gratitude to Saint Louis University for providing the academic environment and resources that made this research possible. We extend our appreciation to the School of Accountancy, Management, Computing, and Information Studies, specifically the Department of Computing and Information Studies, for continuous support and guidance. Our deepest thanks go to Dr. Randy Domantay for leadership and encouragement, and to our research adviser, Dr. Josephine S. Dela Cruz, for expert guidance, insightful feedback, and unwavering support throughout the research. We also acknowledge the

collaboration and contributions of our partner researchers from the SONAHBS Department of Pharmacy, whose expertise significantly enriched this study.

Appendix A: *Proof of Submission to a Journal Publication*



NUARIN, GEORCELLE LEISLEY <2234664@slu.edu.ph>
to repo@bsu.edu.ph

12:46 AM (8 minutes ago)

Subject: Manuscript Submission: Machine Learning Models for TB Treatment Outcome Prediction

Dear Editor,

I am submitting our manuscript **"Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the DOH-CAR TB-DOTS Program"** for publication in the *Mountain Journal of Science and Interdisciplinary Research*.

Authors: Aaron Kyle M. Aguilar, Mark Lestat C. Agustin, Keanu Sonn M. Fortaleza, Benny Gil A. Lactaotao, Marven Joffre G. Luis, Georcelle Leisley U. Nuarin, Sean Drei A. Octavo, and Gebreyl Isaac P. Rabang

Affiliation: Department of Computing and Information Studies, Saint Louis University, Baguio City, Philippines

This research develops machine learning models with SHAP-based explainability to predict treatment failure among TB patients, embedded in a Clinical Decision Support System (CDSS) for early intervention. The work is original, unpublished elsewhere, and follows all MJSIR formatting guidelines.

Attachments:

- AMALGAM.pdf (manuscript with figures and tables)
- AMALGAM.docx (editable version)

Sincerely,

Georcelle Leisley U. Nuarin

2 Attachments • Scanned by Gmail Add all to Drive

AMALGAM.docx AMALGAM.pdf

References

Alruhaymi, A., & Kim, C. (2021). Why can multiple imputations and how (MICE) algorithm work? Open Journal of Statistics, 11(5), 759–777. https://doi.org/10.4236/ojs.2021.115045

Amante, T. D., & Abdosh, T. (2015). Risk factors for unsuccessful tuberculosis treatment outcome (failure, default and death) in public health institutions, Eastern Ethiopia. Pan African Medical Journal, 20. https://doi.org/10.11604/pamj.2015.20.247.3345

Department of Health Republic of the Philippines. (2022). DOH-CAR, USAID recalibrate plan to detect and mitigate tuberculosis hot spots. https://caro.doh.gov.ph

Department of Health [DOH]. (2014). Manual of procedures for the national tuberculosis control program (5th ed.). Department of Health.

Hinay, A. A., Jr., Makinano, J. C., Obenieta, P. J. M., & Cuizon, C. G. (2025). Factors associated with treatment failure among TB patients in Davao City, Philippines. Western Pacific Surveillance and Response Journal, 16(1).

Kaur, G., & Sharma, A. (2021). Machine learning approaches for tuberculosis risk prediction: A systematic review. Journal of Biomedical Informatics, 115, 103689.

Kossakov, R., Yergaliyev, T., & Bekova, A. (2024). Interpretable machine learning for tuberculosis outcome prediction using SHAP. Scientific Reports, 14, 14702.

Lu, J. Y., Zhu, J., Zhu, J., & Duong, T. Q. (2022). Long-short-term memory machine learning of longitudinal clinical data accurately predicts acute kidney injury onset in COVID-19: A two-center study. International Journal of Infectious Diseases, 122, 802–810. https://doi.org/10.1016/j.ijid.2022.07.034

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765–4774.

Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Peck, J. M., Rebala, S., Kunapuli, R. K., Halpern, Y., Fong, R., Lage, I., Soulas, C., Khoshnevisan, S., Lee, S.-I., Caruana, R., & Just, H. A. (2020). From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence, 2(1), 56–67.

Nayebi, A., Tipirneni, S., Foreman, B., Ratcliff, J., Reddy, C. K., & Subbian, V. (2022). Recurrent neural network based time-series modeling for long-term prognosis following acute traumatic brain injury. https://pmc.ncbi.nlm.nih.gov/articles/PMC8861707/

Noubiap, J. J., Nansseu, J. R., Nyaga, U. F., Nkeck, J. R., Nkengfack Nkoke, G., & Bigna, J. J. (2019). Prevalence of diabetes mellitus in people living with HIV: A systematic review and meta-analysis. Diabetes & Metabolism, 45(4), 327–334.

Peng, A., Kong, X.-H., Liu, S., Zhang, H., Xie, L., Ma, L., Zhang, Q., & Chen, Y. (2024). Explainable machine learning for early predicting treatment failure risk among patients with TB-diabetes comorbidity. Scientific Reports, 14(1), 7567. https://doi.org/10.1038/s41598-024-57446-8

Rajpurkar, P., Chen, E., Banerjee, O., & Topol, E. J. (2022). AI in health and medicine. Nature Medicine, 28(1), 31–38.

Tobin, E. H., & Tristram, D. (2024). Tuberculosis overview. In StatPearls. StatPearls Publishing. http://www.ncbi.nlm.nih.gov/books/NBK441916/

World Health Organization. (2023). Global tuberculosis report 2023. World Health Organization. https://www.who.int/teams/global-tuberculosis-programme/tb-reports

Appendix D: Proof of Presentation